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ENGINEERS &
SCIENTISTS

CADILLAC AREA GROUNDWATER CONTAMINATION
REMEDIAL INVESTIGATION
CADILLAC, MICHIGAN

FINAL DRAFT

EPA Region 5 Records Ctr.



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FINAL DRAFT

Prepared for:

MICHIGAN DEPARTMENT OF NATURAL RESOURCES
LANSING, MICHIGAN

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EXECUTIVE SUMMARY

The industrial park area in Cadillac, Michigan, consists of nearly 40 industrial facilities. Past uncontrolled releases of contaminants to the ground by several of these facilities contaminated groundwater in aquifers underlying the park. The contaminants of concern are volatile organic compounds (VOCs) (chlorinated and aromatic hydrocarbons such as trichloroethene and xylene) and hexavalent chromium (derived from chromium plating wastes). The presence of contaminants within the upper aquifers created concern for potential impact on a lower aquifer, which supplies potable water to the city from a wellfield located near the center of the industrial park. To address these concerns, the Michigan Department of Natural Resources (MDNR) contracted E.C. Jordan Co. (Jordan) to conduct a Remedial Investigation/Feasibility Study (RI/FS) of the industrial park area groundwater.

The Cadillac industrial park includes two Superfund sites: the Northernnaire Plating Company (Northernnaire) site and the Kysor of Cadillac, Inc. (Kysor) site. The Northernnaire site and associated metals contamination were the subjects of an RI/FS, also performed by Jordan for MDNR. The Northernnaire FS operable unit for the groundwater will be performed upon completion of the Cadillac RI and included with the Cadillac FS. Groundwater contaminants identified by the Northernnaire study were shown to be moving at a rate and in a direction that have no immediate risk to human health. With multiple and overlapping plumes from the Superfund sites and other sources, the MDNR and U.S. Environmental Protection Agency (USEPA) agreed it would be more appropriate to include assessments of Kysor-related and other VOC contamination in a comprehensive groundwater study of the area.

MDNR began the Cadillac Area RI with state 307 Program funds. However, it was anticipated that USEPA would elect to participate in the project because of the Northernnaire Superfund site. Therefore, the RI was conducted under USEPA guidance available at the time of field work. The scopes for the risk assessment (RA) and FS were subsequently modified, as required under recent USEPA guidance issued as a result of the Superfund Amendments and Reauthorization Act (SARA) of 1986. The multiple contaminant sources and plumes within the study area make this RI/FS somewhat unique. Furthermore, one potentially responsible party (PRP), Four Winns, Inc. (Four Winns), elected to participate in the study with its own consultant Aquatic Systems, Inc. (ASI). This RI includes information gathered by ASI with guidance from the MDNR/Jordan Work Plan, as well as relevant portions of previous investigations by Jordan and others.

This document presents the methods and findings of the intensive study of the geology, hydrology, and industrial contamination at the Cadillac Industrial Park. This Cadillac Area study consists of three major parts - an RI, a baseline RA, and a feasibility study (FS) evaluating proposed site remedial alternatives. The RI and RA portions of the Cadillac study are presented in this document; the FS will be presented as a separate document. The findings and conclusions of the RI and RA are discussed in the remainder of this Executive Summary.

REMEDIAL INVESTIGATION

The major portion of the RI involved subsurface explorations to determine areal and vertical extent of groundwater contaminants, suspected contaminants in soils, and site geology and hydrogeology. More than 80 monitoring wells and piezometers were installed, ranging in depth from approximately 50 to 300 feet below ground surface. In most explorations, vertical sampling of groundwater was conducted with field screening for VOCs to aid in well screen placement and to describe the vertical contaminant profile. Other methods including soil gas surveys, soil borings, water level measurements, laboratory physical and chemical testing, single-well permeability tests, pump tests, and geophysics (i.e., gamma-ray and resistivity logging) were used to characterize site conditions and as guidance to indicate subsequent exploration locations during the program.

Site geology consists of five strata: three sand aquifers alternately separated by two clay aquitards. The sands are typically fine- to medium-grained, with occasional gravel; the clay may contain some silt and gravel. The upper aquitard, ranging in thickness from zero to about 20 feet, slopes upward from the southwest to the northeast, until it pinches out along a northwest-southeast trending line halfway across the study area. The lower aquitard appears to be a clay-sand-clay sequence, which is thickest (i.e., 30 to 35 feet) in the immediate vicinity of the city wellfield. This lower aquitard, previously thought to be extensive and thick, was shown to be thick only below the wellfield, becoming substantially thinner (i.e., 3 to 8 feet) short distances in all directions away from the wellfield. Such a thin aquitard may be providing only limited protection from contaminant migration to the underlying municipal water supply aquifer.

Groundwater must be considered separately in each of the three aquifers, although there is vertical seepage downward between aquifers. The sand above the upper aquitard defines the shallow aquifer, in which groundwater flows north to northeast in the contaminated part. An average groundwater flow rate of about 200 ft/yr was estimated, based on hydraulic gradients of 0.001 to

0.003 ft/ft and hydraulic conductivities of about 1×10^{-2} to 3×10^{-2} cm/sec. The intermediate aquifer is confined in areas where the upper aquitard exists and is unconfined elsewhere; flow direction is north-northwest.

Where the shallow and intermediate aquifers merge, the latter dominates and groundwater flow turns northwest. The hydraulic conductivity of the intermediate aquifer is similar to the shallow aquifer; however, the gradient ranges only from 0.001 to 0.0017 ft/ft, resulting in average groundwater flow rates of about 80 ft/yr. The municipal water supply aquifer lies below the lower aquitard. The soils are similar to those in the aquifers above, but may locally contain more gravel. Only three monitoring wells and seven city production wells penetrate the lower aquitard and aquifer. The flow direction east of the wellfield is westward toward the city wells (probably under their pumping influence); regional flow directions and gradients are not known.

The exploration program was designed to (1) complete the description of contaminant distribution in the shallow and intermediate aquifers where it was only partially defined; and (2) identify other plumes and sources in suspect areas.

The program further delineated known sources at Northernnaire, Kysor, Four Star Corporation (Four Star) (former), and Joynts (former). New plumes were defined in the vicinity of Rexair, Inc. (Rexair) and Four Winns. Suspected groundwater contamination from past activities at the Ingraham property could not be clearly separated from the Kysor and Four Winns plumes. Only one suspected source is located horizontally upgradient of the city wellfield (i.e., Rexair). Therefore, the city wellfield becomes a convenient means by which to differentiate contaminant plumes and their potential environmental impact; those located upgradient (east) and those located downgradient (west).

Much of the shallow and intermediate aquifers is contaminated with large plumes, extending northward from Kysor to just beyond Thirteenth Street. High concentrations of total VOCs in the shallow aquifer were noted near Kysor (115 parts per million [ppm] in June 1987), Four Star (37 ppm in December 1983), and Four Winns (18 ppm in June 1987). The principal contaminants are trichloroethene (TCE) at Kysor and Four Star, and 1,1,1-trichloroethane (TCA) at Four Winns. The large areal extent of the Kysor plume in the shallow aquifer resulted in this plume merging with the Four Star and Four Winns plumes. The VOC plume in the intermediate aquifer, resulting primarily from leakage through the upper aquitard, has a maximum concentration of about 12 ppm. Lesser contamination by VOCs (0.59 ppm) was detected north of Thirteenth Street beneath a residential neighborhood; this represents the most downgradient extent of contamination. No VOC contamination was detected in either aquifer upgradient of Kysor.

The Northernnaire plume, containing up to approximately 1 ppm of hexavalent chromium, is much less extensive areally, but lies within the same general area of the intermediate aquifer as the VOC plume. Recent analyses for hexavalent chromium in the intermediate aquifer show northward movement of the plume and possibly some dilution. Hexavalent chromium detection in the shallow aquifer is sporadic. Detections downgradient from Northernnaire were interpreted as a small plume released when the facility closed.

The VOC plume near the Rexair facility is distinct from the other plumes, and contains up to 153 ppm of total VOCs (principally TCE). This plume, the only one identified horizontally upgradient of the city wellfield, shows evidence of being drawn downward to the lower aquifer under the influence of wellfield pumpage. Concentrated TCE will also tend to sink through an aquifer since it has a density greater than water and the sandy soils at the site have a high hydraulic conductivity which infers a low entry pressure for the solvent to begin to penetrate the aquifer. While no residual source of VOCs in soils was located in samples collected on Rexair property, the following factors indicate that it is highly probable the plume originated there: (1) location of the plume; (2) groundwater flow direction and velocity; (3) historic use of TCE at the facility; and (4) absence of contamination in groundwater upgradient of the property. The municipal wells have not yet displayed evidence of impact by this plume. Rexair is continuing investigation of this contamination with additional soil gas studies and monitoring well installations. Initial soil gas studies by Rexair's consultants have indicated an area of elevated TCE concentration on the eastern side of the plant.

Only one area of soils, the former solvent disposal area behind Kysor, was identified as having significant residual VOC contamination. These soils contained VOCs, primarily TCE and xylene, at up to 750 ppm and to depths of 25 feet. The contaminants were left at depth after a partial remediation by excavation of the top 6 feet of soil in 1981. These VOCs appear to be a continuing source of groundwater contamination in the shallow and intermediate aquifers west of the city wellfield.

The principal mechanism for decreasing contaminant concentrations in groundwater is the process of dispersion as the plume travels within and between aquifers. This mechanism is important because the presence of downward vertical hydraulic gradients indicates that the contaminants may eventually reach the lower aquifer and be captured by the city wellfield cone of influence. It was not a purpose of this RI to directly investigate the likelihood of contaminants downgradient of the city wellfield reaching the lower aquifer, causing subsequent impact to the municipal water supply. However, future potential risk must be evaluated for the intermediate and shallow aquifers in the RA, because controls are not established for aquifer use and domestic wells could be installed unknowingly in areas of contamination. The proximity of the highly contaminated plume near Rexair, situated upgradient of the city wellfield aquifer, raises concern for possible significant impact to the municipal water supply. For this reason, a separate drilling phase in this RI investigated the lower aquifer east of the wellfield, and assessed the potential for contamination to reach it.

In summary, five contaminant sources and seven plumes in groundwater were identified in the study area. The five source locations are (1) Northernnaire, (2) Kysor, (3) Rexair (probable), (4) Four Winns, and (5) Four Star/Joynts (former).

The seven plumes are the following:

1. A hexavalent chromium plume in the intermediate aquifer situated just above the lower clay aquitard. Northernnaire was the source; present concentrations of hexavalent chromium ranging up to 1.0 ppm were detected (Northernnaire Supplemental RI, January 1987).
2. A small hexavalent chromium plume in the shallow aquifer. This plume was detected in 1986 in a well that had not shown contamination in the previous two years. This plume, interpreted as a small slug released from Northernnaire, should eventually merge with the intermediate aquifer and be further diluted.
3. A large, concentrated VOC plume (up to 115 ppm) in the shallow aquifer caused primarily by past solvent dumping behind Kysor. This plume merged with other VOC releases from Four Star/Joynts (former) to the northeast and Four Winns to the northwest.
4. A large but less concentrated VOC plume (up to 12 ppm) in the intermediate aquifer resulting from Kysor-related VOCs migrating from the shallow aquifer through the upper aquitard.

5. A small VOC plume (less than 0.1 ppm) at a depth of 55 feet in the North Park Subdivision. This plume may be a result of releases from the Four Winns/Joynts and Four Star plants.
6. Another small VOC plume (up to 0.59 ppm) in the North Park Subdivision at a depth of approximately 135 feet, which may also be from the Four Winns/Joynts and Four Star areas.
7. A highly concentrated VOC plume (up to 153 ppm) at a depth of 140 feet, located in the intermediate aquifer between Rexair and the city wellfield.

Only one plume (near Rexair) is located horizontally upgradient of the city wellfield. It therefore has the greatest potential to impact humans by migrating to the lower aquifer. Both a thin aquitard, as evidenced by deep borings, and increased downward vertical gradients across the lower aquitard resulting from city well pumping create potential for contamination to reach this lower aquifer. Vertical groundwater sampling and analysis during screened auger drilling indicated that the plume has already reached considerable depth (over 140 feet) within a short distance from Rexair.

The remaining of the contaminant plumes are located horizontally downgradient of the city wellfield and appear to pose a less immediate threat (if any) to the city water supply. It is believed that all potential downgradient receptors (i.e., domestic wells) were replaced by municipal water supplies. However, there are no controls to prevent present or future use of the contaminated aquifers through existing or future well installations.

BASELINE RISK ASSESSMENT

The risk assessment (RA) establishes a baseline of no-action (contaminant plumes are unremediated) risk posed by the study area groundwater contamination in which current and future potential risk to human health and the environment are assessed. The groundwater plumes evaluated for the Cadillac Area study include:

- o Shallow Plume
- o Intermediate Plume
- o Northernnaire Plume
- o Rexair Plume

These major plumes were further subdivided, where possible, into discrete plumes for the discussion of fate and transport.

The following compounds have been identified in the RA as chemicals of concern based on their inherent toxicity, frequency of detection, and concentrations:

1,1,1-Trichloroethane
Acetone
Toluene
Trans-1,2-Dichloroethene

Trichloroethene
Chloroform
1,1-Dichloroethene
1,2-Dichloroethane
Methylene Chloride
Tetrachloroethene
Hexavalent Chromium
Cyanide

Not all chemicals of concern are present in every plume. Each of these chemicals is potentially hazardous to human health. Also, concentrations of chemicals of concern in all four plumes do not meet applicable health-based requirements for these chemicals.

The exposure assessment identifies the populations and environments which are at risk of exposure to the chemicals in groundwater. Various modeling techniques were used to project contaminant migration in each of these plumes to predict chemical concentrations at hypothetical groundwater receptors.

The public health exposure assessment identifies the human populations that may live in and around the site. Currently, no domestic supply wells intercept the contaminated groundwater plumes. However, since there are no restrictions for use of the shallow and intermediate aquifers, the potential for impact to the populations does exist. Also, a potential for impact to the municipal wells is present.

The environmental exposure assessment describes the terrestrial systems present that are potential ecological receptors of contamination. There are no potential routes of exposure identified for terrestrial organisms because current soil contamination lies below ground surface.

The risk to humans and environments potentially exposed to chemicals identified in the groundwater has been characterized. Virtually all the plumes pose a risk to human health, both at current contaminant concentrations and predicted future concentrations. Therefore, remedial actions are necessary to reduce risks to acceptable levels. Although many uncertainties and limitations are inherent in the risk assessment process, it is believed that this risk characterization presents an accurate representation of the present and future potential risk to public health.

1.0 INTRODUCTION

E.C. Jordan Co. (Jordan) was retained by the Michigan Department of Natural Resources (MDNR) to conduct a Remedial Investigation/Feasibility Study (RI/FS) of groundwater contamination beneath the industrial park area in Cadillac, Michigan. This RI was funded by MDNR through the state 307 Program under Contract Number 1525. While this RI was undertaken as a state-funded project, it was anticipated that the U.S. Environmental Protection Agency (USEPA) would become directly involved because the Northernnaire Plating Company (Northernnaire) Superfund site is within the study area. Nominating the entire Cadillac study area as a National Priorities List (NPL) site was also considered. Accordingly, the 1986 RI/FS Work Plan was modeled as close to USEPA format as possible with existing guidance to enable the study to become a joint MDNR/USEPA project with minimal changes.

Jordan's association with the Cadillac Area site began in 1984 with the investigation of the Northernnaire Superfund site. Since then, Jordan and MDNR have worked cooperatively on several field investigative programs within the study area, ranging from surficial soil sampling and building inspection (of the Northernnaire facility) to the installation of 300-foot-deep monitoring wells adjacent to the city wellfield for the Cadillac Area Groundwater Study. Several interim and final reports describing site and contaminant characteristics were issued by Jordan for MDNR.

Also, beginning in the early 1980s, several limited hydrogeologic studies were conducted for some facilities that released volatile organic compounds (VOCs) to the soils and groundwater in the industrial park area. These studies did not fully determine the degree and extent of VOC contamination in the aquifers. However, it was apparent from these studies that VOC contamination was present at several locations and that contaminant plumes from separate sources were probably merging. Thus, a study involving soil and groundwater contamination of the entire industrial park area was required to fully assess the environmental fate and impact of the contaminants present, and to arrive at a comprehensive recommendation for remediation.

1.1 PURPOSE

The primary purpose of the Cadillac Area Groundwater Contamination RI (Cadillac Area RI) was to investigate the distribution of VOC and chromium contamination in groundwater beneath the Cadillac industrial park and adjacent areas in Haring Township, Michigan. Secondly, investigations were made to identify residual contaminants in soils at suspected source locations. Groundwater contamination in the industrial park was first detected in the late 1970s. Several investigations ranging in size and scope were conducted for both individual facilities within and residences north of the park, which had reported contamination in private wells. The current Cadillac Area RI is intended to provide a comprehensive investigation of the industrial park groundwater contamination through review of previous investigation reports, combined with an extensive field program of soil borings, monitoring well installations, and chemical analyses of groundwater and soil samples.

The field program provided geologic, hydrologic, and groundwater quality data necessary to assess the degree of contamination, identify sources, describe migration pathways, and identify possible receptors. A baseline risk assessment (RA) of the environmental fate and impact of the contamination will be made based on findings of the RI. Finally, the baseline RA will be used to conduct an FS to identify remedial alternatives for aquifer restoration based on clean-up objectives.

1.2 SCOPE

The scope of the Cadillac Area RI consists of characterizing VOC contamination in two aquifers beneath the industrial park and assessing the potential for this contamination to reach a third, lower aquifer, from which the city of Cadillac draws its municipal water supply. Prior investigations and historical information indicate that significant contamination of the aquifers is by VOCs and metals only. The findings from this RI are used in conjunction with a previous study by Jordan (i.e., the Northernnaire Site Supplemental RI), which assessed the extent of groundwater contamination by hexavalent chromium. Additional relevant studies conducted by MDNR and others were also reviewed. The FS of the Cadillac Area RI report will include evaluation of clean-up alternatives that are cost-effective and environmentally acceptable for both VOCs and hexavalent-chromium contamination.

Jordan performed several work tasks for MDNR as part of the scope of services for the Cadillac Area RI. These tasks were organized and completed as three separate phases: (1) Phase I - Initial Activities; (2) Phase II - Draft RI/FS Report; and (3) Phase III - Final RI/FS Report. Phase I included the following activities:

- o preparation of the Work Plan, Health and Safety Plan (HASP), and Quality Assurance Project Plan (QAPP)
- o preparation of a report (i.e., Current Site Description, September 1986) formulating and summarizing data gathered from the review of existing studies concerning the geology, hydrogeology, and groundwater quality of the Cadillac industrial park area
- o establishment of an on-site field office for conducting field operations and support services
- o provision of support, as needed, to MDNR in community relations programs for the site investigation

Phase II included the following field exploration programs to collect geologic, hydrologic, and groundwater quality data for site characterization:

- o a "pilot" program of testing the shallow, unsaturated soil zone for the presence of VOCs by soil-gas surveys (if successful, this technique would be used to screen areas for well installation locations)

- o installation of soil borings, monitoring wells, and piezometers by hollow-stem auger drilling to provide (1) soil samples for geologic classification and physical and chemical laboratory analyses; and (2) groundwater samples to field screen for the presence of VOCs
- o a supplemental program of drilling mud-rotary borings and installing monitoring wells to provide geologic data of the lower aquitard and piezometric data in the municipal water supply aquifer
- o sampling of groundwater from the new and selected existing monitoring wells by standard USEPA protocols. Groundwater samples were sent to USEPA Contract Laboratory Program (CLP) laboratories for chemical analysis of VOCs. Groundwater from 10 selected monitoring wells was also analyzed for dissolved total metals and hexavalent chromium
- o aquifer permeability testing by conducting single-well, rising-head tests in selected monitoring wells
- o a long-term pump test (supervised by MDNR) using the city pumping wells and monitoring responses in adjacent wells screened in the intermediate and lower aquifers to assess the effectiveness of the lower aquitard as a hydraulic barrier to contaminant transport between the aquifers.
- o collection of groundwater levels in the monitoring wells and piezometers to determine horizontal groundwater flow directions and vertical hydraulic gradients between and within aquifers
- o compilation and interpretation of available data to provide a comprehensive hydrogeologic evaluation and contaminant assessment of the study area, focusing on (1) vertical and horizontal distribution of contaminants in the groundwater; (2) identification of the probable contaminant sources and migration pathways, and fate of the contaminant plume(s); and (3) potential impact of the contaminants on the city water supply aquifer and other private wells
- o development of remedial alternatives and initial screening of these alternatives based on acceptable engineering practice, effectiveness, and cost, followed by a detailed evaluation of alternatives to determine feasibility and costs of implementation and maintenance
- o preparation of a draft report presenting the findings and conclusions of the RI and FS

Phase III will include the following activities:

- o conceptual design of selected remedial alternative(s), including cost estimates and information needed for initiation of subsequent remedial activities
- o incorporation of comments on the Draft RI/FS by MDNR and other reviewers into a Final RI/FS report

(Note: Certain tasks may be redefined due to the need to produce a risk assessment and FS under the Superfund Amendments and Reauthorization Act [SARA] guidance.)

1.3 PAST AND CURRENT SITE OPERATIONS

The Cadillac industrial park encompasses an approximate 1-square-mile area along the northern boundary of the city of Cadillac (Figure 1-1). The park is bounded by Thirteenth Street to the north, Mitchell Drive (Route 131) to the east, Wright Street to the south, and Leeson Avenue to the west. Extensive development of the park began in about 1955 and continues today. The city of Cadillac water supply wellfield is located near the center of the study area. Several private residences, including a trailer park, exist within the industrial park. Another residential neighborhood within the study area, referred to as the North Park Subdivision, is located adjacent to the northern boundary of the park, across Thirteenth Street, in Harring Township.

Currently, approximately 40 industries operate in the industrial park, many of which manufacture rubber and plastic products and other specialty items for the automotive industry. Several other types of manufacturers exist, including a fiberglass boat builder and a garment company. The majority of industries do not use hazardous compounds or materials in the manufacturing process. However, a few companies have used (and still use) chemicals such as organic solvents and metals in their operations.

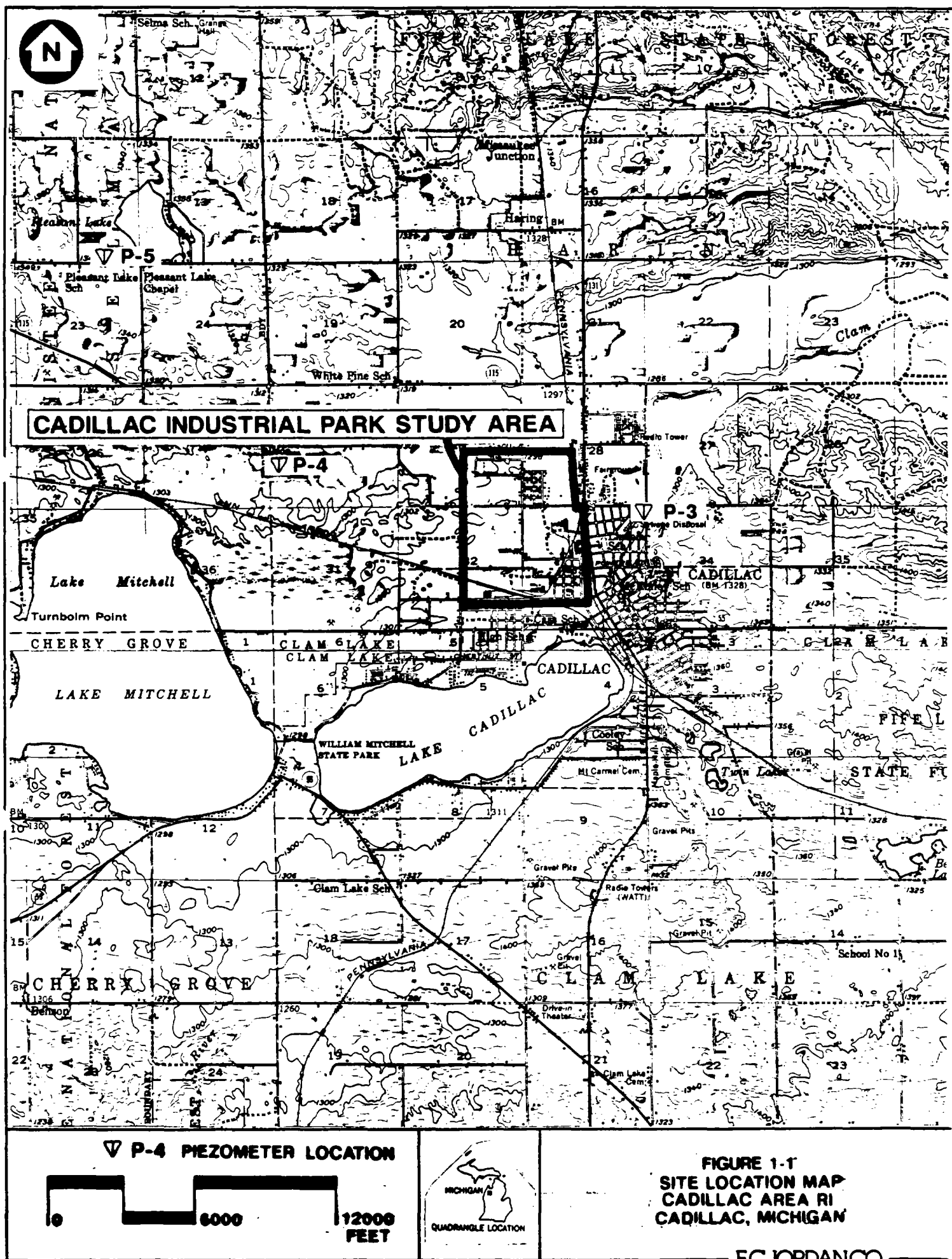
Following is a list of industries and businesses within the study area known to have used solvents and other hazardous materials. Approximate dates of operation are included in parentheses. The facility locations are presented in Figure 1-2.

- o Northern Plating Company (1972 to 1980)
- o Four Star Corporation (former) (1967 to 1979)
- o Kysor, Inc. (1955 to present)
- o Four Winns Company (Frisbie Street) (1975? to present)
- o Joynts (former)/Four Winns (Sixth Avenue) (1973 to present)
- o Rexair, Inc. (1969 to present)
- o Leo Ingraham Excavating Company (1970? to present)
- o Gene's Truck Service (1984? to present)
- o Wedin Corporation (1959 to present)
- o Cadillac Molded Rubber Company (1960 to present)

The degree to which each company may have contributed (if at all) to the existing groundwater contamination varies widely. The Cadillac Area RI focused on identifying the probable primary contributors and determining contaminant distribution in groundwater.

1.4 PRIOR INVESTIGATIONS

Groundwater contaminant studies were conducted for several industrial park facilities. Table 1-1 lists the investigative report, the year of issue, and



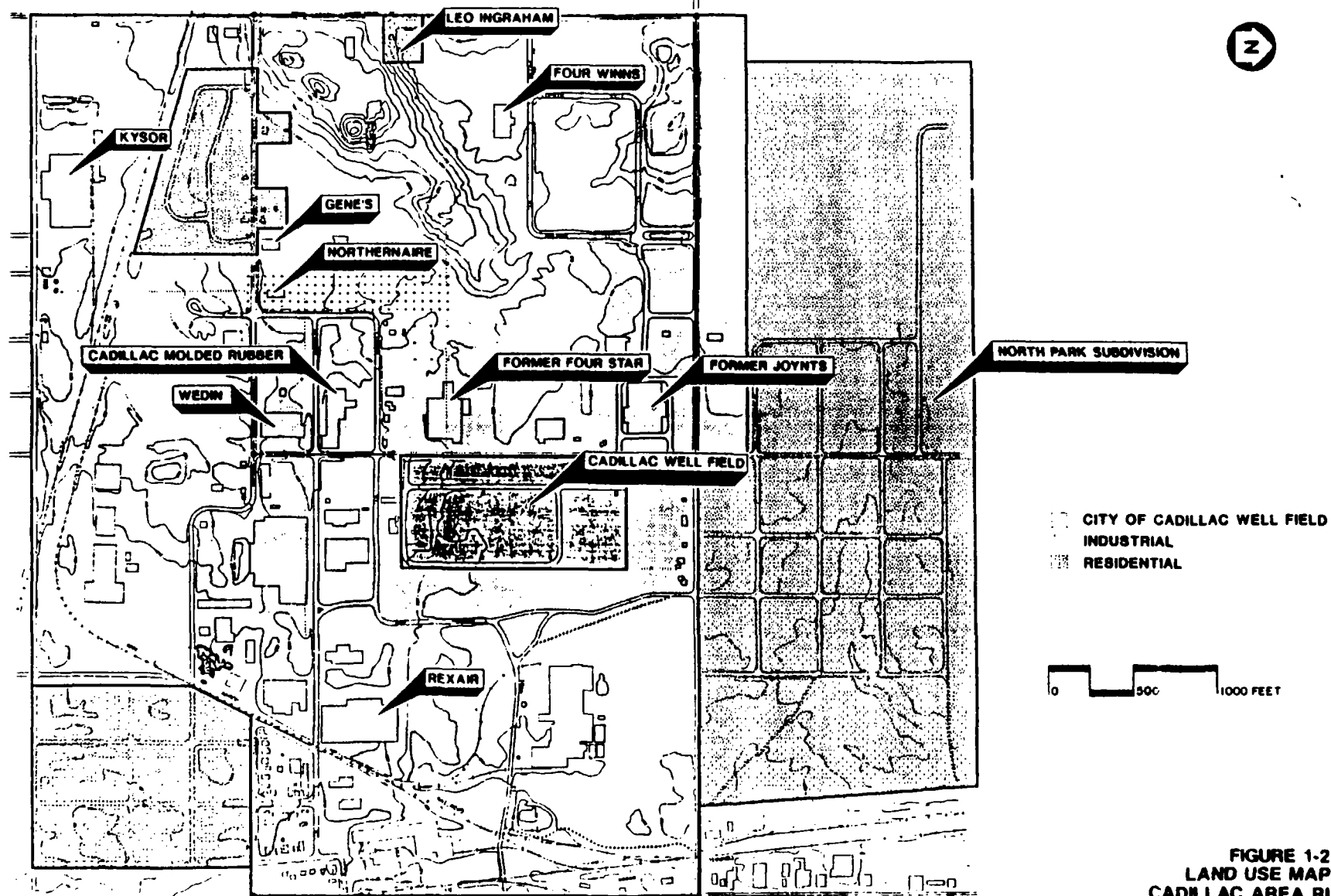


FIGURE 1-2
LAND USE MAP
CADILLAC AREA RI
CADILLAC, MICHIGAN

TABLE 1-1

PREVIOUS STUDIES AND
INFORMATION SOURCES

Chemistry Data of Domestic Wells; Cadillac Department of Public Works (DPW); 1978-1984.

Residential Well Chemistry Data; Michigan Department of Health; 1981-1985.

Letter Reports, Soil Excavation Behind Kysor; Fishback, Thompson, Carr, and Huber, Inc. (FTCH), and Michigan Department of Natural Resources (MDNR); 1981.

Kysor of Cadillac, Inc., Cadillac, Michigan, Hydrogeological Study; FTCH; Phase I, March 1982; Phase II, August 1983.

Groundwater Contamination at Brooks and Perkins' Plant in Cadillac, Michigan; Environmental Data, Inc.; March 1982; Phase II, February 1983.

Remedial Action Master Plan (RAMP); CH₂M Hill; May 1983.

Hydrogeologic Investigation Report for Four Star Corporation, Cadillac, Michigan; Gosling-Czubak Associates (GCA); August 1983; Progress Report, December 1983.

Preliminary Investigation TCE Spill, Cadillac Molded Rubber, Cadillac, Michigan; J.L. Wilcox and Associates, Inc. (Wilcox); November 1984; Addendum, January 1985; Addendum No. 2; Addendum No. 3 (not dated).

Remedial Investigation, Northernnaire Site, Cadillac, Michigan; E.C. Jordan Co. (Jordan); Phase II, May 1985.

A Focused Feasibility Study of Proposed Source Control Measures for the Northernnaire Site, Cadillac, Michigan; Jordan; May 1985.

Pump Test Data, Cadillac Molded Rubber Cleanup; Kraus and Kriscunas, P.C.; August 1985.

Industrial Survey Reports; MDNR; August 1985.

Hydrogeology of the Cadillac Industrial Park; MDNR; August 1986.

Cadillac Industrial Park Remedial Investigation, Task 2 Report, Current Site Description; Jordan; September 1986.

Supplemental Remedial Investigation, Northernnaire Site, Cadillac, Michigan; Jordan; January 1987.

Hydrogeologic Report, 4 Winns Boats, Cadillac, Michigan; ASI; January 1987.

TABLE 1-1
(continued)

PREVIOUS STUDIES AND
INFORMATION SOURCES

Cadillac Area Groundwater Contamination, Technical Memorandum, Information Relative to the City Water Supply Well Field and Rexair, Inc.; August 1987.

Well Field Evaluation, City of Cadillac; Keck; December 1987.

Four Winns, Inc., Hydrogeological Investigation Report, ASI; January 1988.

Technical Information Review and Soil Gas Study, Rexair Inc.; Cadillac, Michigan; EDI; February 1988.

NOTE: This table is not intended to be an all-inclusive list of relevant information, but rather lists the primary sources of information used in this RI.

the consultant conducting the study. To date, the most extensive studies have been for Northernair; Kysor of Cadillac, Inc. (Kysor); and the former Four Star Corporation (Four Star). These three facilities were identified in the studies as principal contributors to the groundwater contamination. The studies are summarized in the following subsections. Also, other known instances of waste disposal or spills in the industrial park are briefly described.

Leo Ingraham, Sr., had accepted waste materials from Kysor, and had either disposed of or stored the wastes on his property on Leeson Avenue. Contaminated soils, identified by visual inspection, were reportedly excavated and removed in 1981, at the same time Kysor was performing partial soils remediation at their Wright Street facility.

A trichloroethene (TCE) spill occurred while filling a storage tank at the Cadillac Molded Rubber Company (CRC) in 1984. The spill was believed to be relatively small. Affected soils were removed shortly afterwards, and monitoring wells and a recovery well were installed downgradient of the spill. The monitoring wells showed only very low concentrations of VOCs. MDNR permitted CRC to cease clean up measures after demonstrating that cleanup was effective and complete. No significant impacts from this incident are anticipated.

In 1985 and 1986, the Wedin Corporation investigated past disposal of sludge at its facility located between Sixth and Seventh streets. Soil samples in the disposal area had relatively low levels of some chlorinated and aromatic solvents. However, a sample of groundwater taken from a monitoring well located in the middle of the disposal area showed no detectable VOCs. Extraction Procedure (EP) Toxicity tests were done for leachability of metals in the sludge. The test values indicate probable low mobility of the metals; therefore, they are defined as non-hazardous waste by this particular test. The sludge materials (approximately 470 cubic yards) were subsequently excavated and removed in 1986. The excavated material was visually inspected and identified as a swarf with metal shavings from grinding and milling machine operations. The excavation was lined with plastic sheeting and backfilled with clean soil.

In 1986, MDNR personnel conducted an on-site inspection of the Four Winns facility on Frisbie Street. During this inspection, MDNR observed improper storage of solvent drums and indications of spills. Soil samples were taken and analyzed by the MDNR laboratory; results showed relatively high concentrations of VOCs, principally 1,1,1-trichloroethane (TCA). There was also concern for possible ketone spills (including acetone) from the storage and handling practices observed. MDNR requested that Four Winns initiate an investigation of the problem. Some investigations were done by Aquatic Systems, Inc. (ASI) for Four Winns which have been described in a January 1987 and a January 1988 report. The findings and limitations of these reports are discussed in Section 5.5.2 of this RI report. Solvent storage and handling procedures at Four Winns were improved, but no remediation is believed to have been conducted. Four Winns, subsequently identified as a Potentially Responsible Party (PRP) for the Cadillac Area RI, elected to participate in the RI, and again retained ASI to perform further investigations.

1.4.1 Northernnaire Plating Company Remedial Investigation

Northernnaire, located at the northwestern corner of Sixth Street and Seventh Avenue, operated from 1972 to 1981, providing custom chromium- and nickel-plating finishes to automobile and other metal parts. Improper waste handling and faulty sewer systems are believed responsible for releasing toxic compounds (including hexavalent chromium, cadmium, and cyanide) to the surrounding soils. As a result of leaching through soils to the groundwater, the highly soluble hexavalent chromium was detected in private wells downgradient of Northernnaire in 1978. The city revoked Northernnaire's wastewater discharge permit and plugged their discharge pipe, forcing Northernnaire to seek other means of disposing their wastewater.

Also in 1978, MDNR ordered Northernnaire to conduct hydrogeologic studies of the problem. In 1979, Northernnaire submitted a plan for the requested study, but later stated no funds were available for the investigation. A 1981 MDNR inspection of the site found the facility closed and apparently out of business. Further inquiries revealed that Northernnaire's assets had been purchased by Top Locker Enterprises of Florida, which had also ceased operations. Two children playing in the area received chemical burns from the hazardous wastes stored in containers outside the Northernnaire building. Emergency response measures were conducted by USEPA and MDNR shortly thereafter in 1982, including chemical testing; removal of waste drum and liquids from process tanks, decontaminating tanks, and building surfaces (with hypochlorite for treatment of cyanide); and removal of a section of the discharge sewer line.

The site was included on the NPL primarily based on concerns for possible impact of wastes on the municipal wellfield (about a quarter-mile northeast and apparently downgradient of Northernnaire) and the potential for direct contact hazards. A Remedial Action Master Plan (RAMP) was completed for Northernnaire in 1983. As of July 1987, the Northernnaire site is in Group 2 of the NPL and 59th overall of 770 listed sites.

For the Northernnaire RI (1984-1986), a series of borings and monitoring wells were installed by hollow-stem auger and screened auger sampling techniques to determine site stratigraphy and the vertical and horizontal distribution of site contaminants. During the initial phase of the RI, groundwater samples were analyzed for metals, cyanide, and VOCs. Cyanides were not detected; the VOCs noted were not attributable to Northernnaire. VOCs were detected at similar concentrations in both upgradient and downgradient wells at Northernnaire. There were no historical records or indications of significant solvent usage at Northernnaire.

Elevated levels of some metals were found in the groundwater; most were believed to derive from salting of area roads (some unpaved) for dust control in summer and ice control in winter. Hexavalent chromium, attributed to Northernnaire, was detected in two wells in the intermediate aquifer, which led to a second phase of well installations to determine its distribution. However, knowing that VOC contamination was the focus of the future Cadillac Area RI, the screened auger groundwater samples were analyzed for TCE in addition to hexavalent chromium. The well screens for the Northernnaire study monitoring wells (identified by the designation "MW") were placed in the zone of highest

hexavalent chromium concentration, which usually did not correspond to the zone of highest TCE contamination.

A remedial design for removal of contaminated soils was completed by Jordan for MDNR in 1987. Remediation requires limited soils excavation and removal on the western side of the building and beneath building drains, and removal of the sewer line and sediments. The design has not yet received final approval, but it is expected to be implemented in 1988.

1.4.2 Kysor of Cadillac, Inc. Investigation

The Kysor facility, located on Wright Street on the southern edge of the industrial park, is a large manufacturer of truck parts. Use of solvent cleaner and degreasers is common; past disposal practices included dumping barrels of spent solvent directly on the soils behind the plant.

A series of borings and wells was installed for the Kysor facility hydrogeologic studies conducted by Fishbeck-Thompson-Carr and Huber in 1980 and 1983. A total of 24 borings with monitoring wells is widely distributed over the industrial park with several in the vicinity of the city wellfield. These wells (designated by "K") monitor both the shallow and intermediate aquifers (see Sections 4 and 5). The well installations consisted principally of 4-inch galvanized steel casings with 2- to 5-foot-long stainless steel screens. Vertical screening for VOCs in the groundwater was conducted during installation. Analysis of groundwater samples from K-4 in 1983 found elevated levels of TCA, TCE, tetrachloroethene (PCE), 1,1-dichloroethene (DCE), 1,2-dichloroethane (DCA), xylene, toluene, ethylbenzene, and chloroform. Specific details were not reported on the monitoring well installation techniques. However, boring logs indicate that the borehole annulus was not typically grouted after installation of the well. Without a grout seal (usually powdered bentonite) in the annulus, the effectiveness of the upper clay aquitard to restrict downward groundwater flow is essentially eliminated at the well point location. A conduit for contaminant migration from the shallow to the intermediate aquifer exists under the impetus of measured downward hydraulic gradients.

The 1983 hydrogeologic study for Kysor also included a 24-hour pump test of City Well Number 7. Water levels in the intermediate aquifer were monitored in nearby K-24. The report concluded that there was little or no apparent influence of pumping in the municipal well aquifer on groundwater flow in the intermediate aquifer. However, MDNR and Jordan are concerned that (1) the test was inadequate because it was of limited duration; and (2) the apparent effectiveness of the aquitard at the wellfield might not extend throughout the study area.

Partial remediation of the contaminated soils at the disposal area was conducted by Kysor in 1981. Excavation depth averaged approximately 6 feet in two primary removal areas. Approximately 700 cubic yards of soil were removed; MDNR observers noted strong odors remaining in the completed excavation pit. Samples taken by MDNR later showed some relatively high concentrations (i.e., greater than 100 parts per million [ppm]) of TCE, TCA, PCE, ethylbenzene, and xylenes remaining in soils. The excavation was backfilled with clean material.

1.4.3 Four Star Corporation Investigations

Four Star, located on Sixth Avenue and directly west of the municipal well-field, manufactured polished aluminum and stainless steel trim for the automobile industry from 1965 to 1979. A vapor degreaser, involving TCE and PCE, was used at the facility from 1978 to 1979. Apparently, releases and spills of these compounds resulted in significant groundwater contamination, which continues to migrate northward. The facility closed in 1979; Mi-Cad Manufacturing is currently located there.

Two series of monitoring wells were installed to define contaminant distribution at the Four Star site. Four Star agreed to an order for site investigations by MDNR; two phases of a hydrogeologic study were performed by Gosling-Czubak Associates in 1983. The study consisted primarily of shallow monitoring wells installed in the general direction of groundwater flow between the Four Star building and Thirteenth Street. Soil samples near the facility were also taken for geologic description and chemical analyses. The 26 Four Star monitoring wells (designated by "FS") do not extend more than 20 feet below the water table (i.e., 52 feet below ground surface). Concerned that the plume was inadequately defined by these shallow wells, MDNR installed additional wells (the "DNR" series) to further define the Four Star TCE/PCE plume and investigate contamination in the North Park Subdivision. The DNR-series monitoring wells generally extend about 67 feet below ground. No remedial actions are believed to have been conducted at the Four Star site.

1.4.4 Summary

The following paragraphs briefly summarize the findings of major previous investigations relative to the Cadillac Area RI. Table 1-1 lists the reports on file with MDNR.

The geology of the site study area generally consists of a sequence of three sand aquifers separated by two clay aquitards. The upper two aquifers consist predominantly of brown, fine- to medium-grained, well-sorted quartz sands. The upper aquitard consists of a brown to red-brown, firm to stiff clay, but is often silty. The lowermost sand and clay units were not penetrated during these earlier investigations; therefore, geologic information was limited to interpretations of city water supply well logs.

The uppermost sand unit is a water table aquifer whose bottom is defined by a clay unit, referred to as the upper aquitard. Groundwater flow in this aquifer is generally to the northeast. The surface of the upper clay aquitard rises from southwest to northeast, generally becoming thinner and pinching out near the center of the industrial park. The upper aquitard thickness varies locally and is thinnest northeast (i.e., hydrologically downgradient) of the Northernaire site.

Where the upper aquitard is present, the intermediate aquifer is confined; where the upper aquitard pinches out, the intermediate aquifer merges with the shallow aquifer and becomes unconfined. Flow is generally north-northwest. (The geology and hydrogeology, with graphic illustrations, are discussed in greater detail in Sections 4 and 5, respectively.)

A municipal water supply wellfield is located about a quarter-mile northeast of the Northernair building (Figure 1-2). Water is pumped from an aquifer beneath the intermediate aquifer. Previous hydrogeologic investigations focused on assessing groundwater contamination and potential migration to the municipal well aquifer. However, the lower aquitard was not penetrated in any of these studies. Also, during the late 1970s when investigations first began, nearby private water supply wells tapped the intermediate aquifer. North Park residences are now served by the city water supply.

The Northernair investigation determined that both the shallow and intermediate aquifers are contaminated by hexavalent chromium. Hexavalent chromium is the principal form of chromium used for plating at Northernair. Primary chromium releases at plating facilities occur through rinsing of parts withdrawn from the plating solution, spills of plating solution, and uncontrolled venting of chromic acid mists released during plating operations. At Northernair, the sewer system piping is known to have permitted wastewater leakage to surrounding soils. Chromium also reached the ground west of the building where supplies and wastes were stored and the air scrubber system was located. The highly water-soluble hexavalent chromium, originating at the Northernair facility, seeped and/or leached through the porous soils to the shallow water table aquifer. The contamination then traveled northeastward with the groundwater, as well as downward due to vertical hydraulic gradients within the aquifer. Where the upper aquitard becomes thin or the downward vertical gradients across the aquitard are steep, the contamination seeps with the groundwater from the shallow to the intermediate aquifer and continues to move downward.

In the intermediate aquifer, the contamination has a horizontal component of flow generally to the northwest. It was determined, therefore, that the city water supply did not appear to be in immediate danger of contamination by hexavalent chromium. However, without knowing the thickness and permeability of the lower clay aquitard, and the hydrologic effects of the city wellfield pumping on the intermediate aquifer (particularly away from the immediate vicinity of the wellfield), the future potential for contamination of the lower aquifer was still unknown and of concern.

Several investigations found that VOC contamination is widespread in both the shallow and intermediate aquifers. Some probable sources of the contamination were identified, but adequate definition of the contaminant distribution, assessment of other possible sources, identification of the migration pathways, and determination of environmental fate have not yet been developed. These data gaps defined the purposes of the Cadillac Area RI.

2.0 PHYSICAL SETTING

This section presents information regarding the site setting. This information is useful in appreciating potential health and environmental risks posed by the site, surrounding land use, climate (which may affect some migration potentials), and natural resources of the area (which might also be impacted by the site).

2.1 POPULATION

The current population of the city of Cadillac is approximately 10,000 people. Population density is greatest along the eastern shore of Lake Cadillac, in which the major downtown commercial district is located. The immediate shorelines of Lakes Cadillac and Mitchell are also densely populated with both year-round and seasonal residences. Other smaller and less densely populated residential neighborhoods are located away from the commercial and lake areas.

Companies with products related to the automobile industry are the major employers. Many companies are located in the Cadillac industrial park near the northern city limits. A significant work force is also employed in tourism and outdoor recreation activities, including boating, hunting, and snowmobiling.

2.2 CLIMATOLOGY

The climate of Cadillac is considered continental with some impact from Lake Michigan, primarily in the winter in the form of lake-effect snowfall. Due to the area's inland location, the impact from Lake Michigan on temperature is minor. Mean annual temperature is 42.9°F, with the lowest and highest average temperatures (18.6°F and 67.4°F) occurring in February and July, respectively.

Annual precipitation averages 31.17 inches, with the lowest and highest average monthly precipitation occurring in February (1.59 inches) and September (3.53 inches), respectively. No specific data are available for Cadillac regarding the maximum 24-hour precipitation; however, Houghton Lake (located about 35 miles east in a similar setting) recorded a total 5.18 inches of rain in a 24-hour period in July 1957. It is assumed that the peak 24-hour storm for Cadillac would be of similar magnitude. Average annual snowfall ranges from 60 to 80 inches, with 1 inch of snow remaining on the ground for an average of 100 days per year. Monthly precipitation averages are listed in Table 2-1. Prevailing winds are from the west and southwest. The winds are calm only 16 percent of the time. The average growing season is approximately 100 days.

2.3 LAND USE

The city of Cadillac is located in Wexford County in northwestern Michigan, approximately 60 miles east of Lake Michigan (see Figure 1-1). Cadillac itself is located in a very flat-lying area of glacial outwash deposits. Two large lakes adjacent to the city, Lake Cadillac and Lake Mitchell, are used widely as

TABLE 2-1
AVERAGE MONTHLY PRECIPITATION
RAINFALL EQUIVALENCE (inches)

January	1.83
February	1.59
March	2.01
April	2.68
May	2.97
June	3.15
July	2.78
August	3.01
September	3.53
October	2.88
November	2.89
December	1.85

vacation and recreation attractions. An approximate 1-square-mile area of Cadillac is used as an industrial park. A municipal wellfield for public and industrial water supply is located within the park. Before 1986, drinking water for most residences north of Thirteenth Street was supplied by private wells screened in the intermediate aquifer. These residences are currently serviced by the municipal water system. A county airport, which handles private and small commercial airplane traffic, is situated just north of the city limits.

In the areas surrounding Cadillac, much of the land is used for tree farming for the Christmas tree industry. Some dairy and livestock farms are also present. Otherwise, the land is rural and undeveloped. Figure 1-2 shows land use in the immediate area of the industrial park.

2.4 NATURAL RESOURCES

Natural resources in Cadillac include the two lakes, the land itself for foresting, and extensive, high-yielding groundwater aquifers. As previously discussed, the lakes and forestlands are used for recreation and cultivating, respectively. The city uses a deep groundwater aquifer for its entire potable water supply; therefore, it is environmentally important to protect. Currently, the municipal water supply is being drawn at the approximate rate of 2 million gallons per day (gpd) from a confined aquifer approximately 300 feet below ground surface. The aquifer is overlain by up to two other aquifers, both of which contain significant contamination by toxic compounds released in the industrial park. Protection of the supply aquifer is currently limited to a clay aquitard of variable thickness and unknown permeability.

Wetlands abut Lake Mitchell along its northern, eastern, and southern sides. Lake Mitchell drains into Lake Cadillac, which discharges into the Clam River. The Clam River flows through the city of Cadillac in manmade channels toward the northeast, where natural channels occur. The Clam River also receives discharge from the city's municipal wastewater treatment plant, located near the northeastern city limits.

Wildlife in the immediate site vicinity is limited due to the lack of protective cover. Surrounding areas have high populations of deer, game birds, and other wildlife. The outlying areas have a sufficient amount of protective cover and food sources; therefore, wildlife movement into or through the industrial park area is limited.

3.0 SITE INVESTIGATION PROGRAM

This section presents general descriptions of procedures and protocols used during the site investigation program. More specific information for soils and groundwater investigations is presented in Sections 4 and 5, respectively. The size and complexity of the Cadillac Area site, which contains several plumes, necessitated the inclusion of several modifications to the original Work Plan scope and the involvement of other firms in the site exploration program. A brief overview of the tasks, sequence of events, and involvements is presented herein as the framework for clarifying future references.

The final Cadillac Area RI Work Plan was approved and issued in November 1985. Jordan was the primary subcontractor to MDNR for Work Plan completion and implementation. In turn, Jordan subcontracted Keck Consulting Services (Keck) of Williamston, Michigan, to conduct subsurface drilling.

In the Work Plan, PRPs were given the opportunity to comment and/or participate on portions of the RI. The only responses received were comments from Kysor and indications of a desire to participate from Four Winns. Rexair had not yet been identified as a PRP. Four Winns selected ASI as its consultant. ASI would perform most of the field investigation task items in the Work Plan, as they pertained to the Four Winns facility at Leeson Avenue and Frisbie Street.

The site investigation program focused on an extensive phase of drilling and monitoring well installations. As the Cadillac drilling program proceeded, it became apparent that the hollow-stem auger drilling technique would not yield sufficient data to define the depth of the upper surface of the lower aquitard. The surface of the lower aquitard appeared to be dropping rapidly in all directions from the municipal wellfield area, below the 200-foot depth achievable by the auger rig. Because the presence or absence of the lower aquitard was a key element of the site assessment, it was decided to sacrifice some of the planned auger wells so MDNR/Jordan and Keck could drill four deep mud-rotary borings at the site perimeters. This would determine the presence and depth of the aquitard surface and, if possible, enable samples of the aquitard material to be obtained. The absence of clay in one hole (taken to a depth of 276 feet), and the presence of only a 3-foot-thick layer of clay in another (taken to a depth of 316 feet), increased concern about the capability of the lower aquitard to protect the municipal wellfield. This concern was further increased by the discovery of a contaminant plume of VOCs east-southeast and hydraulically upgradient of the wellfield.

The combined discoveries were the basis for defining an imminent threat to the wellfield. MDNR obtained additional funds to further investigate the aquitard and wellfield by installing three deep wells (using mud-rotary techniques) at the level of the municipal wells to evaluate hydraulic gradients, stratigraphy, and water quality. Two additional mud-rotary borings were installed to further define the clay surface and thickness. These were installed by Keck under Jordan's supervision.

Finally, the original boring budget for the project had been consumed by the planned wells and the first four mud-rotary borings. However, because so many

wells were required to define the plume east and southeast of the wellfield, two or three wells and 10 soil borings were still regarded as necessary for the site assessment. MDNR could not obtain further funds for Jordan to perform this work under the present contract, but could under a separate contract with Warzyn Engineering, Inc. (Warzyn), another MDNR environmental consultant.

Arrangements were made for Warzyn to supervise the final installations, while Keck was retained as drilling subcontractor. To maintain consistency in conduct, Warzyn operated under a Work Plan drafted by Jordan (with slight modifications by Warzyn) and approved by MDNR. Two deep wells were installed, one upgradient of Kysor and the other upgradient of Rexair. In addition, 10 borings were drilled to obtain soil samples for the quantification of residual VOCs near the former disposal area behind Kysor. With the new deep mud-rotary wells, a pump test using the city supply wells appeared worthwhile. This test, performed by Keck under MDNR direction, was included in the final scope modification.

3.1 INVESTIGATIVE TECHNIQUES

Several techniques were used to investigate the subsurface to provide the necessary data with which to assess site geology, hydrology, and distribution of groundwater contamination. Both the soil and groundwater were investigated, and various drilling and other techniques were required to provide data, primarily dependent on depth below the ground surface. The following techniques were used:

- o soil-gas surveys and split-spoon sampling of the shallow soils (zero to 30 feet deep)
- o screened-auger groundwater sampling with hollow-stem augers and monitoring well installations in the shallow and intermediate aquifers (30 to 200 feet deep)
- o mud-rotary drilling and well installations combined with downhole gamma-ray and electrical resistivity logging to investigate the lower aquitard and water supply aquifer (200 to 300 feet deep)

Other exploration techniques used to investigate hydrogeologic parameters included the following:

- o single-well aquifer permeability tests
- o water level data collection from monitoring wells and piezometers
- o soil grain-size distribution analyses
- o chemical analysis of soil and groundwater samples
- o construction of a numerical groundwater model for the site

Keck provided rigs, operators, and other support services to conduct the subsurface drilling investigation program. The hollow-stem auger borings and screened auger sampling were accomplished using both a Gus Pech Brat 22R and a Bantam drill rig, and 7.4-inch OD, 6.25-inch ID augers. With plumbing and drive-head modifications, the Brat rig was also used with the mud-rotary

casing. Other technical services for which Keck supplied both equipment and operators included grain-size distribution analyses, soil-gas surveys, portable gas chromatograph (GC) operations, single-well permeability tests, and gamma-ray and electrical resistivity logging.

3.1.1 Soil-gas Survey

Soil-gas studies are generally used as a pre-drilling method of detecting areas of high VOC contamination. Probes are driven into the upper levels of the unsaturated soil. A vacuum pump draws a soil-gas sample through the probe to be analyzed by a portable GC. When the method is effective, the degree of contamination in the soil-gas is generally proportional to the contamination in soils or the groundwater and, therefore, is capable of providing locations for subsequent boring and/or monitoring well installations.

A pilot study to determine the effectiveness of the soil-gas technique on the soils and contamination of the Cadillac industrial park was conducted at the beginning of the RI. Inconsistent results in areas of known groundwater contamination near the Four Star facility precluded the use of the technique for much of the study. However, after some modifications, the technique was used behind the Kysor plant in an area of relatively high-level contamination in shallow groundwater to avoid extensive subsurface drilling where access was limited. Section 4.1 describes the investigative methodology.

3.1.2 Hollow-stem Auger Drilling

The hollow-stem auger/screened-auger method of aquifer investigation was used for most of the data collection, for several reasons:

- o The relatively clean soils (i.e., few fine-grained silts or clays) were conducive to auger drilling and screened auger sampling methods.
- o It was a relatively fast means by which to obtain depths 180 to 200 feet below ground surface in the sandy soils at Cadillac.
- o Split-spoon samples were readily obtained in the unsaturated zone.
- o The depth to the different stratigraphic units at the Cadillac site were identified with sufficient accuracy by noting changes in auger torque and penetration rate. The soils were then visually logged as they circulated up the outside flights at the augers.
- o Representative groundwater samples were obtained at regular intervals using screened augers from the shallow and intermediate aquifers without the need to add foreign drilling muds or water to the formation.
- o Chemical analysis of groundwater samples in the field allowed for determination of the optimal well screen placement and defined both the horizontal and vertical extent of the plume(s).

- o Monitoring wells were installed in soils unaffected by drilling fluids.
- o A low-permeability bentonite-grout seal was placed in each borehole annulus to maintain hydraulic barriers between aquifers and prevent downward contaminant migration through the disturbed zone adjacent to the boreholes.

The use of screened augers precluded the opportunity to more accurately define the geology through split-spoon soil samples with depth. However, because the identification of geologic units throughout the study area is relatively straightforward, the slight loss of direct geologic definition was considered much less important than the vertical definition of groundwater contamination. Also, the large number of borings performed allowed refinement of driller and geologist interpretations. Furthermore, downhole geophysical methods were used at some locations to confirm interpretations.

3.1.3 Mud-rotary Drilling

Mud-rotary drilling was necessary to attain the approximate 300-foot depth to investigate geology, groundwater quality, and groundwater flow characteristics. This drilling method is relatively fast, but requires the use of a thick drilling mud to circulate in the borehole and flush out drill cuttings. This precludes acquisition of groundwater samples during drilling and, after well installation, necessitates extensive well development to clear the well screen of drilling mud. The interpretation of the geology encountered in the borehole was refined using two geophysical methods: gamma-ray and electrical resistivity logging. These methods can differentiate between sand and clay layers based on differing radioactive and electrostatic properties.

3.1.4 Aquifer Permeability Tests

Single-well rising-head tests were conducted in several monitoring wells to determine permeability of the shallow and intermediate aquifers across the site. The test was conducted by lowering a length of metal pipe, closed at both ends (i.e., a slug), below the water level in the monitoring well, and then allowing the water level to reach equilibrium (i.e., the original height of water in the well). Once equilibrium was attained, the slug was quickly removed, instantly lowering the head (i.e., height of water) in the monitoring well. The subsequent rise in head, as the well approached equilibrium again, was monitored by a pressure transducer, which recorded the change in head versus time. Using the Hvorslev Method, the permeability was then calculated from the response to the induced change in head.

3.1.5 Groundwater Elevation Data

Several sets of groundwater elevation data were obtained from most of the monitoring wells at various times to determine horizontal flow patterns in the aquifers, as well as vertical hydraulic potentials between aquifers. Water level measurements were obtained to the nearest 0.01 foot with an electronic water level meter. Measurements were taken relative to elevations established

for each well by an MDNR survey crew and referenced to the U.S. Geological Survey (USGS) datum.

3.1.6 Soil Grain-size Distribution

Grain-size analyses were conducted on the split-spoon samples collected during hollow-stem auger drilling. Based on the analyses and visual observations, the soils were classified according to the Unified Soils Classification System (USCS). These data provided information for assessing changes in soil characteristics in the unsaturated zone.

3.1.7 Chemical Analyses of Soil and Groundwater

Chemical analyses of the split-spoon soil samples for VOCs were conducted in the field and the laboratory. Section 4.4 discusses results of the analyses. Two rounds of groundwater samples were collected from nearly all monitoring wells installed for the Cadillac Area RI and from selected existing wells of previous investigations. Based on previous data and historical information, analytical parameters were limited to VOCs and metals. Sampling methods are discussed in Section 5.4.

3.1.8 Groundwater Modeling

Groundwater modeling of the site was performed by MDNR using the USGS numerical model by MacDonald and Harbaugh. The model was prepared as a tool to help understand the hydrogeologic setting, assess potentials for contaminant migration across the lower aquitard, and help design and evaluate the effect and effectiveness of proposed remedial alternatives that would attempt to control contaminant migration and/or remove and treat contaminated groundwater. The modeling effort is described in more detail in Section 5.

3.2 TECHNICAL APPROACH SUMMARY

The approach to investigating soil and groundwater contamination at the Cadillac industrial park was twofold: (1) combine the subsurface investigative techniques to define the site stratigraphy in terms of clay aquitards and sand and gravel aquifers; and (2) determine the horizontal and vertical extent of various contaminant plumes. Plume configurations, historical data, and definition of the hydrogeologic regime would indicate probable sources and potential receptors. Environmental fate of contaminants would be a function of the geohydrology, the contaminant type and concentration, and the actions of available attenuative mechanisms.

Previous investigations identified contamination sources, but did not completely define contaminant distributions in groundwater. The Cadillac Area RI drilling program began near these industries to complete the definition. Borings and monitoring wells were installed in a line, progressing hydraulically downgradient from the suspect facility, at approximately 150- to 250-foot intervals until the longitudinal dimension of the contaminant plume was determined. The plume width was then defined by placing a line of borings perpendicular to the longitudinal line and centered approximately where contamination

was highest. Additional wells needed to adequately define the plume were then installed. Wells upgradient of a particular industry were also installed to further indicate that the suspect industry was the contamination source.

Placement of the monitoring wells were subject to property owner approval and physical barriers such as roads, overhead and underground utilities, buildings, and railroad tracks. Once a particular plume was defined with an acceptable amount of detail, the drilling operations were shifted to another area of the industrial park. Monitoring wells and piezometers were also installed along the perimeter of the industrial park and beyond to obtain regional data of the geology, groundwater flow potentials, and groundwater quality.

While it was originally planned that drilling would be done with hollow-stem auger methods, the depth to the lower clay aquitard in much of the study area required a distinct phase of mud-rotary drilling to be conducted. The primary purpose of this phase was to determine depth, thickness, and areal extent of the lower clay aquitard. These data would aid in assessing the potential for contaminants in the intermediate aquifer to migrate to the water supply aquifer under influence of the pumping wells. Six deep borings, widely spaced across the study area, provided important geologic data concerning areal extent and thickness of the lower aquitard. In addition, three monitoring wells were installed at depths of 300 feet near the wellfield to monitor the pumping effects of the aquifer and to provide piezometric data and groundwater samples.

The compilation of geologic, hydrologic, and groundwater quality data obtained from the tasks outlined in Section 3.1 is the basis for a detailed site assessment of the industrial park area. This draft RI report includes discussions of the probable contamination sources, contaminant distributions, migration pathways, and potential receptors. A baseline risk assessment will be developed to evaluate the relative effectiveness of remedial action alternatives considered in the FS.

3.3 IDENTIFICATION OF ARARs

As required by the National Contingency Plan (NCP) of 1985 and the Comprehensive Environmental Response, Compensation, and Liability Act (CERCLA) Section 121, added by SARA, Applicable or Relevant and Appropriate Requirements (ARARs) were developed for the Cadillac Area site. ARARs are used as a guide to (1) evaluate the appropriate extent of site cleanup; (2) scope and formulate remedial action alternatives; and (3) govern the implementation and operation of a selected action. ARARs are identified and considered so that CERCLA responses are consistent with state and federal environmental laws.

3.3.1 Definition and Development of ARARs

CERCLA Section 121 establishes clean-up standards for remedial actions under Sections 104 and 106 of CERCLA, and requires that remedial actions comply with all federal and state ARARs.

The NCP defined ARARs as federal statutory or regulatory requirements (40 CFR Part 300, November 20, 1985). Although state regulatory requirements were

considered, they were given less weight than federal requirements. With the enactment of SARA, state requirements, standards, or criteria that are promulgated and more stringent than federal requirements must generally be attained in remedial response actions. An exception is that SARA prohibits the application of any state requirement that results in a statewide ban on land disposal unless certain conditions are met. Also, state requirements that have not been consistently applied can be discounted.

According to SARA, requirements may be waived by USEPA under six specific conditions, provided that protection of human health and the environment is still assured. These conditions are as follows:

- o The selected remedial action is an interim remedy.
- o Compliance with such requirements will result in greater risk to human health and the environment than alternative options.
- o Compliance with such requirements is technically impracticable from an engineering perspective.
- o The selected remedial action will provide an equivalent standard of performance using another approach.
- o The requirement is a state requirement that has been inconsistently applied.
- o Attainment of the ARAR would entail extremely high costs relative to the added degree of reduction of risk afforded by the standard.

To properly consider ARARs and, more importantly, to clarify their function in the RI/FS and remedial response processes, the NCP defines two ARAR components. These definitions were revised to incorporate applicable portions of SARA; they are discussed in the following paragraphs.

Applicable Requirements. Applicable requirements are those federal and state clean-up standards, standards of control, and other environmental requirements, criteria, or limitations that could be legally applied to the response action if that action were not pursuant to Section 104 or 106 of CERCLA.

Requirements that are applicable to and have jurisdiction over the given situations are considered "applicable requirements." An example of an applicable requirement would be Maximum Contaminant Levels (MCLs) for a site that exhibits groundwater contamination entering a public water supply.

Relevant and Appropriate Requirements. Relevant and appropriate requirements are those federal and state clean-up standards, standards of control, and other environmental requirements, criteria, or limitations that, while not "applicable," are designed to apply to problems sufficiently similar to those encountered at CERCLA sites that their application is appropriate.

Relevant and appropriate requirements are intended to carry the same weight and consideration as applicable requirements. Once a requirement is identified as

relevant and appropriate, it is applied in the same manner as an applicable requirement. The term "relevant" was included so that a requirement initially screened as nonapplicable because of jurisdictional restrictions would be reconsidered and, if appropriate, included as an ARAR for the site. For example, MCLs would be a suitable nonapplicable but relevant and appropriate requirement for a site that exhibited groundwater contamination in a potential drinking water source.

To-Be-Considered (TBCs). Although not legally required, nonpromulgated advisories or guidance issued by federal or state government are to be considered during development and implementation of remedial alternatives. TBCs do not have the status of potential ARARs, but in the absence of regulatory requirements, TBCs may have to be considered and used when appropriate.

3.3.2 Approach to ARAR Development

Under the description of ARARs set forth in SARA, many state and federal environmental requirements must be considered. In general, these ARARs fall into three categories:

- o chemical-specific (i.e., governing the extent of site remediation)
- o location-specific (i.e., pertaining to existing site features)
- o action-specific (i.e., pertaining to proposed site remedies and governing implementation of the selected site remedy)

Chemical-specific ARARs are usually health- or risk-based standards that limit the concentration of a chemical that can be found in or discharged to the environment. They govern the extent of site remediation by providing either actual clean-up levels or the basis for calculating such levels. For example, drinking water standards may provide the necessary clean-up goals for the Cadillac Area site. Chemical-specific ARARs for the site are presented in Section 3.3.4. These ARARs will be used to refine remedial response objectives, and to modify and formulate potential remedial alternatives.

Location-specific ARARs must be considered as they pertain to a site location and features. Site features governed by location-specific ARARs may include natural features such as wetlands, floodplains, and sensitive ecosystems. In addition, places of historical or archeological significance, and landfill or disposal areas, may be governed by environmental requirements. These ARARs provide a basis for assessing existing site conditions, which subsequently aid in assessing potential remedies. Federal sole-source aquifer requirements may be pertinent at the Cadillac site. Further discussion of location-specific ARARs follows in Section 3.3.3.

Action-specific ARARs are based on the limitations or features of remedial technologies and site activities. These ARARs are used during evaluation, selection, and design of site remedial alternatives. For example, if groundwater is treated and discharged to surface water, federal and state requirements for pollution discharge elimination systems must be attained. Potential action-specific ARARs are identified in Section 3.3.5.

3.3.3 Identification of Location-specific ARARs

Based on a study of potential ARARs listed in the NCP preamble and Michigan state regulations, Jordan identified potential location-specific ARARs for the Cadillac Area site. After reviewing site features, Jordan narrowed this list of potential ARARs using a checklist, which includes explanations for eliminating any potential ARAR from consideration (see Appendix A, Table 1). The selected ARARs concern only the aquifer in the site area, and are presented in Table 3-1 and described in the following paragraphs.

Aquifer. The Safe Drinking Water Act (SDWA) Sole-Source Aquifer Demonstration Program and the USEPA Regulations on Sole-Source Aquifers are potential ARARs for the Cadillac Area site. Since the Cadillac aquifer is a sole or principal source of drinking water for the city, it could be potentially so designated. Pursuant to Section 1424(e) of the SDWA, if USEPA determines on its own initiative or upon petition that an area has an aquifer that is the sole or principal drinking water source for the area and that, if contaminated, would create a significant hazard to public health, USEPA may designate the aquifer as sole-source.

If the designated sole-source aquifer is "particularly vulnerable" to contamination due to hydrogeologic characteristics of the recharge area, and is the source of drinking water for at least 75 percent of the population in the aquifer service area, and if the cost of replacing the water supply from the sole-source aquifer would cause supply costs to exceed 0.7 percent of mean annual income, then the recharge area of the sole-source aquifer may be designated as a "critical aquifer protection area."

Section 1427 of the SDWA sets forth procedures for developing, implementing, and assessing a program for a critical aquifer protection area. The protection program may include establishment of a planning entity, hydrogeologic assessment, comprehensive management plan, procedures for public participation for review of the plan, an assessment of the relationship between land surface activities and groundwater quality, requirements to maintain groundwater quality, land use management, and special techniques such as clustering or transfer of development rights sufficient to protect the area.

Although the Cadillac Area site is not presently designated as a sole-source aquifer or critical aquifer protection area, components of a protection program and comprehensive management plan are relevant and appropriate to development of remedial alternatives and long-term management of the area.

3.3.4 Identification of Chemical-specific ARARs

As discussed in Section 3.3.2, chemical-specific ARARs are usually health- or risk-based values that limit the concentration of a chemical found in or discharged to the environment. As in the case of location-specific ARARs, federal and state regulatory and nonregulatory requirements are considered.

After reviewing Cadillac Area site contaminants, the list of potential ARARs was narrowed using the checklist presented in Appendix A, Table 2. The checklist includes reasons for eliminating a potential ARAR from consideration.

TABLE 3-1
POTENTIAL LOCATION-SPECIFIC ARARS
CADILLAC AREA RI

SITE FEATURE	REQUIREMENT	REQUIREMENT SYNOPSIS	CONSIDERATION IN THE RI
RCRA Facilities			
Federal Regulatory Requirements	RCRA - Standards for Owners and Operators of Permitted Hazardous Waste Facilities (40 CFR 264.10 - 264.18)	General facility requirements outline waste analysis, security measure, and training requirements.	For RCRA Subtitle C requirements to have jurisdiction and therefore be applicable for Cadillac, RCRA listed hazardous waste must be present onsite. Because this regulation outlines waste analysis requirements, it will be considered when determining the jurisdiction of general Subtitle C requirements.
	RCRA - Preparedness and Prevention (40 CFR 264.30 - 264.37)	This regulation outlines safety equipment and spill control requirements for hazardous waste facilities. Part of the regulation includes a requirement that facilities be designed, maintained, constructed and operated so that the possibility of an unplanned release which could threaten human health or the environment is minimized.	RCRA requirements must be considered when evaluating the design of potential alternatives.
	RCRA - Contingency Plan and Emergency Procedures (40 CFR 264.50 - 264.56)	This regulation outlines requirements for emergency procedures to be used following explosions and fires. This regulation also requires that threats to public health and the environment be minimized.	RCRA requirements must be considered when evaluating the design of potential alternatives.
	RCRA - Groundwater Protection (40 CFR 264.90 - 264.109)	Under this regulation, groundwater monitoring and corrective action requirements are outlined.	A groundwater monitoring system must be installed as part of any alternative. During site characterization, the location and depth of monitoring wells will be evaluated for use in this monitoring program.
	RCRA - Closure and Postclosure (40 CFR 264.110 - 264.120)	This requirement details the specific requirements for closure and postclosure of hazardous waste facilities.	Potential alternatives must be evaluated for attainment of these requirements.
	RCRA - Underground Storage Tanks (40 CFR 264.190)	This requirement outlines notification and installation of underground storage tank procedures.	Because underground storage tanks exist on site, this regulation should be considered during site characterization.
	SDWA - Sole Source Aquifer Program (Section 1424(e))	This section authorizes EPA Administrator to determine that an aquifer is the "sole or principal source of drinking water" for an area. After such determination, EPA will review Federal financially assisted projects planned for the area to determine their potential for contaminating the aquifer.	It should be determined if the Cadillac aquifer may be considered a sole source aquifer.

TABLE 3-1 (continued)
POTENTIAL LOCATION-SPECIFIC ARARS
CADILLAC AREA RI

SITE FEATURE	REQUIREMENT	REQUIREMENT SYNOPSIS	CONSIDERATION IN THE RI
State Regulatory Requirements	EPA Regulations on Sole Source Aquifers, Subpart A (40 CFR 149)	This subpart provides criteria for identifying critical aquifer protection areas, pursuant to Section 1427 of the SDWA.	These criteria should be considered when determining if the Cadillac aquifer is a sole source aquifer.
	Michigan Hazardous Waste Management Act (Act No. 64, P.A. 1979)	This regulation outlines general requirements for management of hazardous waste facilities in Michigan.	During the implementation of any site activities, these requirements will be considered and followed when appropriate.
	Michigan Underground Storage Tank Act (Act No. 423, P.A. 1984)	This regulation requires the registration and permitting of underground storage tanks.	Underground storage tanks should be identified and requirements should be considered when developing response objectives and remedial actions for source control.

ARARs that remained after completion of the checklist are presented in Table 3-2 and discussed in this section.

Chemical-specific ARARs are described by the medium requiring remediation (i.e., groundwater and soil). Because treated groundwater could potentially be discharged to surface water or a publicly owned treatment works (POTW), chemical-specific ARARs for such discharges are also discussed.

Groundwater. Aquifers that are hydraulically connected to groundwater beneath the Cadillac Area site currently provide a drinking water source for the city; the city wellfield is located near the middle of this site. Therefore, the SDWA's MCLs are applicable clean-up levels for these aquifers. MCLs are legally enforceable federal drinking water standards which are based on adverse health effects of a contaminant and which reflect the technical and economic feasibility of removing the contaminants from the water supply. The SDWA's Maximum Contaminant Levels Goals (MCLGs) are nonenforceable health goals established by USEPA, set at levels that would result in no known or anticipated adverse health effects with an adequate margin of safety. CERCLA Section 121 states that remedial actions shall attain MCLGs where they are relevant and appropriate under the circumstances of release. MCLGs will be used as a clean-up standard at the Cadillac Area site for non-carcinogens when an extraordinary risk is present and no MCLs are available. Federal Ambient Water Quality Criteria (AWQC) are also potentially relevant and appropriate standards under CERCLA Section 121. Because the designated use of the water is for drinking, AWQC adjusted for drinking water are relevant and appropriate. In addition, Michigan Groundwater Discharge Rules are applicable; those more stringent than federal standards must be attained unless specifically waived under the conditions of SARA.

Federal nonregulatory criteria to be considered when ARARs are not available for specific contaminants include USEPA Health Advisories, USEPA Risk Reference Doses, and USEPA Carcinogen Assessment Group Potency Factors.

If a remedial alternative involves discharging treated wastewater back into the groundwater, the wastewater must meet the more stringent of the state or federal standards. Resource Conservation and Recovery Act (RCRA) groundwater protection guidance will also be considered in the development of remedial alternatives. The procedures for developing alternative concentration limits (ACLs) for site-specific conditions are outlined under RCRA (Subpart F, Groundwater Protection Standards).

Discharge to a POTW. If a groundwater treatment system is installed at the Cadillac Area site, and the discharge of this system is sent to a POTW, then pretreatment standards under the Clean Water Act (CWA) apply (i.e., 40 CFR 403). If the POTW has an approved pretreatment program, then discharge from the site must comply with this approved program.

The Michigan Water Resources Rules also contain a provision for pretreatment requirements. These requirements are consistent with CWA requirements; therefore, compliance with the CWA pretreatment standards will result in compliance with applicable Michigan Water Resources Rules.

TABLE 3-2
CHEMICAL-SPECIFIC ARARS
CADILLAC AREA RI

SITE FEATURE	REQUIREMENT	REQUIREMENT SYNOPSIS	CONSIDERATION IN THE RI
GROUNDWATER			
Federal Applicable Requirements	SDWA - Maximum Contaminant Levels (MCLs) (40 CFR 141.11 - 141.16)	MCLs have been promulgated for a number of common organic and inorganic contaminants. These levels regulate the concentration of contaminants in public drinking water supplies, and are considered applicable for groundwater aquifers used for drinking water.	When the risks to human health due to consumption of groundwater are assessed, concentrations of contaminants should be compared to their MCLs.
	RCRA - Subpart F Groundwater Protection Standards Alternative Concentration Levels (ACLs)	ACLs are one of three possible standards (other than MCLs and background concentrations) available under Subpart F for setting a clean-up level for remediating groundwater contamination from a RCRA facility.	ACLs may be applicable if certain conditions relating to transport and exposure are met. ACLs may need to be determined by USEPA. Procedures for developing ACLs are outlined in RCRA Subpart F, Section 264.94.
Federal Relevant and Appropriate Requirements	SDWA - Maximum Contaminant Level Goals (MCLGs)	MCLGs are health-based criteria that are relevant and appropriate for drinking water sources, as a result of SARA, when extraordinary risks exist. These goals are available for a number of organic and inorganic contaminants.	Groundwater concentrations of noncarcinogenic contaminants should be compared to their MCLGs when extraordinary risks exist.
	Federal Ambient Water Quality Criteria (AWQC)	Federal AWQC are health based criteria that have been developed for 95 carcinogenic and noncarcinogenic compounds.	AWQC should be used in characterizing human health risks due to contaminant concentrations in groundwater.
State Applicable Requirements	MWRL, Act 245, R 323 Part 22, Groundwater Discharge Rules	Draft proposed rules to establish allowable discharge levels to protect Michigan groundwater quality in all usable aquifers as potential water supplies.	The draft revisions to the existing Part 22 Rules are still in the approval process; it is unknown when they will take effect.
Federal Advisories and Guidance to be Considered	Health Advisories (USEPA Office of Drinking Water)	Health advisories are estimates of risks due to consumption of contaminated drinking water; they consider noncarcinogenic effects only.	Health advisories should be considered for contaminants in groundwater that is used for drinking water.
	USEPA Risk Reference Doses (RfDs)	RfDs are dose levels developed by USEPA for noncarcinogenic effects.	USEPA RfDs are used to characterize risks due to exposure to contaminants in groundwater, as well as other media.
	USEPA Carcinogen Assessment Group Potency Factors	Potency Factors are developed by USEPA from Health Effects Assessments (HEA) or evaluations by the Carcinogen Assessment Group.	USEPA Carcinogenic Potency Factors are used to complete the individual incremental cancer risk resulting from exposure to contaminants.

TABLE 3-2 (continued)
CHEMICAL-SPECIFIC ARARS
CADILLAC AREA RI

SITE FEATURE	REQUIREMENT	REQUIREMENT SYNOPSIS	CONSIDERATION IN THE RI
	Acceptable intake - Chronic (AIC) and Subchronic (AIS) - USEPA Health Assessment Documents	AIC and AIS values are developed from RfDs and HEAs for noncarcinogenic compounds.	AIC and AIS values are used to characterize the risks due to several noncarcinogens in various media.
	USEPA Office of Water Guidance - Water Related Fate of 129 Priority Pollutants (1979)	This guidance manual gives transport and fate information for 129 priority pollutants.	The manual is used to assess the transport and fate of a variety of contaminants.
<u>DISCHARGE TO PUBLICLY-OWNED TREATMENT WORKS</u>			
Federal Applicable Requirements	CWA - Pretreatment Standards (40 CFR 403)	Discharges to a POTW must comply with the POTW's USEPA-approved pretreatment requirements.	POTWs in the area with approved pretreatment programs will be identified. The discharge must be treated to those levels required by the program.
State Applicable Requirements	MWRL, Act 245, R 323.2163 Pretreatment Program	Discharges to a POTW must comply with the POTW's pretreatment requirements.	POTWs in the area with approved pretreatment program will be identified. The discharge must be treated to those levels required by the program.
<u>DISCHARGE TO SURFACE WATER/MERRIMAC RIVER</u>			
State Applicable Requirements	Michigan Water Quality Standards, Act 245, R 323.	Standards have been developed for dissolved oxygen; temperature standards for specific contaminants are developed on a case-by-case basis.	These standards will be used for setting limits for discharge to surface water.
Federal Relevant and Appropriate Requirements	Federal Ambient Water Quality Criteria (AWQC)	Federal AWQC are health-based criteria that have been developed for 95 carcinogenic and noncarcinogenic compounds.	AWQC may be used for setting limits for discharge to surface water.

Discharge to Surface Water. The CWA governs discharge of priority pollutants into surface water. All such discharges must comply with the federal CWA National Pollution Discharge Elimination System (NPDES) and the Michigan Water Quality Standards. Discharge limits set in the NPDES permit are determined on a case-by-case basis, but use AWQC for guidance. The Michigan Water Quality Standards provide specific standards for dissolved oxygen and temperature.

Soils. No federal or state regulatory requirements currently govern the extent of cleanup for contaminants in soil. Target levels for cleanup of contaminated soil are developed based on results of the RA and the potential for the soil to continue contaminating the groundwater.

3.3.5 Identification of Action-specific ARARs

Action-specific ARARs are technology- or activity-based requirements or limitations that control remedial actions taken at CERCLA sites. The remedial technologies trigger certain action-specific ARARs that regulate how and if those technologies may be implemented at the Cadillac Area site. For example, action-specific ARARs may dictate how remedial actions (e.g., extraction or treatment) are conducted. Additionally, action-specific ARARs may indicate if a remedial action (e.g., land disposal) may be implemented.

Identification of preliminary action-specific ARARs is performed using a checklist developed to ensure that all potential ARARs have been considered. The checklist indicates the federal and state regulations considered ARARs, and the federal criteria, advisories, procedures, and guidance documents to be considered during evaluation (see Appendix A, Table 3). Regulations identified as ARARs, for consideration of potential alternatives, are presented in Table 3-3.

RCRA Subtitle C. Many RCRA Subtitle C requirements apply because: (1) the site contains RCRA-listed hazardous waste; and (2) some of the proposed technologies will probably constitute treatment, storage, or disposal of RCRA-listed hazardous waste.

RCRA requirements that apply to potential alternatives involving treatment, storage, or disposal of RCRA waste include standards for owners and operators of permitted hazardous waste facilities; preparedness and prevention; contingency plan and emergency procedures; recordkeeping and reporting; groundwater monitoring; and closure and post-closure requirements. All alternatives must also meet corrective action requirements for either RCRA or the Hazardous and Solid Waste Amendments (HSWA) of 1984, Section 3004(u) and (v). RCRA Subpart F corrective action provisions apply to releases of RCRA waste to groundwater from RCRA facilities, and require groundwater cleanup within the site boundary to either the background level, an SDWA MCL, or the ACL for each hazardous constituent. The HSWA requirements, which are only now being developed by USEPA, are expected to apply to all CERCLA remedial actions that seek to abate releases of RCRA hazardous waste to any medium, whether on- or off-site, regardless of when the waste was received. When HSWA corrective action requirements are developed, USEPA expects that it will be more appropriate to comply with them rather than any other Subtitle C requirements. Meanwhile,

TABLE 3-3
POTENTIAL ACTION-SPECIFIC ARARs
CADILLAC AREA RI

ARARs	REQUIREMENT SYNOPSIS	ACTION TO BE TAKEN TO ATTAIN ARARs
FEDERAL ARARs		
RCRA - Standards for Owners and Operators of Permitted Hazardous Waste Facilities (40 CFR 264.10 - 264.18)	General facility requirements outline general waste analysis, security measures, inspections, and training requirements.	Facilities will be constructed, fenced, posted, and operated in accordance with this requirement. Workers will be properly trained. Wastes will be evaluated for the characteristics of hazardous waste to assess further RCRA requirements.
RCRA - Preparedness and Prevention (40 CFR 264.30 - 264.31)	This regulation outlines requirements for safety equipment and spill control.	Safety and communication equipment will be installed at the site; local authorities will be familiarized with site operations.
RCRA - Contingency Plan and Emergency Procedures (40 CFR 264.50 - 254.56)	This regulation outlines the requirements for emergency procedures to be used following explosions, fires, etc.	Plans will be developed and implemented during site work, including installation of monitoring wells and implementation of site remedies. Copies of the plans will be kept on-site.
RCRA - Manifesting, Recordkeeping, and Reporting (40 CFR 264.70 - 264.77)	This regulation specifies the recordkeeping and reporting requirements for RCRA facilities.	Records of facility activities will be developed and maintained during remedial actions.
RCRA - Groundwater Protection (40 CFR 264.90 - 264.109)	This regulation details requirements for a groundwater monitoring program at the site.	A groundwater monitoring program is a component of all alternatives. RCRA regulations will be considered during development of this program.
RCRA - Closure and Post-closure (40 CFR 264.110 - 264.120)	This regulation details specific requirements for closure and post-closure of hazardous waste facilities.	Those parts of the regulation concerned with long-term monitoring and maintenance of the site will be considered during remedial design.
OSHA - General Industry Standards (29 CFR Part 1910)	This regulation specifies the 8-hour, time-weighted average concentration for various organic compounds.	Proper respiratory equipment will be worn if it is impossible to maintain the work atmosphere below the concentrations required in the regulations.
OSHA - Safety and Health Standards (29 CFR Part 1926)	This regulation specifies the type of safety equipment and procedures to be followed during site remediation.	Appropriate safety equipment will be on-site. In addition, OSHA safety procedures will be followed during on-site activities.
OSHA - Recordkeeping, Reporting, and Related Regulations (29 CFR 1904)	This regulation outlines the recordkeeping and reporting requirements for an employer under OSHA.	These requirements apply to all site contractors and subcontractors and must be followed during all site work.
RCRA - 40 CFR 268 USEPA Regulations on Land Disposal Restrictions	This regulation outlines land disposal requirements and restrictions for hazardous wastes.	Regulations to be phased in over the next few years require contaminated soils to be treated to the Best Demonstrated Available Technology levels before being placed or replaced on the land. Hazardous waste cannot be stored except for accumulation for recovery, treatment, or disposal.
RCRA - Surface Impoundments (40 CFR 264.220 - 264.249)	This regulation details the design, construction, operation, monitoring, inspection, and contingency plans for a RCRA surface impoundment. Also provides three closure options for CERCLA sites.	At closure, owner must remove or decontaminate all waste or eliminate free liquid, stabilize remaining waste, and cover impoundment with a cover that complies with the regulation. Integrity of cover must be maintained, groundwater system monitored, and runoff controlled.

TABLE 3-3 (continued)
POTENTIAL ACTION-SPECIFIC ARARs
CADILLAC AREA RI

ARARS	REQUIREMENT SYNOPSIS	ACTION TO BE TAKEN TO ATTAIN ARARs
RCRA - Landfills (40 CFR 264.300 - 264.339)	This regulation details the design, operation, monitoring, inspection, recordkeeping, closure, and permit requirements for a RCRA landfill.	Disposal of contaminated materials from the site must be to a RCRA-permitted facility that complies with all RCRA landfill regulations.
CWA - 40 CFR Parts 122, 125	Any point source discharges must meet NPDES permitting requirements, which include compliance with applicable water quality standards; establishment of a discharge monitoring system; and routine completion of discharge monitoring records.	If groundwater that has been treated by on-site treatment processes is discharged to on-site surface water, treated groundwater must be in compliance with applicable water quality standards. In addition, a discharge monitoring program must be implemented. Routine discharge monitoring records must be completed.
CWA - 40 CFR Part 403	This regulation specifies pretreatment standards for discharges to a publicly owned treatment works (POTW).	If a collection system is installed and the discharge is sent to a POTW, the POTW must have an approved pretreatment program. The collection system water quality must be in compliance with the approved program. Prior to discharging, a report must be submitted containing identifying information, list of approved permits, description of operations flow measurements, measurement of pollutants, certification by a qualified professional, and a compliance schedule.
CAA - NAAQS for Total Suspended Particulates (40 CFR 129.105, 750)	This regulation specifies maximum primary and secondary 24-hour concentrations for particulate matter.	Fugitive dust emissions from site excavation activities will be maintained below 260 $\mu\text{g}/\text{m}^3$ (primary standard) by dust suppressants, if necessary.
Protection of Archeological Resources (32 CFR Part 229, 229.4; 43 CFR Parts 107, 171.1-171.5)	This regulation develops procedures for the protection of archeological resources.	If archeological resources are encountered during soil excavation, work will stop until the area has been reviewed by federal and state archeologists.
DOT Rules for Transportation of Hazardous Materials (49 CFR Parts 107, 171.1-171.5)	This regulation outlines procedures for the packaging, labeling, manifesting, and transportation of hazardous materials.	Contaminated materials will be packaged, manifested, and transported to a licensed off-site disposal facility in compliance with these regulations.
Michigan ARARs Michigan - Hazardous Waste Management Act (Act No. 64, P.A. 1979)	This regulation provides a comprehensive program for the handling, storage, and recordkeeping at hazardous waste facilities. They supplement RCRA regulations.	Because these requirements supplement RCRA hazardous waste regulations, they must also be considered at the site.
Air Pollution Control Commission General Rules (R 336, Part 9)	This regulation outlines the standards and requirements for air pollution control in the State of Michigan. All provisions, procedures, and definitions are described.	Particulate matter emissions from site excavation activities must be maintained at an annual geometric mean of 75 $\mu\text{g}/\text{m}^3$ and a maximum 24-hour concentration of 40 $\mu\text{g}/\text{m}^3$ (primary standards).
Underground Storage Tank Act, Act No. 423, P.A. 1984	This regulation requires the registration and permitting of underground storage tanks.	Alternatives requiring the removal or installation of underground storage tanks would require this regulation to be considered.

TABLE 3-3 (continued)
POTENTIAL ACTION-SPECIFIC ARARs
CADILLAC AREA RI

ARARS	REQUIREMENT SYNOPSIS	ACTION TO BE TAKEN TO ATTAIN ARARs
Environmental Response Act, Act No. 307, P.A. 1982	This regulation requires that risk assessments be conducted for hazardous waste sites.	This regulation should be considered for all remedial alternatives.
MDNR-MWRC Groundwater Discharge Rules, Act 245, R 323 Part 22	Draft proposed rules to establish allowable discharge levels to protect Michigan groundwater quality in all usable aquifers as potential water supplies.	Discharges to groundwater must not degrade the quality of groundwater and must be treated to drinking water quality. Requirements for recordkeeping and monitoring must be attained.
MDNR-MWRC, Wastewater Reporting and Surveillance Fees Rules, and Wastewater Discharge Permits Act 245, R 323, Parts 9, 21	This regulation outlines the requirements for reporting discharges of wastewater to state waters or POTWs. The discharge permit requirements are compatible with NPDES requirements.	Discharges to surface water must not degrade the quality of the water. Requirements for recordkeeping and monitoring must be attained.
MDNR - Liquid Industrial Waste Disposal Act (Act No. 136, P.A. 1984)	This regulation addresses the removal of liquid industrial waste, and outlines the requirements for licenses, recordkeeping, and transportation.	Removal of liquid industrial waste must be recorded. Appropriate licenses will be obtained. Waste will be transported in accordance with these requirements when more stringent than federal requirements.

CERCLA remedial actions must continue to operate within the bounds of existing RCRA requirements.

RCRA Part 264, surface impoundment and landfill closure rules, currently provides two basic closure options for CERCLA sites. The clean closure option requires removal or decontamination of all hazardous constituents, and includes very stringent groundwater standards of clean-up levels. If all hazardous constituents will not be removed or decontaminated, the landfill closure option may be used. Landfill closure is a containment option requiring a final cover or cap and a post-closure plan that protects human health and the environment. These requirements will be addressed during cleanup of contaminated soils.

For the alternatives that involve transportation, treatment, storage, and/or disposal of hazardous wastes, RCRA Subtitle C disposal requirements, including the Land Disposal Restrictions, will be an ARAR. When RCRA hazardous wastes are moved, disturbed, or physically altered during remedial action, Part 268, Land Disposal Restrictions (i.e., Land Ban), requires that the waste be treated by Best Demonstrated Available Technology (BDAT) to specific treatment levels before the waste can be placed or replaced on land. Land Ban also prohibits storage of hazardous waste, except for accumulation to facilitate recovery, treatment, or disposal. Land Ban regulations are scheduled to be phased in over a number of years. Currently, the restrictions apply to solvent- and dioxin-containing hazardous wastes and California Listed Wastes.

CWA. Several regulations promulgated under the CWA apply when considering alternatives that involve dredging, groundwater treatment, and discharges to either groundwater, surface water, or POTWs. Although on-site CERCLA actions do not require formal permitting, the substantive requirements of the NPDES permit must be met, including the NPDES Best Management Practice Program, compliance with water quality standards, a discharge monitoring system, and records maintenance. Off-site discharges are not exempt from the procedural and administrative permitting requirements, and an NPDES permit must be obtained before an off-site discharge may occur. CWA Pretreatment Standards (40 CFR 403) apply if groundwater is discharged to a POTW.

Clean Air Act. Additional requirements concerning potential excavation and air-stripping activities include the National Air Quality Standards for Total Suspended Particulates under the Clean Air Act.

Occupational Safety and Health Administration (OSHA). Federal OSHA requirements that regulate worker safety and employee records must be followed during all site work. These regulations include safety and health standards for federal service contracts and recordkeeping, reporting, and related regulations.

Department of Transportation (DOT) Rules for Transport of Hazardous Materials. If materials that still contain hazardous wastes are to be transported off-site, the DOT packaging, transport, and manifest requirements apply.

Michigan Regulatory Requirements. In addition to federal ARARs, several Michigan regulations apply to potential alternatives. The MDNR Hazardous Waste Management Act is consistent with RCRA, providing a comprehensive program for

the identification, handling, transportation, and recordkeeping of hazardous waste.

If an alternative involves removing liquid industrial waste, the MDNR Liquid Industrial Waste Disposal Act is applicable. Appropriate licenses should be obtained, and recordkeeping and transportation requirements will be attained.

Alternatives that involve groundwater or surface water discharges require the MDNR Groundwater Discharge Rules and Wastewater Reporting and Surveillance Fees Rules standards to be followed. Alternatives involving soil removal or air-stripping of contaminated water would require consideration of the Air Pollution Control Commission general rules.

Federal Criteria, Advisories, Procedures, and Guidelines. Federal policies, executive orders, procedures, and guidance are available and should be considered for development of various alternatives. These standards, criteria, and advisories are not legally enforceable, but should be used when ARARs do not exist or are not sufficiently protective. Federal criteria, advisories, and guidance to be considered for the Cadillac Area site are listed in Appendix A, Table 3.

3.4 SURVEY AND BASE MAP PREPARATION

The horizontal locations (± 0.1 feet), ground surface elevations (± 0.1 feet), and well riser elevations (± 0.01 feet) of the new and existing monitoring wells were surveyed by MDNR. A USGS benchmark was used and all elevations are based on Mean Sea Level (MSL) datum. MDNR prepared the final base map for use in the RI/FS. The benchmark location and elevation are shown on the base map, included as Plate No. 1 at the end of this report. The surveyed coordinates and elevations of the wells are presented in Appendix B.

4.0 SOILS INVESTIGATION

The stratigraphy and physical properties of the soils in the study area were investigated by several methods. Shallow soil samples above the water table were collected for physical analysis to assess grain-size distribution and, in addition, for chemical analysis to determine possible sources of residual contamination in soils. Also, soil-gas samples were collected by Keck for chemical analysis.

4.1 SOIL-GAS SURVEYS

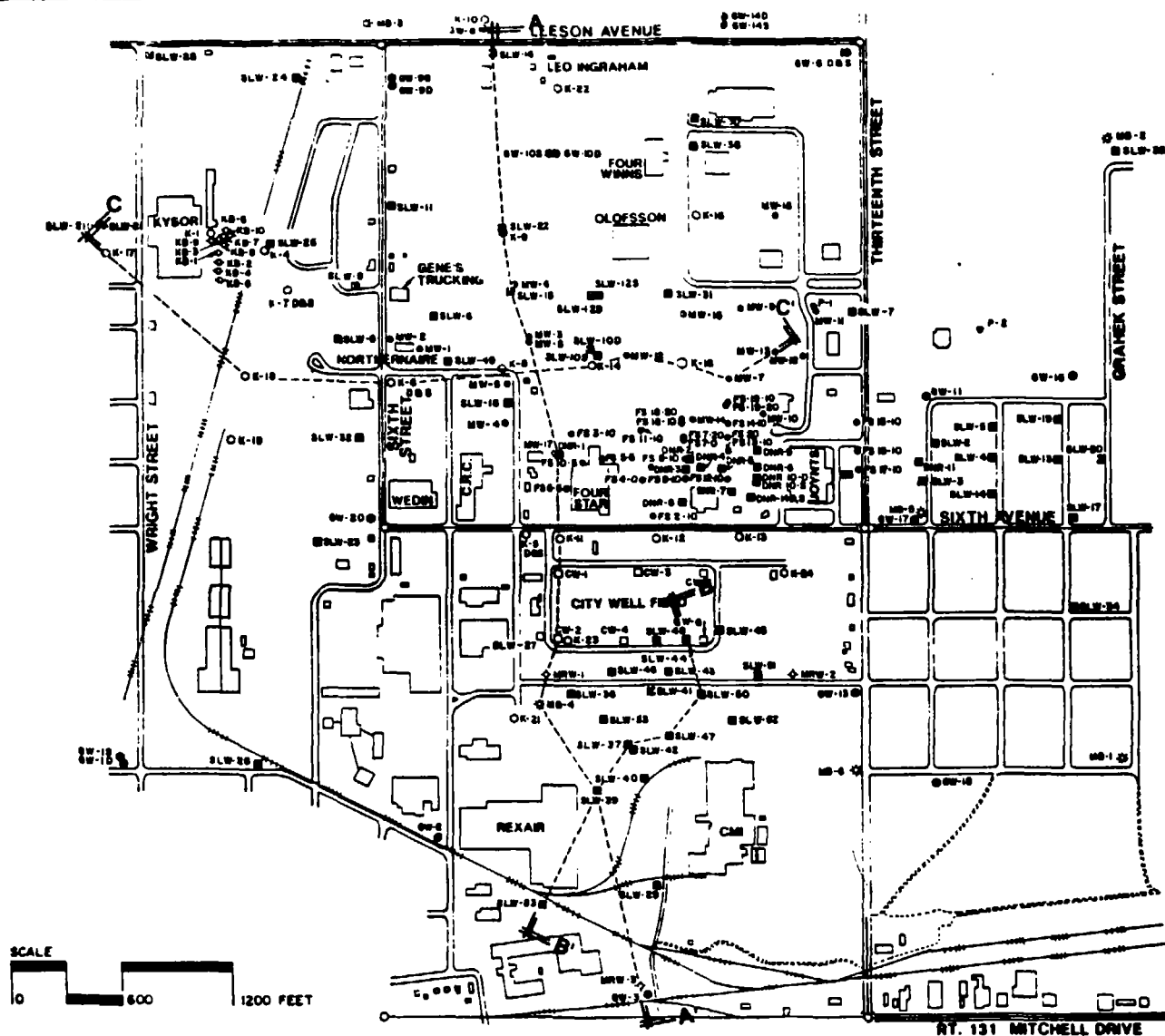
As stated earlier, soil-gas survey techniques proved to be limited in effectiveness and reliability. The survey was applied with confidence only near Kysor, where residual contamination in soils was suspected to be relatively high in concentration and at relatively shallow depths below ground surface (i.e., 10 to 25 feet). A description of the soil-gas methods and analytical procedures was provided by Keck (see Appendix C).

4.2 SOIL BORINGS

The soils were investigated directly through the advancement of borings by hollow-stem auger and mud-rotary casing techniques. The soils were visually inspected and logged as drill cuttings came up the outside auger flights or in the circulating drilling mud. Split-spoon samples were obtained from the unsaturated soils in several borings. Description of the soils was based on the USCS. The boring logs for the total of 98 auger and mud-rotary boreholes, monitoring wells, and piezometers drilled for the Cadillac Area RI are included in Appendix D. Appendix E is a complete summary of subsurface explorations and monitoring wells.

Six different series of borings and monitoring wells were installed for the Cadillac Area RI, each to supply a particular type of information needed for a comprehensive site assessment. Each boring/monitoring well was labeled to identify the series to which it belongs; the location of each is shown in Figure 4-1. The six series include the following:

- o Source Location Wells (SLW), designed to identify sources of the soil and groundwater contamination and to define contaminant distribution
- o General Wells (GW), intended to provide geologic and groundwater quality data on a more regional basis
- o Mud Borings (MB), drilled by mud-rotary techniques solely to provide geologic data to a depth of about 300 feet or more
- o Mud-Rotary Wells (MRW), installed to provide geologic, piezometric, and groundwater quality data at a depth of about 300 feet near the city wellfield



LEGEND

- FOUR STAR WELLS
- D.M.R. WELLS
- KYSOR WELLS
- M.W. WELLS (NORTHERNAIRE)
- CITY WELL HOUSES
- SOURCE LOCATION WELLS
- GENERAL WELLS
- ▼ PIEZOMETERS
- ✱ MUD BORINGS
- ◇ KYSOR SOL. BORINGS
- ◇ MUD ROTARY WELLS

A A' LINE OF GEOLOGIC PROFILE



FIGURE 4-1
MONITORING WELL LOCATION
AND IDENTIFICATION PLAN
CADILLAC AREA RI
CADILLAC, MICHIGAN

ECJORDANCO

- o Piezometers (P), installed to provide groundwater elevation data only
- o Kysor Borings (KB), drilled immediately north of the Kysor facility to investigate soil contamination at or near an original VOC dumping location; no wells were installed in these borings

Monitoring well installations in the borings are discussed in Section 5.1.

An MDNR-identified PRP, Four Winns on Frisbie Street (a fiberglass boat manufacturer), chose to retain its own consultant (i.e., ASI) to conduct a contamination investigation of its property. The 17 borings and monitoring wells installed on or near Four Winns property by ASI represent an additional boring/well series.

Finally, Warzyn was retained by MDNR to assist in completing the drilling phase of the Cadillac Area RI. Warzyn supervised the installation of SLW-21D, SLW-53, and the KB series of 10 borings. Methods used for these borings and well installations were identical to those used by Jordan.

4.2.1 SLW Borings

The placement strategy and rationale for the SLW borings and wells was described in Section 3.1. During drilling of these wells, soil samples were collected for both visual inspection and physical and chemical analysis. A 2-foot split-spoon sampler was used to obtain soil samples at 5-foot intervals from ground surface to the water table. The samples were placed in Teflon-capped glass jars for subsequent semi-quantitative field screening for total VOCs by headspace analysis. Headspace analysis was done using a Photovac Model 10-A-10 GC, which was stationed in the site support trailer for close ambient temperatures control. Based on this field screening, samples were selected for quantitative analysis of VOCs through the USEPA CLP. Also, some soil from selected split spoons was used to conduct grain-size distribution analysis to aid in soil classification.

In the zone of saturation (below the groundwater table), the vertical profile of VOC contamination was determined by groundwater samples collected through the augers in lieu of soil samples. This sampling process is discussed in Section 5.1, which describes the installation of monitoring wells. Changes in stratigraphy from sand to clay could also be determined by noting changes in auger torque and penetration rate. The borehole was advanced until the upper clay aquitard was encountered (where present). In the northern and eastern sections of the study area where the upper aquitard is absent, the borings were terminated at a depth of 90 feet, unless significant groundwater contamination was detected. In this case, the borings were continued until the contamination was at only trace or non-detectable levels.

4.2.2 GW Borings

The GW borings and monitoring wells were installed in a manner similar to the SLW borings. Many of the GW borings were drilled at the study area perimeter to provide hydrogeologic and groundwater quality data and, therefore, were not always near suspected contaminant sources. Split-spoon samples of the shallow

soils were not obtained. It was originally intended for these borings to be drilled to the depth of the lower clay aquitard. However, in a large part of the study area, the clay unit was at a depth greater than could be reached by the hollow-stem auger drilling method. Most borings attained a depth of from 150 to 180 feet below ground surface, depending on formation density and resultant torque on the augers.

4.2.3 MB Borings

Mud-rotary boring techniques were used to attain a greater depth than could be reached by augers to determine the depth to the lower aquitard and, in selected locations, its thickness. Initially, four mud-rotary borings (MBs) were drilled near the four corners of the study area to provide desired geologic information. These borings ranged in depth from 220 to 316 feet, depending on depth to the lower aquitard. Drill cuttings were logged as they circulated out of the borehole in the drilling mud. To confirm the stratigraphic interpretation, the downhole geophysical techniques of gamma-ray logging and electrical resistivity were used in MB-2. Upon completion, each boring was backfilled with a thick bentonite grout.

Due to concern raised by the thinness or apparent absence of the lower aquitard in several areas, as indicated by the initial MB borings, MDNR and Jordan concurred that further information was needed to assess potential risk to the municipal wellfield. Therefore, five additional mud-rotary borings (located away from areas of known contamination) were drilled to a depth of 300 feet to investigate the lower clay aquitard near the city wellfield. Geophysical logging techniques were again used to confirm geologic interpretations; the logs are provided in Appendix F.

4.2.4 MRW Wells

Monitoring wells were emplaced in three of the mud-rotary borings close to the wellfield. The deep monitoring wells were installed to determine:

- o the influence of the pumping of the city wells on the lower aquifer
- o the vertical hydraulic gradients that exist across the lower aquitard which may induce contaminants downward into the lower aquifer
- o groundwater quality in the lower aquifer

These borings were also logged by geophysical techniques, and are provided in Appendix F.

4.2.5 Piezometers

Five piezometers (P) were installed by hollow-stem auger methods solely for groundwater elevation data on a regional basis. These piezometers also provided information needed for calibration of the groundwater numerical flow model, which will be discussed in more detail in Section 5.2.

4.2.6 KB Borings

The KB borings were drilled to collect soil samples for chemical analysis. Ten borings were drilled by hollow-stem auger behind (i.e., north of) the Kysor facility main building where prior dumping of organic solvents was known to have occurred. The borings were drilled to the water table and soil samples were collected at 5-foot intervals. Headspace analyses were conducted on all samples by field GC; two samples from each boring were selected for transport to a CLP laboratory for VOC analysis. The intent of this boring and sampling was to determine the degree of residual soil contamination.

4.2.7 Four Winns Borings

The boring logs and monitoring well installation records for the 17 wells installed by ASI for Four Winns are included in Appendix G. These wells were installed specifically to characterize the contaminants at the Four Winns Frisbie Street property.

4.2.8 Additional Boring Series

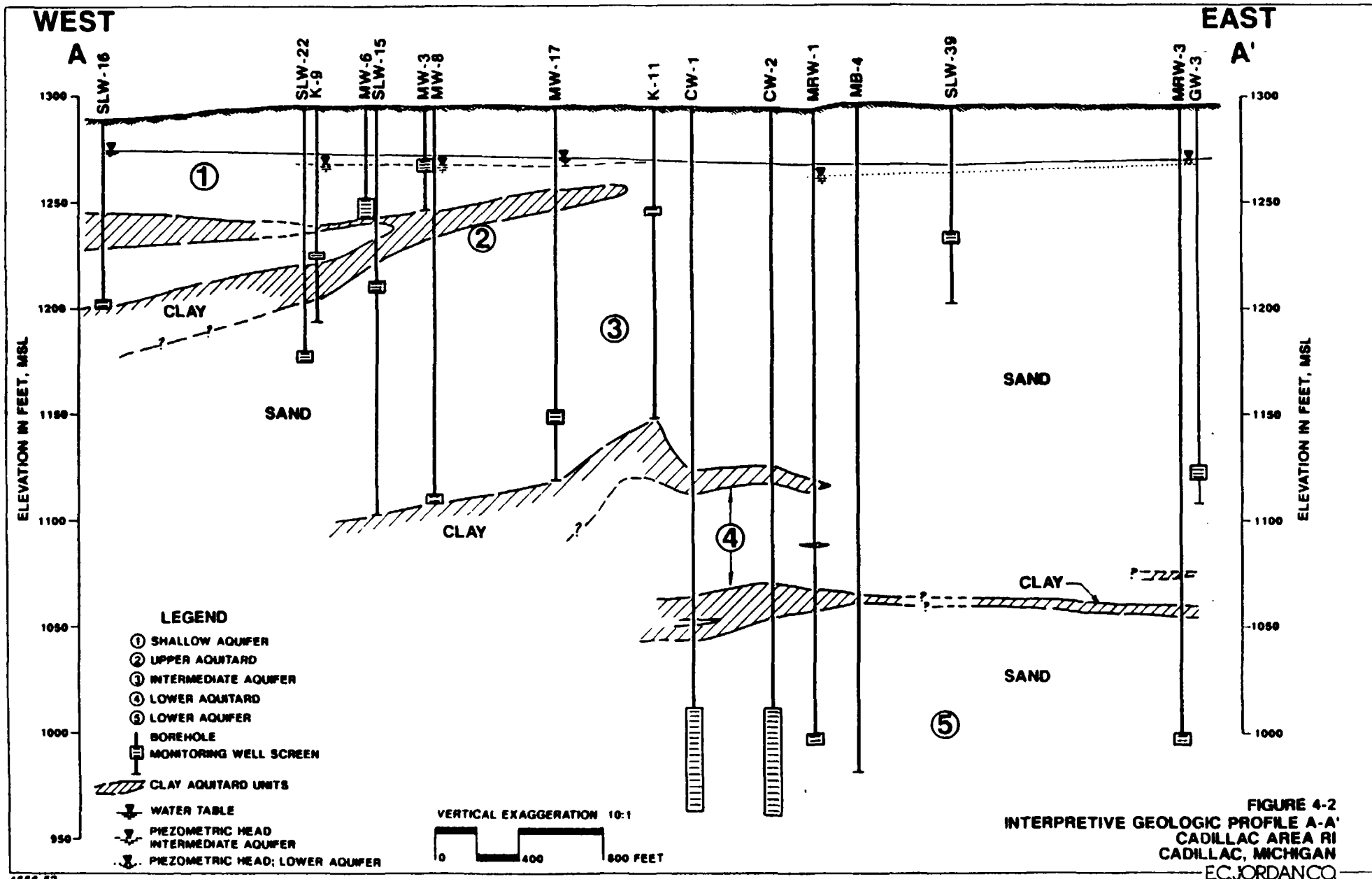
Four additional boring and monitoring well series were installed during separate investigations of individual plant sites. These include the MW, K, FS, and DNR series discussed in Section 1.4. Boring logs from these series were reviewed for geologic interpretation. These borings are located in Figure 4-1; Appendix E includes drilling and installation summary tables.

4.3 GEOLOGY

Regionally, the geologic formations in the Cadillac Area are unconsolidated glacial deposits of the Wisconsin and older glacial episodes. Cadillac itself lies in a flat-lying glacial outwash area. Topographic relief in the form of glacial moraines surround this area. The Lake Border moraine exists to the north, east, and west; the Valparaiso moraine exists to the south. Two large, shallow glacial lakes (i.e., Lakes Mitchell and Cadillac) exist in the Cadillac outwash plain. The Clam River, which is the only significant surface drainage feature in the area, drains the plain to the northeast through a water gap in the moraines (MDNR, 1986).

Interpretation of the Cadillac industrial park geology was developed from data obtained while drilling over 185 monitoring wells and borings. Much information came from previous hydrogeologic investigations, including installation of the city water supply wells. Conclusions from these prior investigations are discussed in Sections 1.4 and 2.3. The current Cadillac Area RI has added data regarding the areal extent and thickness of the various aquifers and aquitards, particularly the lower clay aquitard overlying the city water supply aquifer.

The geology pertinent to the Cadillac Area RI consists primarily of five hydrogeologic units: three aquifers and two aquitards. In order of stratigraphic succession from top to bottom, these units (shown in geologic profile in Figure 4-2) are referred to as the following:



- o The shallow aquifer, present only in the southwestern half of the study area.
- o The upper aquitard, a clay stratum of varying thickness, present only in the southwestern half of the study area. This unit dips to the southwest and defines the bottom of the shallow aquifer.
- o The intermediate aquifer, a thick sand unit between the upper aquitard and a lower aquitard. This aquifer is confined in the southwestern portion of the site, and is under water table conditions where the upper aquitard pinches out and the shallow and intermediate aquifers merge.
- o The lower aquitard, which defines the bottom of the intermediate aquifer. Limited data exist on the extent and thickness of this clay unit, but it is believed that near the municipal wellfield, this unit is actually two clay layers separated by a sand unit. Total clay thickness vary from about 2 to 40 feet. The sand unit may, in fact, be a separate aquifer; however, insufficient data exist on a regional scale to define it as such. In the wellfield vicinity, the sand varies from 10 to 80 feet in thickness. Due to extreme variability in the thickness of this sand and two adjacent clay layers, these three units are most easily discussed as a single aquitard unit.
- o The lower aquifer, immediately below the lower aquitard and the municipal water supply source for the city of Cadillac.

As would be expected in a glacio-lacustrine environment, there are localized variations in geology from this generalized scheme, including the following:

- o a clay-sand interbedding of the shallow aquifer in the west and southeast portions of the study area
- o gravel zones in the North Park Subdivision and other areas
- o occasional small clay lenses

One of the more substantial clay lenses was encountered in the shallow aquifer along Sixth Street between Leeson Avenue and Northernnaire. This layer lies directly beneath a stormwater infiltration lagoon at the corner of Leeson and Sixth, which explains why the lagoon frequently retains water for a relatively long time after storms, while another infiltration lagoon on Thirteenth Street allows for rapid infiltration.

4.3.1 Shallow Aquifer

These deposits of sand and clay are glacio-lacustrine in origin. The sands in the shallow aquifer are generally a light brown, well-rounded, well-sorted, fine- to medium-grained quartz with some feldspar. Locally, the sand may contain small percentages of coarse sand and fine gravel. Grain-size analyses reveal a fine brown sand with generally less than 3 percent fines (passing a U.S. #200 sieve). Occasional samples contained as much as 17 percent fines.

The USCS designation for the sand is SP. Grain-size distribution analyses are presented in Appendix H.

The bottom of this aquifer is defined by the upper clay aquitard, which exists only in the southwestern portion of the study area (see Section 4.3.2). The shallow aquifer increases in thickness from 35 to 105 feet in a southwestern direction, reflecting the dip in the upper aquitard. The depth of the water table varies from 20 to 30 feet, from the southwestern corner of the study area to the edge of the upper aquitard.

4.3.2 Upper Aquitard

Samples of the upper clay aquitard were obtained from the auger flights during borehole advancement. The clay is generally brown to red-brown, firm to stiff, and may contain from zero- to 30-percent silt. The thickness and extent of this stratum varies across the site. Figure 4-3 is a surface contour map of the top of the upper aquitard. The aquitard rises from the southwest to the northeast, and pinches out along a northwest-southeast trending line near the middle of the study area. Figure 4-4 is an isopach (i.e., thickness) map of the clay. The clay is thickest (about 22 feet) near the former Olofsson facility and SLW-12. The upper aquitard becomes very thin (about 2 feet) downgradient of the Northernaire facility, where the clay forms a "trough" paralleling Seventh Avenue just east of Northernaire.

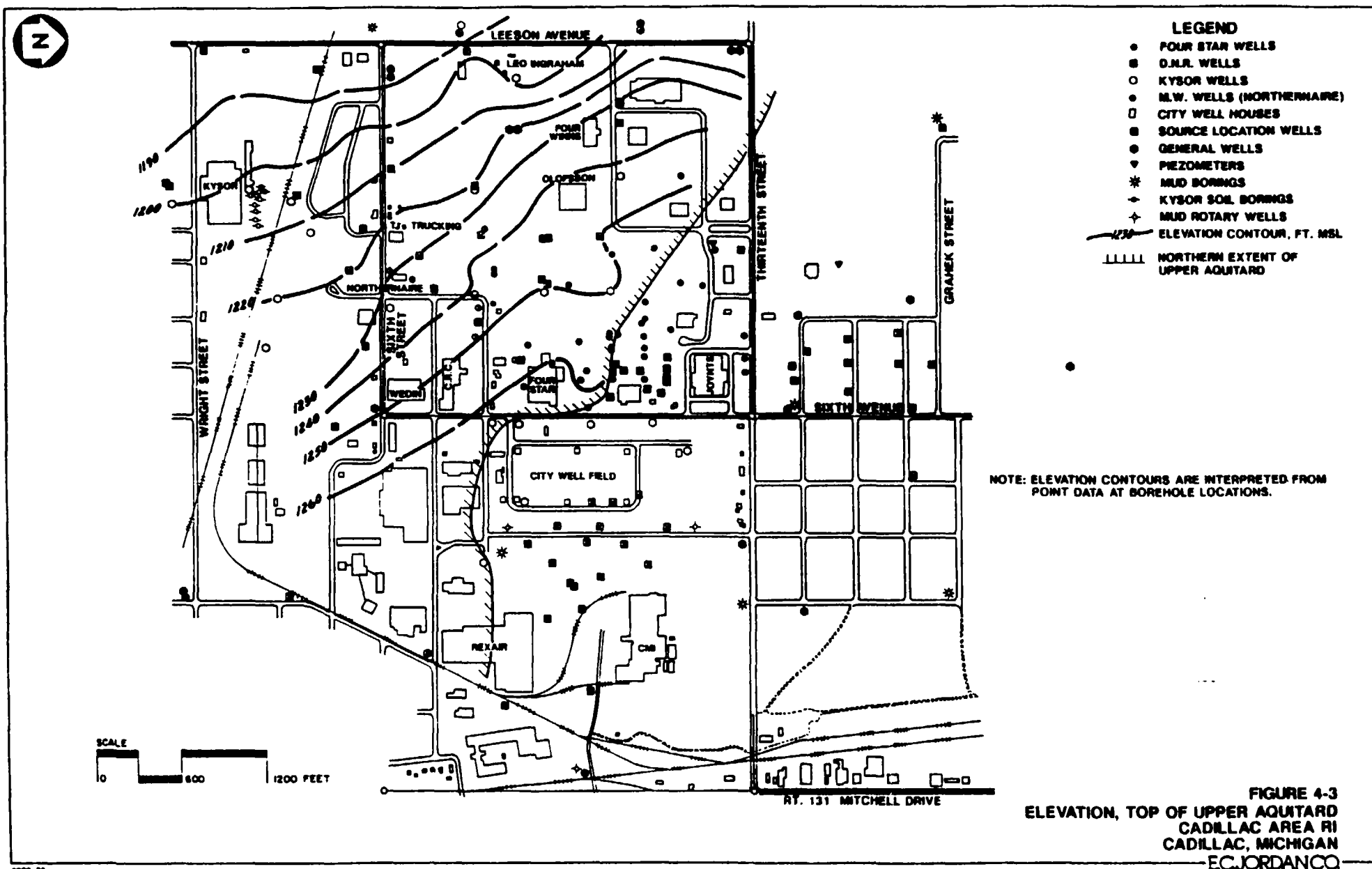
4.3.3 Intermediate Aquifer

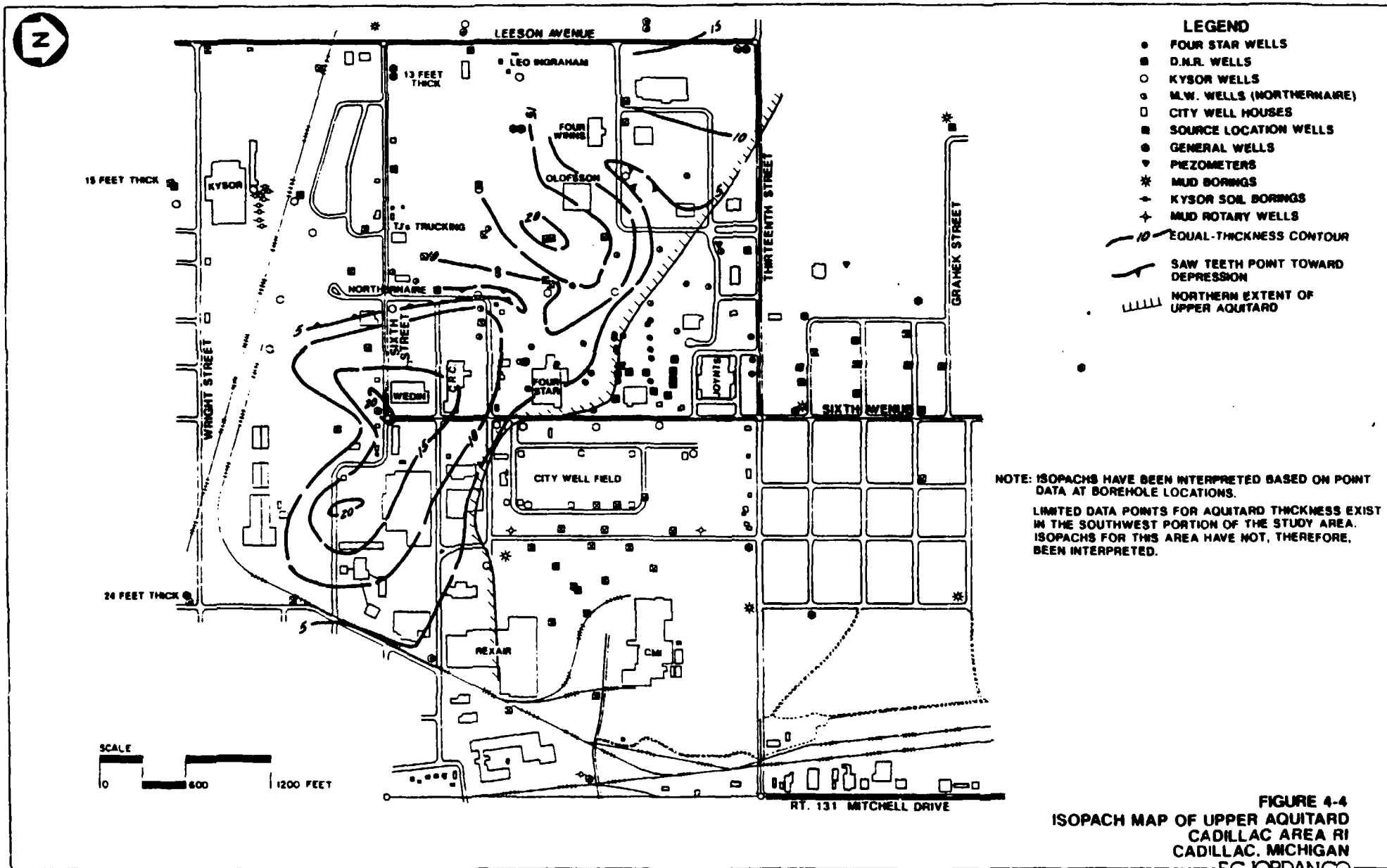
The intermediate aquifer ranges in thickness from approximately 100 to 220 feet. It is thickest in the northern and eastern portions of the site, where the aquifer is unconfined and the upper aquitard is absent. While the character of the sand is fairly homogeneous across the site, variations in the grain-size distribution occur in local areas. The most notable change is between the Rexair facility and the city wellfield where, at a depth of about 130 feet, a very dense gravel and silt zone was encountered. This zone was encountered in SLW-41, SLW-42, SLW-43, SLW-45, SLW-47, SLW-50, SLW-51, and SLW-52, and was too dense to penetrate more than 10 to 20 feet with augers.

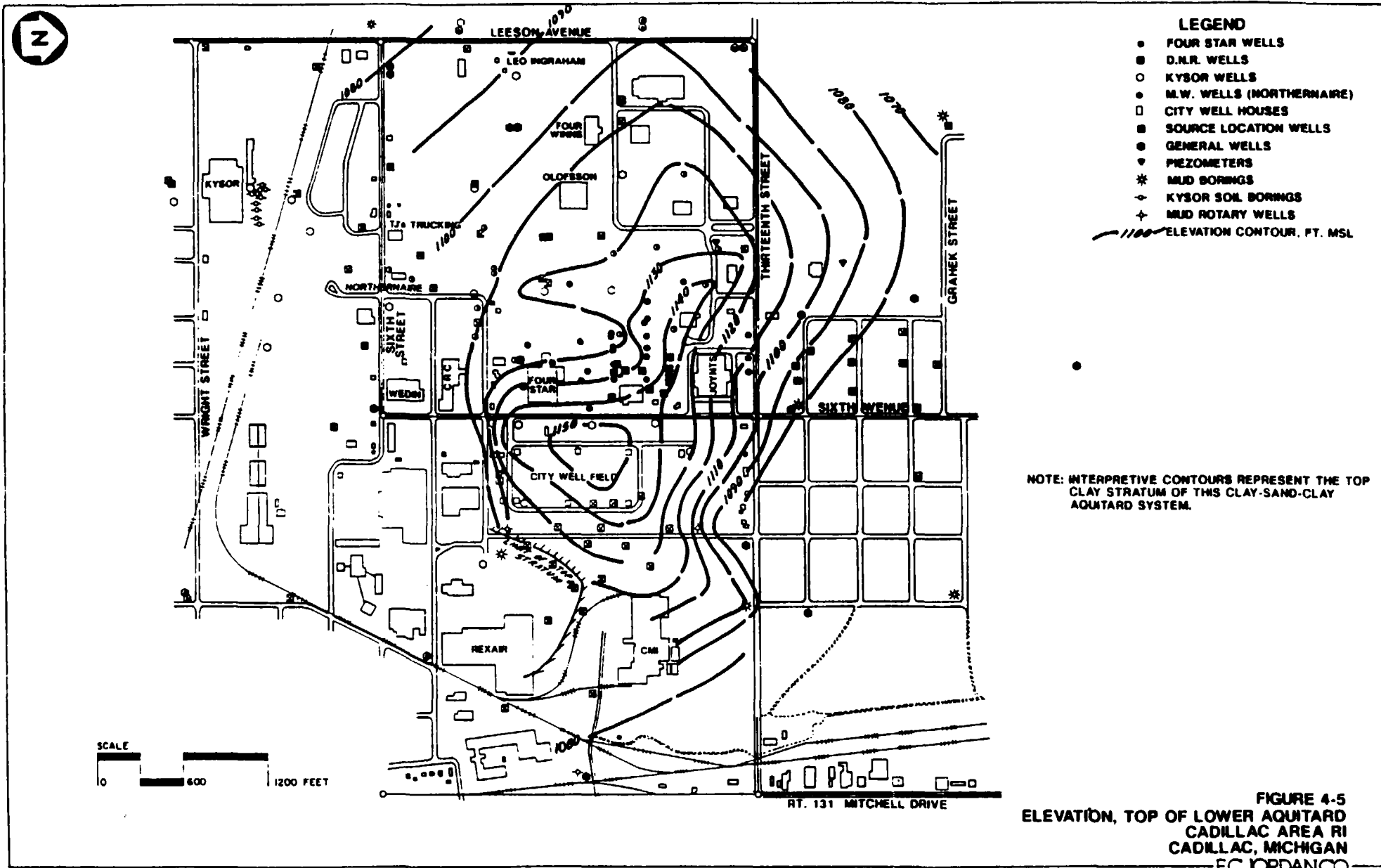
Another variation in grain size of the intermediate aquifer materials was noted in monitoring wells in the North Park Subdivision along Fourteenth and Fifteenth streets. At depths of 50 to 100 feet, coarse well-rounded gravel was encountered and occasionally comprised more than 50 percent of the formation.

4.3.4 Lower Aquitard

Geologic interpretation of the lower aquitard is based on only a few borings that penetrated this stratum, most of which are near the city wellfield. Figure 4-5 shows the interpreted surface elevation contours of the top of this unit. The lower aquitard is a clay-sand-clay sequence, with widely varying thicknesses in each stratum. The top clay stratum of the lower aquitard is encountered at a depth between 140 and 170 feet below ground surface, and ranges in thickness from zero to 19 feet at MB-4 and CW-3, respectively. The lower clay stratum ranges in thickness from 2 to 22 feet, while the intervening sand formation thickness ranges from 10 to 80 feet. (As stated previously,







this sand unit may be a separate aquifer. However, data do not exist on a sitewide basis to assess the extent of this aquifer. Including this unit as part of the lower aquitard does not affect the interpretation of groundwater and contaminant movement in the city wellfield vicinity.) Interpretations of the lower aquitard stratigraphy from boring log information were refined by results of the downhole geophysical logging in MB-2, MB-5, MB-6, MRW-1, MRW-2, and MRW-3. (See Appendix F for results of the downhole geophysics.)

It appears that the clay layers of the lower aquitard become very thin just east of the city wellfield, while the upper clay layer of the aquitard sequence vanishes altogether. Boring MB-4 encountered only 3 feet of the bottom clay layer from a depth of 231 to 234 feet (see Figure 4-2). This is significant because this clay aquitard is the only hydraulic barrier protecting the city water supply from contamination in the overlying aquifer. Based on the boring logs of city wells CW-3, CW-5, and CW-7, the lower aquitard clay layers appear to be thickest along the western edge of the wellfield. A total of 33 feet of lower aquitard clay was encountered in CW-3 and 38 feet in CW-7. As shown in Figure 4-5, a "hill" of clay forming the lower aquitard exists directly below the city wellfield. Also shown in Figure 4-5, however, is a sharp decrease in the elevation of the clay, which corresponds to the decline in clay thickness to the north, east, and south of the wellfield. For example, over the 200 feet separating CW-2 and MRW-1 to the east, the clay thickness decreases from 24 to 13 feet; within an additional 200 feet to MB-4, the clay thickness decreases still further to only 3 feet. To the north, the clay aquitard thickness decreases from 30 feet in CW-6 to 8 feet in MRW-2 over a distance of about 600 feet.

West of the city wellfield, the clay aquitard elevation dips more gradually, giving the hill of clay an elongated, east-west axis (see Figure 4-5). Less detail on the aquitard thickness is available west of the wellfield. The lower aquitard clay layers are a total of 7 feet thick in MB-5; further northwest in MB-2, the clay layers total about 8 feet thick. Due to the large distance between these borings, the horizontal continuity of this lower clay aquitard cannot be accurately assessed.

4.3.5 Lower Aquifer

Little geologic information was obtained from the lower aquifer. Based on drill cuttings and review of city well logs, the lower aquifer is predominantly sand, similar to other upper aquifer materials, but contains more gravel.

4.4 CHEMICAL ANALYSES

As described in Section 4.2, soil samples were collected in the unsaturated zone by split-spoon samplers. These samples were placed in appropriate containers for subsequent field screening for VOCs. Selected samples for VOC analysis were also sent for quantification to Versar, Inc. (Versar), a CLP-approved laboratory. The CLP Caucus on Organic Protocols (COP) were used to analyze the samples for Hazardous Substance List (HSL) VOCs.

Field screening for VOCs was accomplished by storing the jarred soil samples for approximately 15 to 20 minutes to allow VOCs in the soil to volatilize into the air in the jar. The jar headspace was analyzed with an HNU photoionization (PI) meter. This procedure provides a relative indication of contamination and was used to select soil samples to send to Versar.

The headspace analyses were conducted in several SLW boreholes, including some in known areas of high contamination. It was found that, if present, VOC contamination in the soils produced a concentration in the headspace less than the minimum detection level of the HNU (approximately 1 ppm). Results of this screening are provided on appropriate boring logs in Appendix D.

Analysis of the soils from SLW borings sent to the laboratory were similar to the field-screening results; that is, little contamination was detected. Inasmuch as the soils of the unsaturated zone contain very little silt or clay-sized particles and are very porous, small quantities of VOCs would not remain in this zone for an extended period of time. Also, two mechanisms (i.e., the downward vertical gradients of the groundwater and the downward push of rainfall infiltration) drive the contaminants below the water table where they cannot volatilize into the soil pore spaces. Appendix I contains the original and validated soil analyses from Versar.

Dumping of solvents directly to the soils is known to have occurred behind the Kysor facility. For this reason, in 1981 MDNR ordered Kysor to conduct a clean-up program involving the excavation of approximately 700 cubic yards of soil from a pit averaging about 6 feet deep. The excavation area, shown in Figure 4-6, was backfilled with clean soil to original grade. Strong odors were noted during excavation; MDNR analyzed several soil samples taken from the bottom of the pit for VOCs. Tabulated results appear in Appendix J, where relatively high concentrations (i.e., greater than 100 ppm) were shown to have remained.

To determine the extent of residual contamination in the soils, 10 KB borings were drilled 20 to 25 feet deep to the water table for the Cadillac Area RI. Section 4.2.6 describes the drilling method. Headspace analyses were conducted on the soil samples in the field, and two samples from each boring were sent to a CLP laboratory for VOC analysis. Table 4-1 lists the VOCs detected in soils, and Figures 4-1 and 4-6 locate the KB borings. Appendix K provides summary tables and original data sheets listing quantities of contaminants found at depth in each boring.

The highest concentration of contaminants was found to be approximately 150 feet north from the middle of the main Kysor building. Soil samples from adjacent KB-3 and KB-7 contained from 40 to 750 ppm of VOCs, with the highest contamination at a depth of approximately 20 feet (just above the water table). The primary contaminants found included xylene, toluene, ethylbenzene, TCE, and TCA. This contamination is relatively localized, inasmuch as KB-1 and KB-5, located 50 feet to the east and west (respectively), contain less than 0.5 ppm of VOCs. Also, KB-9, just south of KB-3 and KB-7, contains about 0.2 ppm; KB-8 and KB-10, just north of KB-3 and KB-7, contain 1.0 and 0.4 ppm, respectively. KB-2, located 100 feet east of KB-3 and containing over 6.3 ppm total VOCs, is the only other boring containing more than 0.4 ppm (see Figure 4-6).

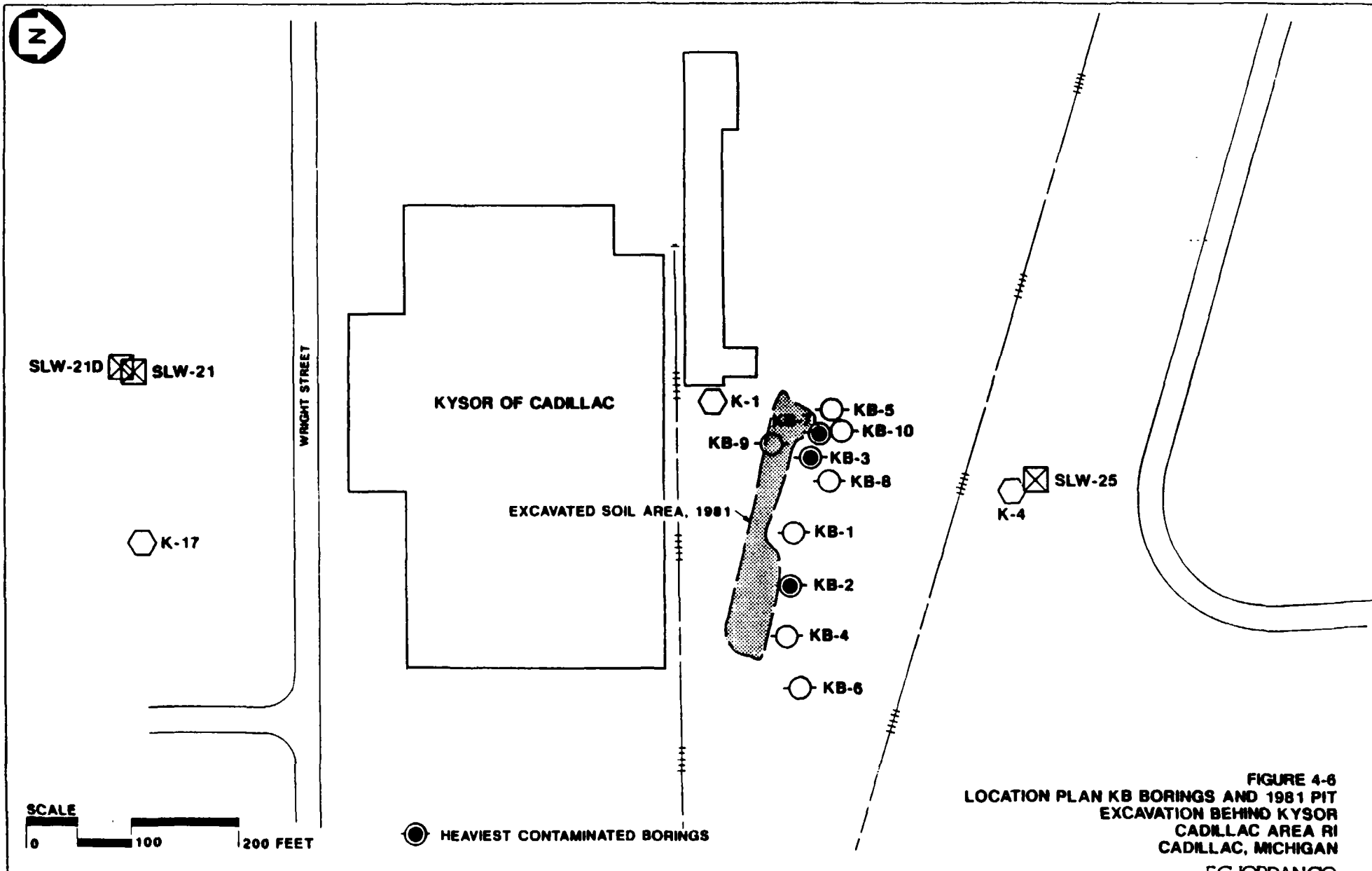


TABLE 4-1
VOLATILE ORGANIC COMPOUNDS DETECTED
IN SOILS BEHIND KYSOR
JULY 1987

COMPOUND	HIGHEST CONCENTRATION	BORING ID	
Xylene (Total)	520	KB-3	15-17 feet
Toluene	95	KB-3	15-17 feet
Trichloroethene	74	KB-3	15-17 feet
Ethylbenzene	68	KB-3	15-17 feet
1,1,1-Trichloroethane	24	KB-3	15-17 feet
Acetone	2	KB-3	10-12 feet
Tetrachloroethene	2.2	KB-3	10-12 feet
2-Butanone	5.7	KB-3	10-12 feet
1,1,2-Trichloroethane	.055	KB-1	25-27 feet
1,2-Dichloroethane	.005	KB-8	25-27 feet

NOTE: Results in parts per million (ppm).

Appendix K provides summary tables and quantitative laboratory analyses.

The contamination levels found in KB-3 and KB-7 indicate that the soils, combined with the infiltration of precipitation, are still a source of ground-water contamination. The clean-up efforts conducted by Kysor extended only to an average depth of about 6 feet, and significant contamination remains at a depth of 15 to 25 feet.

5.0 GROUNDWATER INVESTIGATION

The groundwater investigation beneath the Cadillac industrial park, ongoing since the late 1970s, has consistently concluded that this resource was contaminated from past disposal practices of hazardous substances, particularly VOCs. The Cadillac Area RI incorporated findings from previous investigations with those of the most recent field drilling and sampling program. This field program included the strategic location of monitoring wells to define the vertical and horizontal extent of VOC contamination across the industrial park in vertically adjacent aquifers. Other hydrologic characteristics that were determined include groundwater flow patterns in the three aquifers, vertical gradients between aquifers, and aquifer permeabilities.

5.1 MONITORING WELL INSTALLATION

Three types of monitoring wells were installed during the Cadillac Area RI to characterize groundwater quality:

- o source location wells (SLW)
- o general wells (GW)
- o mud-rotary wells (MRW)

The borings associated with these monitoring wells were described in Section 4.2. After reaching the desired depth of the boring (based on screening or other criteria), the monitoring well was installed through the hollow-stem of the auger or the grouted borehole of the mud-rotary boring. Figure 4-1 shows the boring and monitoring well locations.

5.1.1 SLW and GW Monitoring Wells

The SLW monitoring wells were installed to obtain groundwater quality and elevation data upgradient and downgradient of suspected contaminant sources. The GW monitoring wells were installed to provide these data on a regional basis at the study area perimeters and in areas between groups of SLW wells. The MRW monitoring wells were installed into the lower aquifer, adjacent to the city wellfield.

The SLW and GW monitoring wells were installed using the hollow-stem auger drilling technique with a 5-foot screened auger. The screened auger enables the aquifers to be pumped through the hollow-stem, thereby obtaining fairly discrete groundwater samples with depth as the boring is advanced. This is particularly true for aquifer materials of relatively high hydraulic conductivity, such as those at the Cadillac Area site.

Groundwater samples were collected through the screened auger at 10-foot intervals, beginning approximately 5 to 10 feet below the water table. At each interval, the "auger-well" was developed by air-surfing with a hose and air compressor until the groundwater became clear of fine sediment. A submersible pump was used to purge approximately three well volumes. Once the well was

developed and purged, a groundwater sample was collected in a 40-ml Teflon-capped glass vial.

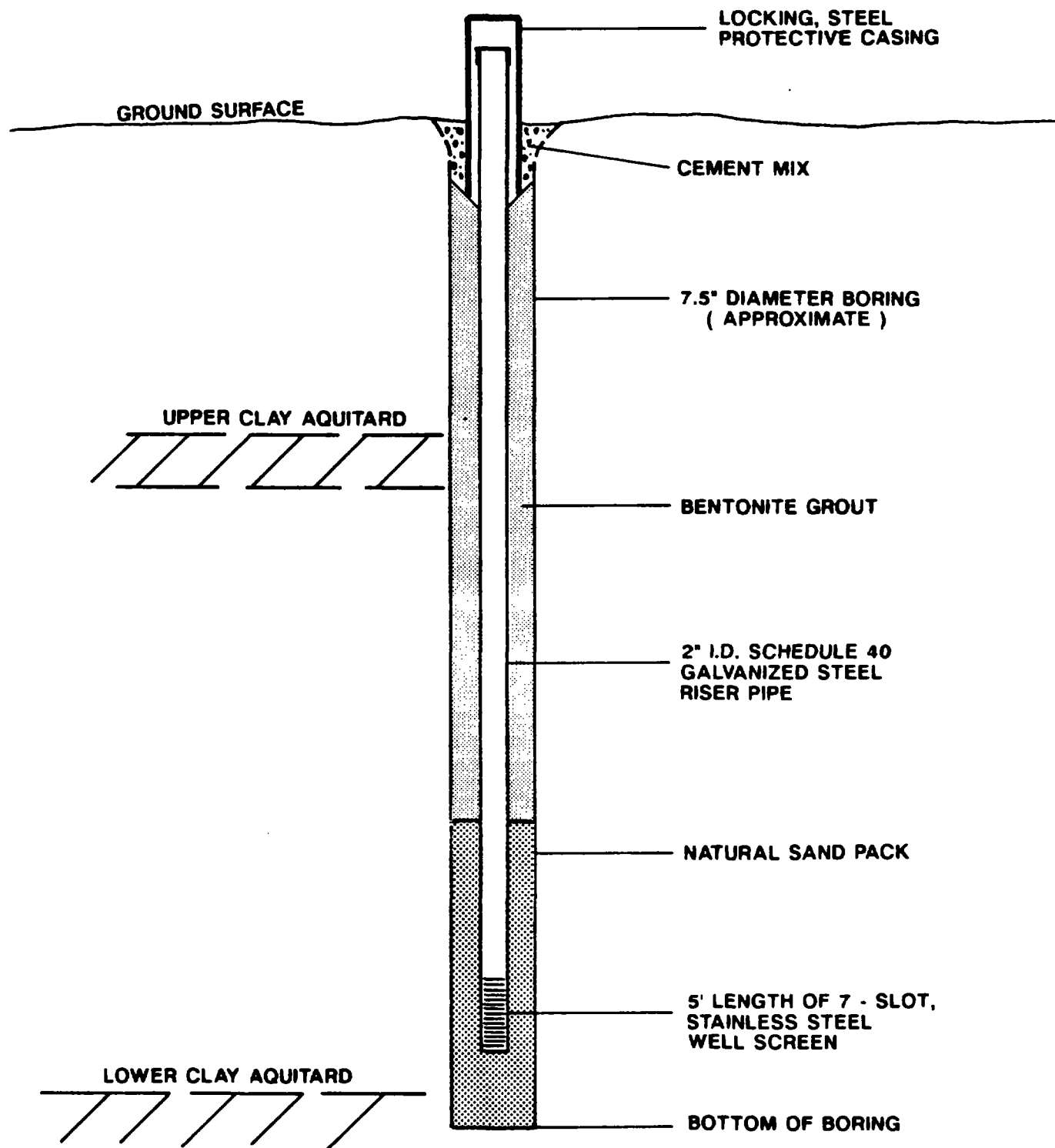
Using a Photovac 10-A-10 portable GC stationed in the site field office, each groundwater sample was then promptly analyzed for up to 10 of the most common VOCs known to be present at the site. These analyses provided an indication of the degree of contamination with depth. The monitoring well screen was then placed in the zone of highest contamination. In some locations where either the contamination was extensive with depth or the upper aquitard was present, two monitoring wells with screens at different depths (i.e., a well cluster) were installed. In these cases, the well screened in the upper aquifer was designated by an "S" (shallow), while the well screened in the intermediate aquifer was designated by a "D" (deep) (e.g., SLW-12S and SLW-12D). Results of the field screening and summary tables are included in Appendix L.

The SLW and GW monitoring wells were installed through the hollow-stems of the augers. First, the augers were pulled back to the depth of highest contamination where the well screen was to be placed. The borehole below this point was collapsed, and filled with natural sand. A 5-foot length of stainless steel well screen was then attached to an appropriate length of galvanized steel well casing to allow for a typical aboveground stick-up of approximately 2 feet. In some locations, particularly the roads of the North Park Subdivision, the well riser was set about 6 inches below ground level to allow construction of a flush-mounted valve box protector. The screen slot size (i.e., opening width) was 0.007 inches; the ID of the monitoring well was 2 inches. Once the casing and screen were in place, 10 to 15 feet of augers were removed, allowing the sand formation to collapse around the well screen and the casing to serve as a natural filter pack. A bentonite grout was then pumped through a tremie line into the annulus between the steel casing and augers to provide a seal, thereby restricting groundwater migration through the disturbed soil of the borehole. The grouting was also necessary to seal a borehole that penetrated the upper aquitard to restrict groundwater transfer between the shallow and intermediate aquifers along the borehole. Grouting was completed to within 1 foot of the surface, at which time a locking, steel protective casing was cemented over the well casing. Figure 5-1 illustrates a typical monitoring well installation. Appendix D provides completed specifications for each monitoring well. Appendix E lists specifications for new and existing monitoring wells.

After installation, the monitoring wells were developed by hydraulic surging with an air compressor until the groundwater became sediment-free. Before drilling the next well, downhole drilling tools (and the rig) were decontaminated with a high-pressure steam generator in a designated decontamination area at the site field office, adjacent to the Northernair facility.

5.1.2 MRW Monitoring Wells

Mud-rotary techniques were used to penetrate the lower aquitard and aquifer, and to install three monitoring wells. This method precludes the acquisition of groundwater samples with depth for field-screening analysis. The screens for the deep monitoring wells were placed between approximately 295 and 300 feet below ground, which is the approximate depth of the city water supply wells (although the city water supply wells have longer screen lengths).



NOTE: SIMILAR CONSTRUCTIONS WERE USED FOR MONITORING WELLS LOCATED IN THE SHALLOW, INTERMEDIATE, AND DEEP AQUIFERS.

**FIGURE 5-1
TYPICAL MONITORING
WELL CONSTRUCTION
CADILLAC AREA RI
CADILLAC, MICHIGAN**

EC.JORDANCO

Once the mud-rotary boring was drilled to the desired depth, the casing was removed, while the thick drilling mud (i.e., bentonite grout) prevented the hole from collapsing. Stainless steel well screens and galvanized well casings used to construct the monitoring wells were lowered through the grout to the bottom of the borings. The grout was removed from around the well screen, and a natural sand filter pack was emplaced by forcing clean water through the well screen using a high-pressure hose connected to the top of the well casing. This water would flow out through the well screen, and approximately 15 to 20 feet of grout would be lifted from the bottom of the borehole. The monitoring well was then extensively developed to remove any remaining drilling mud and other fine-grained material.

5.2 PIEZOMETER INSTALLATION

Three piezometers (i.e., P-3, P-4, and P-5) were installed to provide groundwater elevation data on a regional basis. Figure 1-1, the regional plan, locates these piezometers (P-1 and P-2 were drilled during the Northernnaire study), which were installed with hollow-stem augers to a depth between 35 and 50 feet. The piezometers were constructed with 2-foot sections of stainless steel well screens and appropriate lengths of galvanized steel well casings. The borehole annulus was not sealed with a bentonite grout, but rather backfilled with the drill cuttings, because groundwater quality samples were not to be obtained.

P-3 was installed approximately one-half mile east of the study area on the southwestern corner of Thirteenth and Plett streets. P-4 was installed at the southwestern corner of the municipal airport, north of the industrial park. P-5 was installed approximately 3 miles northwest of the study area, near the southwestern edge of Pleasant Lake. Appendix D includes the boring logs and installation reports for the piezometers.

5.3 HYDROGEOLOGY

The hydrogeology of the Cadillac industrial park is discussed on the basis of the following characteristics:

- o the horizontal groundwater flow patterns in each of the three identified sand aquifers
- o the vertical gradients between aquifers and the apparent effectiveness of the clay aquitards
- o the influence of groundwater pumping from the lower aquifer on groundwater flow in this and overlying aquifers

5.3.1 Horizontal Groundwater Flow

The depth to groundwater was measured in the monitoring wells in the industrial park on several occasions over the past year (see Appendix M). Table 5-1 lists the most complete set of groundwater elevation data obtained June 29, 1987, including the first set from the lower aquifer. These data show consistent

TABLE 5-1

GROUNDWATER ELEVATION DATA
JUNE 29, 1987
CADILLAC AREA RI

Monitoring Well	Groundwater Elevation	Monitoring Well	Groundwater Elevation	Monitoring Well	Groundwater Elevation
SLW-1	1270.11	SLW-39	1271.89	MW-1	1274.88
SLW-2	1269.47	SLW-40	1271.71	MW-2	1277.08
SLW-3	1269.39	SLW-41	1271.44	MW-3	1275.04
SLW-4	1268.99	SLW-42	1271.64	MW-4	--
SLW-5	1269.03	SLW-43	1271.29	MW-5	1274.92
SLW-6	1276.63	SLW-44	1271.14	MW-6	1275.82
SLW-7	1269.86	SLW-45	1270.89	MW-7	1270.51
SLW-8	1277.42	SLW-46	1271.46	MW-8	1271.16
SLW-9	1277.55	SLW-47	1271.42	MW-9	1270.33
SLW-10S	1273.45	SLW-48	1271.25	MW-10	1270.31
SLW-10D	1271.04	SLW-49	1271.47	MW-11	1270.08
SLW-11	1277.76	SLW-50	1271.13	MW-12	1270.87
SLW-12S	1274.21	SLW-51	1270.79	MW-13	1270.25
SLW-12D	1270.92	SLW-52	1271.05	MW-14	1270.69
SLW-13	1268.66	SLW-53	--	MW-15	1270.53
SLW-14	--			MW-16	1269.90
SLW-15	1271.06	GW-1S	1274.00	MW-17	1271.48
SLW-16	1275.11	GW-1D	1272.92	MW-18	1270.10
SLW-17	1268.63	GW-2	1272.63		
SLW-18	1271.42	GW-3	1271.77	K-6S	1274.60
SLW-19	--	GW-6S	1271.74	K-6D	1273.77
SLW-20	1268.52	GW-6D	1269.21	K-8	1271.32
SLW-21S	1279.43	GW-7	1268.06	K-10	1274.40
SLW-21D	--	GW-8	1269.25	K-12	1271.05
SLW-22	1270.86	GW-9S	1277.48	K-13	1270.51
SLW-23	1276.35	GW-9D	1270.64	K-14	1271.05
SLW-24	1278.05	GW-10S	1275.90	K-15	1270.69
SLW-25	1278.32	GW-10D	1270.35	K-16S	NS
SLW-26	1272.91	GW-11	NS	K-16D	1270.21
SLW-27	1271.72	GW-13	1270.12	K-17	1279.42
SLW-28	1278.75	GW-14S	1273.22	K-21	1271.99
SLW-29	1272.05	GW-14D	1269.30	K-23	1271.57
SLW-30	1273.58	GW-16	--	K-24	1270.23
SLW-31	1272.84	GW-17	1268.82		
SLW-32	1276.66	GW-18	1269.74	P-1	1270.14
SLW-33	1271.68	GW-20	1272.01	P-2	1270.17
SLW-34	1268.81			P-3*	1273.74
SLW-35	1268.65	MRW-1	1266.71	P-4*	1273.00
SLW-36	1291.74	MRW-2	1264.91	P-5*	1311.69
SLW-37	1271.63	MRW-3	1270.48	DNR-1	1272.40
SLW-38	1273.55				

NOTES: All elevations in feet above mean sea level.
 * Elevations obtained on August 21, 1987.
 -- No data.
 NS = Well not surveyed.

patterns of horizontal flow directions in the shallow and intermediate aquifers over time. The July 29 data are used to construct the groundwater contour maps in this report.

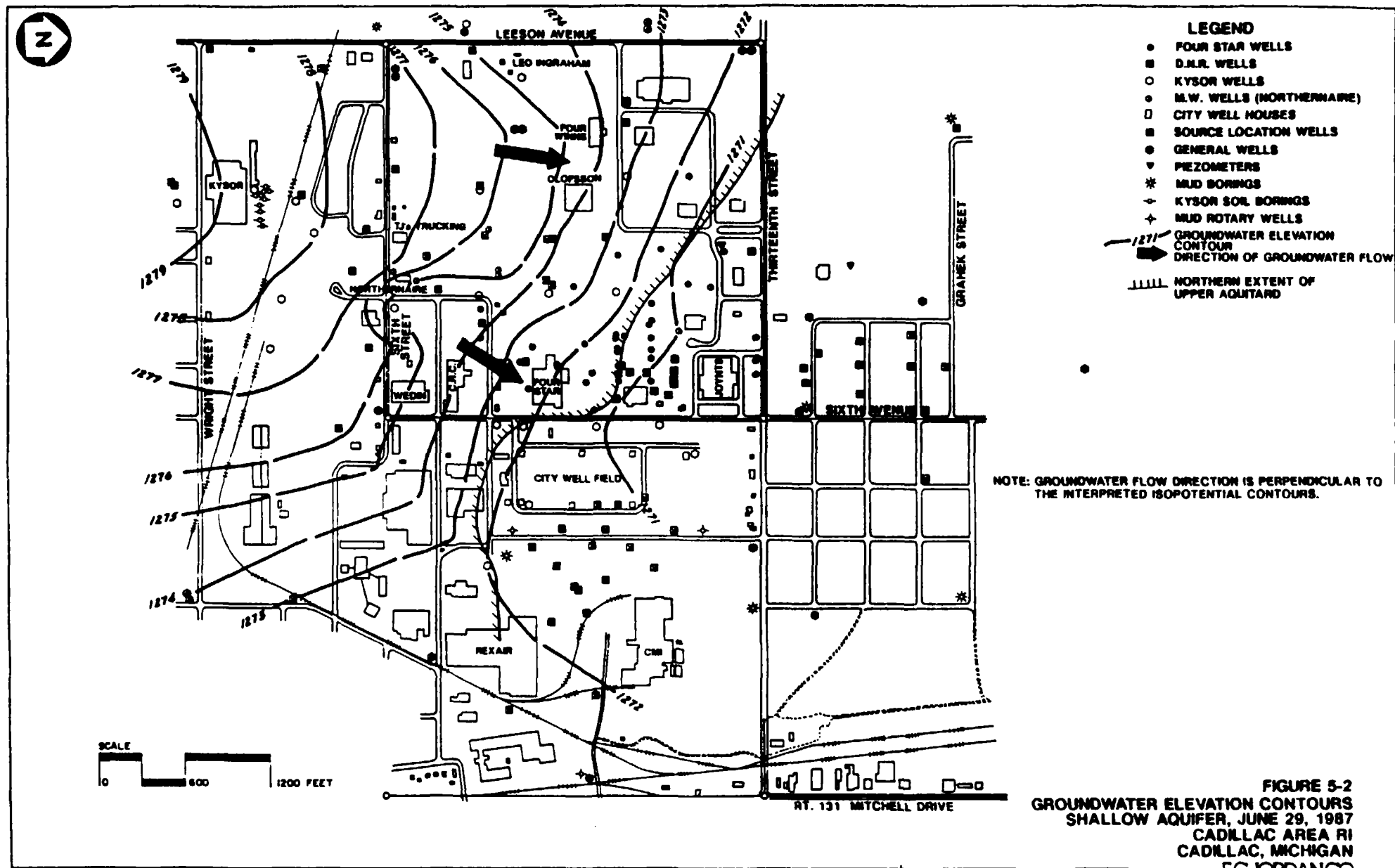
Shallow Aquifer. As discussed in Section 4.3, the shallow aquifer exists in the southwestern half of the study area and consists of a fairly uniform, well-rounded quartz sand. The aquifer becomes thicker to the southwest due to the sloping upper aquitard. The upper aquitard pinches out along a north-west-southeast trending line near the center of the site (see Figure 4-2), as well as where the shallow and intermediate aquifers merge.

The shallow aquifer is under water table (i.e., unconfined) conditions, receiving recharge directly from infiltration of precipitation through the soils. The depth to the water table ranges from 20 feet near the Kysor facility to about 30 feet near the northern edge of the upper aquitard. Seasonal water table elevations at any one location fluctuated as much as 3 feet between the spring high and the late summer low.

Groundwater flow in the shallow aquifer is in a northeastern direction in most site areas. However, toward the western side of the study area, groundwater flow becomes more northerly to northwesterly. Also, along the southern boundary of Wright Street, groundwater flows approximately due east. North of the limit of the upper aquitard, the flow direction is to the northwest, due to influences of the intermediate aquifer. Figure 5-2 illustrates the horizontal flow pattern in the shallow aquifer. This broad, radial pattern suggests that an area of significant groundwater recharge exists south of Kysor beyond the industrial park study area. While the recharge may simply be infiltrating precipitation, the source may be Lake Cadillac. Insufficient water level data between the site and the lake preclude this determination.

A deviation from the general northeastward groundwater flow direction is indicated just east of the Northernaire facility at K-6S. The water table elevation is significantly lower at K-6S than in surrounding wells, resulting in the formation of a small groundwater "sink" (see Figure 5-2). This apparent depression in the water table elevation may reflect the relative thinness of the upper clay aquitard in the area, as shown in Figure 4-4. Due to the thinness of the aquitard, groundwater can flow more easily from the shallow to the intermediate aquifer.

The horizontal gradient in the shallow aquifer increases from approximately 0.001 ft/ft to the south by Kysor, to 0.003 ft/ft to the north near the limit of the upper aquitard. The increasingly steeper gradient above the aquitard is due to the clay sloping upward from southwest to northeast in the direction of groundwater flow, thereby decreasing the saturated thickness of the shallow aquifer. This increase in gradient is necessary to maintain a constant quantity of water per unit time where the cross-sectional area is decreasing. However, the increase in gradient is not in the same ratio as the available cross-sectional area, which implies appreciable seepage through the underlying aquitard. The gradient quickly becomes less steep, toward the northwest where the shallow and intermediate aquifers merge.



The permeability of the shallow aquifer is estimated to be from 1×10^{-2} to 3×10^{-2} cm/sec. In local areas, this may increase or decrease due to silt or gravel content, but the permeability is otherwise apparently quite uniform. Table 5-2 lists the permeability estimates of selected monitoring wells based on single-well, rising-head tests. Appendix N provides the testing method and raw data from which the permeabilities were estimated by Keck.

Based on the hydraulic gradient, the hydraulic conductivity, and an assumed porosity of 0.25 to 0.30, the groundwater flow rate may vary from about 40 to 400 ft/yr. Over much of the site, the average flow rate is about 200 ft/yr.

Intermediate Aquifer. The intermediate aquifer exists across the entire study area. It is a confined aquifer in the southwestern portion of the site where the shallow aquifer and upper aquitard are present, becoming a water table aquifer northeast of the extent of the upper aquitard.

Groundwater flow in the intermediate aquifer is in a north-northwest direction. Figure 5-3 illustrates this flow pattern. The horizontal hydraulic gradient ranges from approximately 0.0010 to 0.0017 ft/ft, and is steepest north of the city wellfield. Also, a distortion of the contour lines is evident in the area north of the wellfield. This distortion and the steeper gradient may reflect an influence on the intermediate aquifer due to pumping in the lower aquifer. Since the aquifer materials are similar to those in the shallow aquifer, the groundwater flow rates are estimated between 40 and 200 ft/yr, averaging about 80 ft/yr for most site conditions. The intermediate aquifer is considerably thicker (about 100 to 130 feet) than the upper aquifer (10 to 20 feet) where they merge.

Lower Aquifer. Geologic and hydrologic data from the lower aquifer was collected primarily in the vicinity of the city wellfield. Few physical properties of the aquifer are known other than its composition is primarily of sand, similar to the shallow and intermediate aquifers, and contains a relatively high percentage of gravel. Various hydrologic properties were estimated from pump test results using the city pumping wells and the MRW wells to monitor changes in hydraulic head (see Section 5.3.3).

Groundwater flow in the lower aquifer is in a west-northwest direction, as shown in Figure 5-4. The horizontal gradient is approximately 0.003 ft/ft. These interpretations are based on data from MRW-1, MRW-2, and MRW-3 (situated east of the city wellfield), which may be strongly affected by pumping of the city supply wells. However, without piezometric data in the lower aquifer to the north, south, and west of the wellfield, or data from the existing wells during times of non-pumping, it is not possible to directly assess the influence of pumping on regional groundwater flow. Actual gradients and flow directions in the general extent of the lower aquifer may vary from those interpreted from the limited data available from these three monitoring wells.

5.3.2 Vertical Gradients

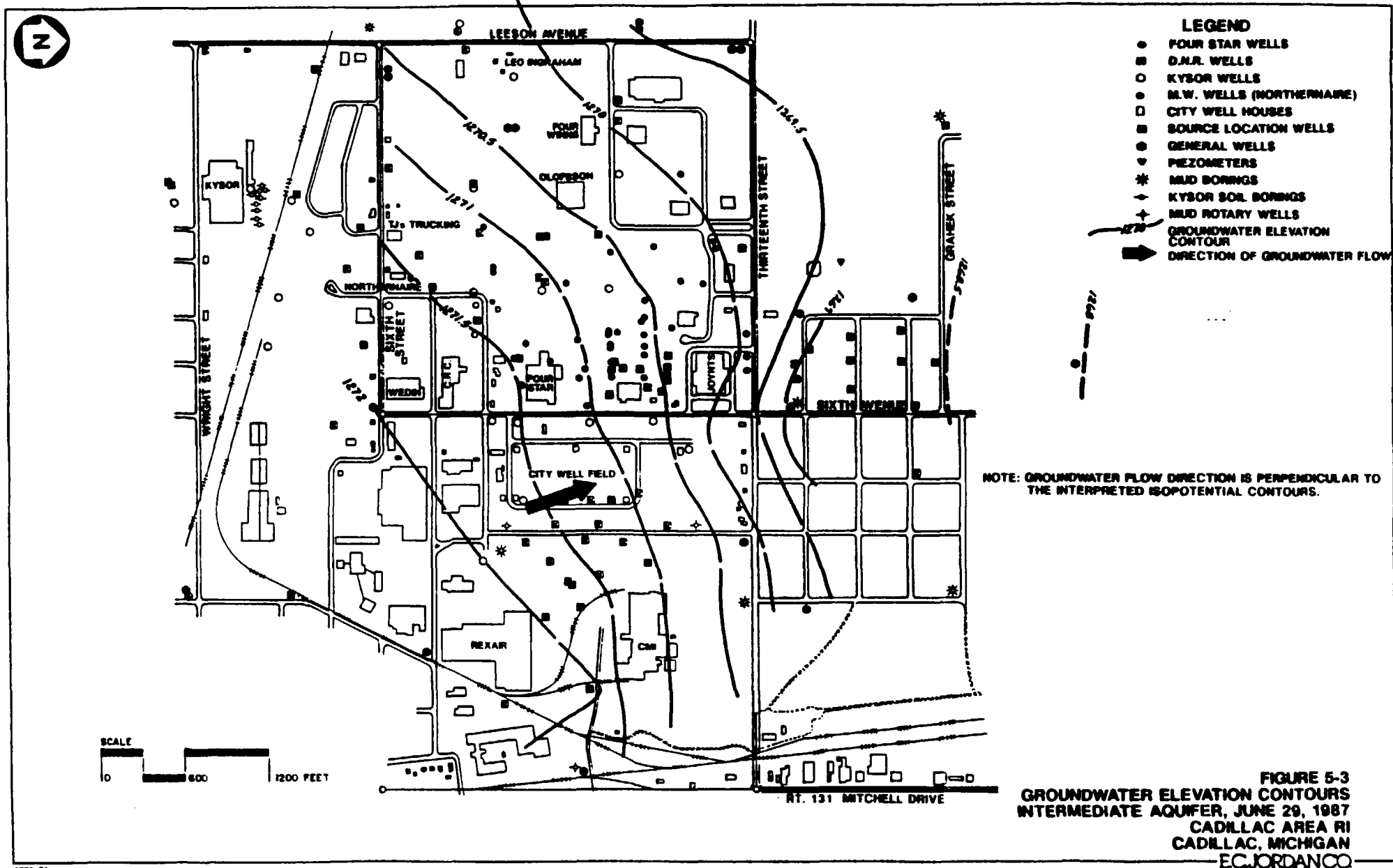
Vertical gradients between the aquifers were determined by comparing piezometric data in closely grouped monitoring-well pairs that are screened in different aquifers. These gradients comprise the force by which groundwater

TABLE 5-2

PERMEABILITY TEST RESULTS OF SELECTED WELLS
CADILLAC AREA RI

MONITORING WELL	PERMEABILITY (CM/SEC)
<u>Shallow Aquifer Wells</u>	
SLW-6	0.040
SLW-9	0.058
SLW-12S	0.034
SLW-16	0.020
SLW-21S	0.048
SLW-23	0.026
SLW-24	0.037
SLW-30	0.013
SLW-32	0.037
GW-1S	0.008
GW-9S	0.034
GW-14S	0.052
MW-4	0.003
<u>Intermediate Aquifer Wells</u>	
SLW-1	0.044
SLW-2	0.030
SLW-10D	0.063
SLW-12D	0.026
SLW-18	0.034
SLW-20	0.079
SLW-22	0.034
SLW-33	0.035
GW-1D	0.041
GW-2	0.044
GW-7	0.055
GW-8	0.044
GW-9D	0.040
GW-11	0.049
GW-14D	0.035
GW-16	0.042
GW-18	0.034
MW-7	0.055
MW-17	0.063

Permeability testing and results provided by Keck Consulting Services, Inc.
See Appendix N.



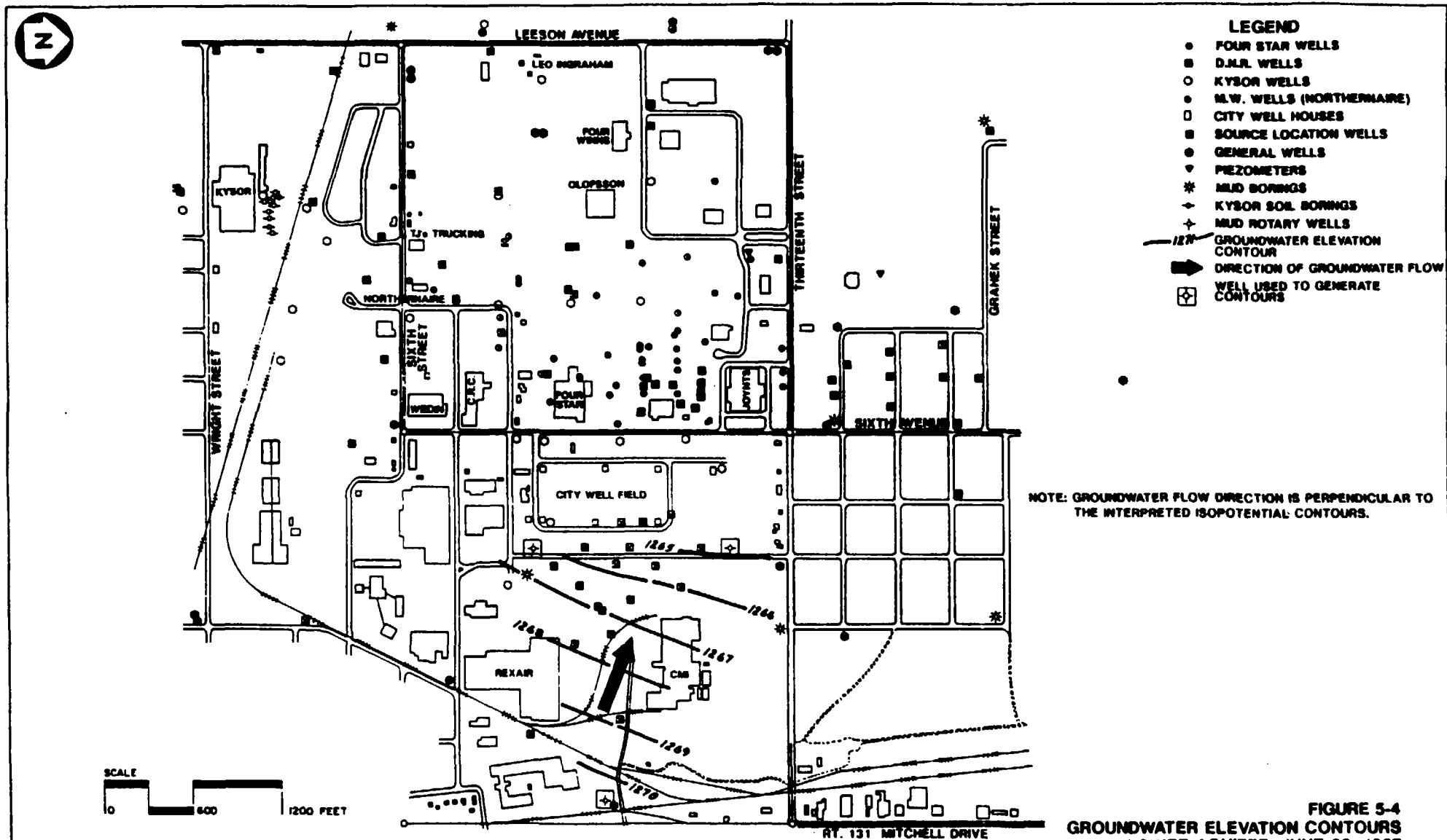


FIGURE 5-4
GROUNDWATER ELEVATION CONTOURS
LOWER AQUIFER, JUNE 29, 1987
CADILLAC AREA RI
CADILLAC, MICHIGAN
EC JORDAN CO

will move vertically downward through the aquifers and aquitards. The effectiveness of the clay aquitards to restrict groundwater mixing is also dependent on thickness and head loss across the aquitard. Vertical gradients between the shallow and intermediate aquifers were measured in 16 well pairs (Table 5-3).

In all cases, the piezometric level in the monitoring well screened in the confined, intermediate aquifer is below the water table level in the unconfined, shallow aquifer. This indicates that a downward vertical gradient exists and groundwater will tend to migrate from the shallow to the intermediate aquifer through the upper aquitard. The vertical component of the gradient is generally larger than the horizontal component (see Section 5.3.1), indicating a general downward seepage throughout the aquifers. This suggests that a primary migration route is between aquifers, which is confirmed by observing contaminants at depth within relatively short distances downgradient from probable or identified sources.

The thinness of the upper clay aquitard in the vicinity of Seventh Avenue, as logged by the drilling and shown in Figure 4-4, is further indicated by the presence of a relatively small downward vertical gradient at monitoring well cluster K-6. Conversely, toward the southwest and west where the clay is thickest, the vertical gradients are generally larger. The area by K-6 (east of Northernnaire) may be an area where groundwater and contaminant seepage is significant, due to the thin aquitard.

Vertical gradients were measured between the intermediate and lower aquifers near the three monitoring wells screened in the lower aquifer (see Table 5-3). As before, vertical gradients are downward. A head difference of 1.3 feet exists on the eastern edge of the study area near MRW-3, while head differences are more than 5 feet adjacent to the city wellfield near MRW-1 and MRW-2. The greater head differences closer to the wellfield are interpreted as a result of pumping in the lower aquifer. The lower aquitard, over which most of the head drop would occur, appears to be approximately the same thickness in the areas of the three MRW wells (see Figure 4-2).

5.3.3 Influence of City Wellfield Pumping

The influence of the city wellfield pumping on groundwater movement in the lower and intermediate aquifers was estimated by studying groundwater flow patterns in these two aquifers, assessing vertical gradients between aquifers, and analyzing pump test data using the city supply wells and adjacent monitoring wells. Pump test methodologies and data acquired by Keck and MDNR are compiled in a report by Keck entitled "Well Field Evaluation, City of Cadillac" dated December 21, 1987.

Currently, several hydrologic observations are attributable to the influence of the wellfield pumping on local groundwater movement:

- o Groundwater in the eastern portion of the lower aquifer is flowing directly toward the wellfield (based on limited data).
- o Vertically downward gradients between the lower and intermediate aquifers are greater closer to the pumping source.

TABLE 5-3

MONITORING WELL PAIRS AND
VERTICAL HYDRAULIC GRADIENTS

Well	Groundwater Elevation	Head (feet)	Gradient	Well	Groundwater Elevation	Head (feet)	Gradient
SLW-10S	1273.45			GW-6S	1271.74		
SLW-10D	1271.04	2.41	0.04	GW-6D	1269.21	2.53	0.02
SLW-12S	1274.21			GW-9S	1277.48		
SLW-12D	1270.92	3.29	0.08	GW-9D	1270.64	6.84	0.11
SLW-37*	1271.20			GW-10S	1275.90		
SLW-42*	1271.16	0.04	5×10^{-4}	GW-10D	1270.35	5.55	0.04
MW-6	1275.82			GW-14S	1273.22		
SLW-15	1271.06	4.76	0.14	GW-14D	1269.30	3.92	0.03
SLW-36	1271.74			MW-3	1275.04		
MRW-1	1266.71	5.03	0.02	MW-8	1271.16	3.88	0.03
SLW-51	1270.79			P-1	1270.14		
MRW-2	1264.91	5.88	0.03	MW-11	1270.08	0.06	5×10^{-4}
GW-3	1271.77			DNR-1	1272.40		
MRW-3	1270.48	1.29	0.01	MW-17	1271.48	0.92	9×10^{-4}
GW-1S	1274.00			K-6S	1274.60		
GW-1D	1272.92	1.08	0.01	K-6D	1273.77	0.83	0.01

NOTES:

1. All elevations obtained on June 29, 1987, except if noted by *, where elevations were obtained on August 21, 1987.
2. Gradient is estimated as:
$$\frac{\text{Head Difference}}{\text{Distance Between Screens}}$$
3. The first well listed in each pair has the shallowest screen.
4. In every instance shown above, the vertical gradient is downward.

- o The vertical and horizontal configuration of a solubilized VOC plume in the intermediate aquifer near the wellfield suggests an influence from the pumping of the lower aquifer (see Sections 5.4 and 6.2).
- o Pump test results could not be analyzed in detail because of the variability of the city wells pumping rates during the testing period. Also due to pumping requirements, it was difficult to keep the same wells pumping in the same part of the wellfield; therefore, affecting drawdown influences. Data from this test and tests run at the time of assessment of the aquifers for suitability as a water supply were used by Keck to estimate the seepage of water from intermediate to the lower aquifers.

5.3.4 Groundwater Modeling

MDNR (Ms. Hope Croskey, principal investigator) prepared a detailed groundwater flow model of the study area. The model was developed based on the modular three-dimensional finite-difference groundwater flow model code by McDonald and Harbaugh of the U.S. Geological Survey.

Due to the absence of nearby natural geologic and hydrogeologic groundwater flow controls, the final model covered approximately 10 square miles with the industrial park area situated near the center of the model. The model area was discretized using a variable grid size into 48 rows and 44 columns, with the smallest grid block being 200 feet by 200 feet, and the largest, 1000 feet by 1000 feet. Four layers were employed in the model. The first (uppermost) represented the water-table aquifer and incorporated the upper aquitard where present. The second layer represented the intermediate aquifer which could behave either as confined or unconfined depending on the presence or absence of the upper aquitard. The third layer represented the lower aquitard, which was relatively thick in areas and which also contained zones of sands layered with the clays. The fourth layer represented the municipal well aquifer, which was treated as confined and of constant transmissivity.

The model was run at steady-state and calibrated against estimated average site conditions based on available water level data. Some piezometers had been installed for the express purpose of providing data for model calibration. Overall mass water balances were utilized as a frame of reference for the model, including an evaluation of the Clam River serving alternately as a sink or recharge area for groundwater flow. Sensitivity analyses were performed, particularly with regard to the municipal well aquifer where data were limited.

The final model reasonably represents the flow system at the site and has been used to infer probable interactions between the intermediate and municipal well aquifers. As more site data become available, the model will continue to be refined. The model may also serve as one facet of the evaluation of remedial alternatives.

5.4 GROUNDWATER QUALITY SAMPLING

Groundwater samples were collected from the monitoring wells installed for the Cadillac Area RI as well as several Northernnaire and Kysor wells to quantify VOC and metals contamination in the aquifers. Two rounds of samples were collected: in June and July/August 1987. Although other data sources may represent past conditions, they were used to describe site conditions and trends, where appropriate. However, the emphasis in the following sections is on current data.

Groundwater sampling was conducted according to strict USEPA sampling protocols. The monitoring wells were developed within a week of drilling and installation. A minimum of two weeks was allowed to elapse between development and groundwater sampling.

Groundwater samples were collected in a specific order so that upgradient (i.e., background) wells were sampled first and downgradient (i.e., contaminated) wells last. Before sampling, each well was checked for proper identification and location. A PI meter was used to measure VOCs at the well mouth and in ambient air. Ambient air was monitored before and after removing the well cap.

Static water level and well depth were measured from the top of the inner well casing with an electronic water level indicator. The volume of water standing inside the well casing was calculated and, with the use of a packer, three times the volume of water below the packer was evacuated from the well. Well evacuation (i.e., purging) was accomplished with a submersible pump and packer. If the concentration of VOCs measured at the well mouth was greater than background, ambient air monitoring continued during purging. Groundwater temperature, pH, and specific conductance were also recorded using a Tripar Analyzer at the start of purging, after purging 1.5 well volumes, and just before sampling. These measurements were used to establish the collection of a representative groundwater sample. If these three parameters were stabilized, it was assumed that stagnant water in the well casing had been removed.

Groundwater samples for VOC analysis from wells evacuated with the pump were collected in two 40-ml Teflon septum-capped glass vials, directly from Teflon tubing connected to the pump. The samples were then preserved on ice, packaged, and sent by overnight courier to Versar, Inc. (Versar) of Virginia. Groundwater samples for metals analysis were filtered and preserved, as appropriate, in the field. Samples to be analyzed for dissolved metals were pressure-filtered with nitrogen in a Geotech 2.4-liter barrel filter using a 0.45-micron membrane filter. Between samples, the filter assembly was rinsed with a dilute nitric acid solution, followed by distilled water. The membrane filter was replaced between samples. Samples to be analyzed for metals were preserved with nitric acid to a pH of less than 2, while samples to be analyzed for cyanide were preserved with sodium hydroxide to a pH of greater than 13. Preservatives were provided by Jordan's laboratory. Sample duplicates and blanks were obtained according to USEPA CLP requirements.

The sampling pump was decontaminated between samples by pumping a solution of trisodium phosphate (TSP) and water through the pump and tubing, followed by a

rinse of clean distilled water. This was done by submerging the pump in a decon tube made of a 4-foot length of 4-inch-diameter PVC pipe sealed at one end. The exterior of the pump and tubing were then cleaned with TSP and water, and rinsed with distilled water. Equipment that came in contact with groundwater (e.g., Tripar probes or water level sensors) was rinsed with distilled water between wells.

5.5 GROUNDWATER ANALYSIS AND CONTAMINANT SOURCE IDENTIFICATION

Chemical analyses of groundwater were conducted by Versar and its affiliate, ESM of Maryland, both CLP laboratories. Table 5-4 is the HSL of VOCs for which each sample was analyzed. Other non-HSL VOC compounds were tentatively identified, if present. Metals, including hexavalent chromium, were analyzed in selected monitoring wells defining the Northernnaire plume. Approximately 105 monitoring wells were sampled in each round. Jordan personnel reviewed and validated the CLP data. Data validation reports are included in Appendix O along with summary tables of the original laboratory data. Due to the large data base, the tables were not presented in the text, but are included in Appendix O. The data were plotted and presented in graphical form for easier comprehension. Figures were based on June 1987 data; the July/August data largely confirm the June round. Significant differences are discussed in the following paragraphs.

Currently, five distinct probable contaminant sources have been identified based on plume definition in groundwater. These probable sources and respective plumes provide a basis for discussing the laboratory analyses. The sources include Kysor, Four Winns (Frisbie Street) and Leo Ingraham Excavating Company, Four Star/Joynts (Sixth Avenue), Rexair, and Northernnaire.

Other smaller sources within the industrial park may have contributed to the groundwater contamination. However, due to the great areal extent and high concentrations in the contaminant plumes attributed to the industries discussed previously, they are not distinguishable based solely on review of the chemical analyses.

5.5.1 Kysor Source

The Kysor plume is the most areally extensive in the industrial park and is found in both the shallow and intermediate aquifers. Figures 5-5 and 5-6 illustrate the horizontal plume dimensions for the shallow and intermediate aquifers, respectively. An arbitrary lower limit contour of 0.1 ppm VOCs was used for the plume definition contours.

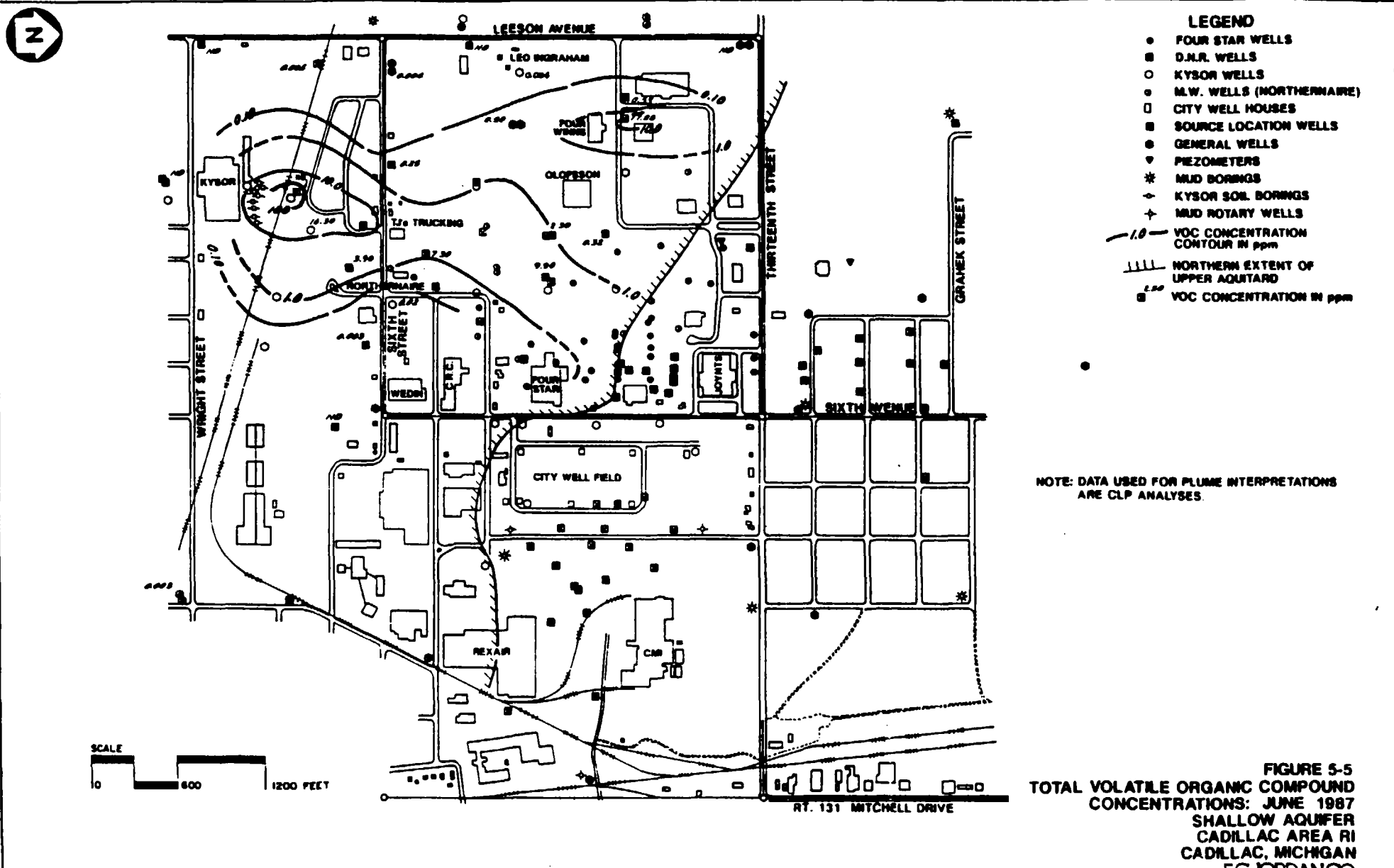
The highest concentration of VOC contaminants encountered in the shallow aquifer occurred approximately 300 feet downgradient (north) of the Kysor plant. Total VOC concentration reached approximately 115 ppm in SLW-25. Table 5-5 lists the compounds found in greatest abundance in the Kysor plume.

As shown in Figure 5-5, the Kysor plume extends longitudinally for over 2,500 feet in the shallow aquifer. Laterally, the plume reaches a width of about 1,500 feet near the Kysor plant. Further downgradient, the plume becomes

TABLE 5-4

HAZARDOUS SUBSTANCE LIST OF VOLATILE ORGANIC COMPOUNDS
FOR WHICH GROUNDWATER WAS ANALYZED

1. Chloromethane
2. Bromomethane
3. Vinyl Chloride
4. Chloroethane
5. Methylene Chloride
6. Acetone
7. Carbon Disulfide
8. 1,1-Dichloroethene (DCE)
9. 1,1-Dichloroethane (DCA)
10. Trans-1,2-Dichloroethene
11. Chloroform
12. 1,2-Dichloroethane
13. 2-Butanone
14. 1,1,1-Trichloroethane (TCA)
15. Carbon Tetrachloride
16. Vinyl Acetate
17. Bromodichloromethane
18. 1,2-Dichloropropane
19. Bis-1,3-Dichloropropene
20. Trichloroethene (TCE)
21. Dibromochloromethane
22. 1,1,2-Trichloroethane
23. Benzene
24. Trans-1,3-Dichloropropene
25. 2-Chloroethylvinylether
26. Bromoform
27. 4-Methyl-2-Pentanone
28. 2-Hexanone
29. Tetrachloroethene (PCE)
30. 1,1,2,2-Tetrachloroethane
31. Toluene
32. Chlorobenzene
33. Ethylbenzene
34. Styrene
35. Xylene (Total)



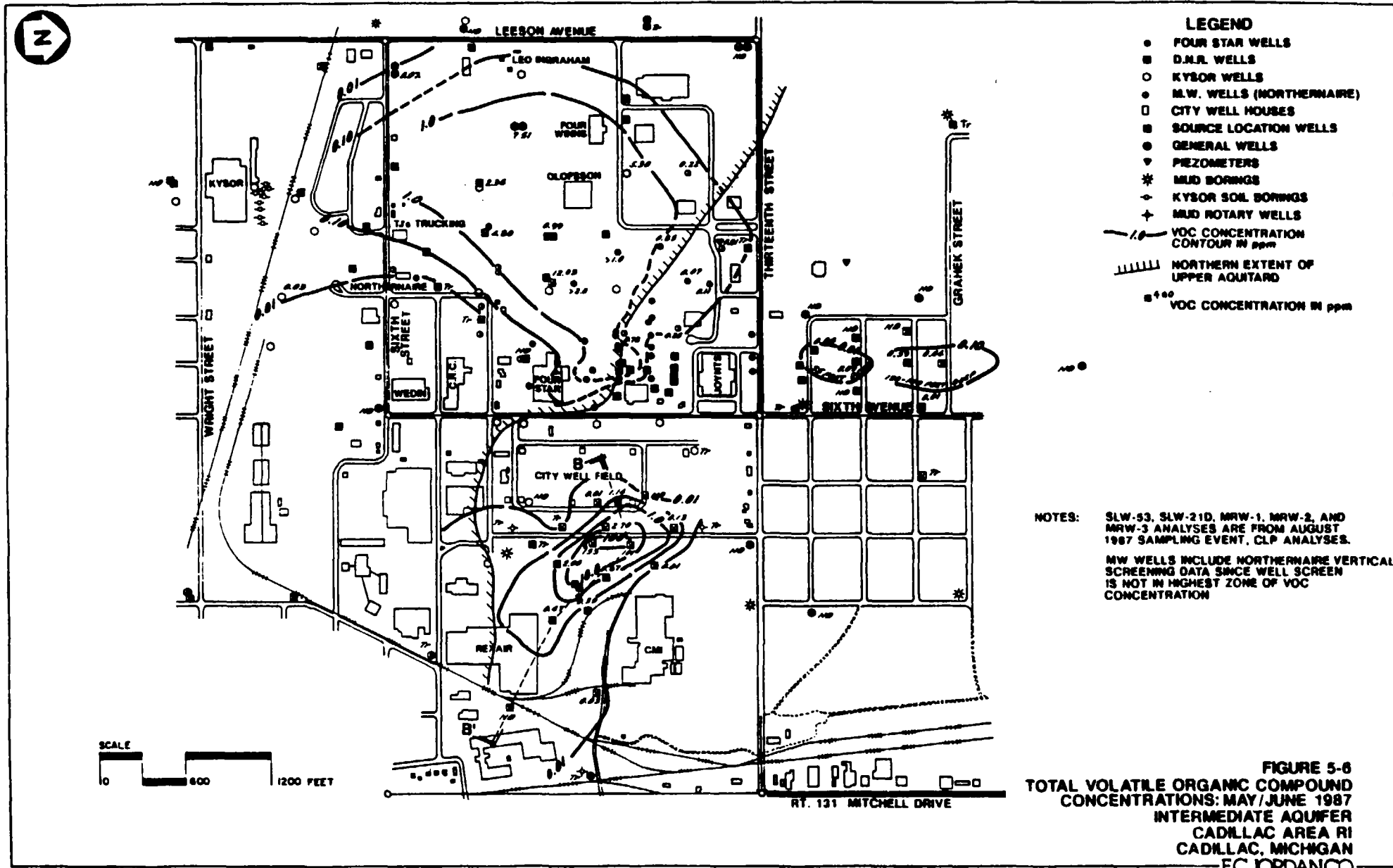


TABLE 5-5

CHARACTERISTIC PLUME CONTAMINANTS
IN GROUNDWATER

Kysor Plume	Four Star/ Joynts Plume	Four Winns Plume	Rexair Plume
TCE	TCE	TCA	TCE
TCA		PCE	
		TCE	
Ethylbenzene		DCE	
Xylene		Acetone	
PCE			
DCE			

Other Compounds Detected

Methylene Chloride
Trans-1,2-Dichloroethene
Chloroform
DCA

wider, which is due in part to contributions from Four Star to the east and Four Winns to the west. Based on field-screening data, the plume is most concentrated at a depth of approximately 45 to 50 feet below ground. "Releases" at Kysor are probably related to the annual variations in net infiltration and water table elevation which flush out contaminants from the soils. Thus, the plume near the facility is being recharged by a series of pulses rather than a continuous source. Concentrations in downgradient wells may be expected to fluctuate with time. Groundwater flow rates in the shallow aquifer are about 200 ft/yr, accounting for short-term fluctuations in concentrations. For example, VOC concentrations in SLW-25 dropped significantly from June (115 ppm) to August (5 ppm), which may be explained by these factors.

A large VOC plume also exists in the intermediate aquifer. Figure 5-6 illustrates the large areal extent of the plume, both in the confined and unconfined portions of the intermediate aquifer. As shown, this plume is bounded by Thirteenth Street (north), Sixth Avenue (east), and Leeson Avenue (west). The southern extent was not defined because monitoring wells are not screened in the intermediate aquifer between the Northernair and Kysor facilities. Due to extreme contamination in the shallow aquifer between these facilities, it was decided not to penetrate the upper aquitard. However, an upgradient well, SLW-21D, was installed in the intermediate aquifer south of Kysor and showed no contamination.

The plume in the intermediate aquifer is less concentrated than in the upper aquifer, reaching a measured maximum of 12 ppm of total VOCs in SLW-10D. There are no known sources above the confined portions of the intermediate aquifer that could directly contaminate this aquifer. Therefore, seepage of VOCs caused by the downward vertical gradients existing across the upper aquitard (see Table 5-3), particularly in areas where the clay is thin, is probably contaminating the intermediate aquifer. Also, it is possible that downward migration of contaminants occurs along the ungrouted borehole annulus of the Kysor monitoring wells penetrating the upper aquitard (see Section 1.4.2). It is not possible, however, to assess the degree to which this mode of contaminant migration is occurring. In the unconfined portion of the intermediate aquifer, other sources of VOC contamination may be contributing to the extent of this plume.

5.5.2 Four Winns (Frisbie Street) and Leo Ingraham Sources

Another source of VOC contamination that can be identified based on plume configuration and compounds present is Four Winns, a fiberglass boat manufacturer. Located in the northwestern portion of the study area, VOCs apparently were released on or near the property. The resulting groundwater contamination broadened to the west the areal extent of the shallow aquifer plume attributable to Kysor. Figure 5-5 illustrates the lateral spreading of the contaminant plume, which occurs primarily by a process called dispersion. Groundwater flow between two points is not direct; it flows along a tortuous pathway through the soil particles. Clay layers and low-permeable soils also force the groundwater to flow around them. With clean water, these effects are not observed. However, with contaminated water, dispersion causes the plume to spread out horizontally and vertically and concentrations to drop as the contaminants mix with clean water.

For the Cadillac Area RI, monitoring wells were installed directly upgradient and downgradient of the Four Winns property. Groundwater sample analyses show significant contamination by TCA in the shallow aquifer downgradient of the site (SLW-38), which is present in much lesser concentrations upgradient (GW-10S). The relative abundance of TCA over TCE is a distinguishing feature of the Four Winns plume. TCA is a compound typically used in fiberglass hardening. Almost 18 ppm of VOCs were detected at a depth of 55 feet in SLW-38. To the east, the plume becomes less concentrated in K-16S (0.28 ppm) and apparently merges with the Kysor plume. To the west, no contamination was detected across Leeson Avenue in GW-14S. Downgradient to the north, the upper aquitard pinches out, and the plume becomes diluted and merges with the Kysor plume in the intermediate aquifer. This is based on vertical screened auger analyses from MW-16 and GW-6D. (Appendix P includes the VOC field-screening data obtained during the drilling of MW-16 for the Northernair study.)

ASI was retained by Four Winns to investigate groundwater contamination in the vicinity of the Frisbie Street plant. Their report suggests that, of four primary contaminants (i.e., TCA, TCE, PCE, and acetone), two are a result of upgradient sources (PCE and TCE) and two are attributable to Four Winns activities (TCA and acetone). Although the monitoring well and sampling procedures are largely undescribed, the groundwater quality data appear to support that TCA and acetone originate at the Four Winns plant, with the possibility that some amount of TCA originated from an upgradient source.

The conclusion that TCE and PCE are from an upgradient source is not entirely supported by the ASI data. The highest PCE concentration was detected at the center of the site in MW-11 and MW-15, and is consistently high in MW-16 downgradient from Four Winns. Southern boundary wells contain lesser PCE concentrations. TCE concentrations are highest in the downgradient wells, MW-9 and MW-16, but TCE is also detected to a lesser degree in some upgradient wells. Therefore, the origin of these compounds is unclear, inasmuch as Four Winns reportedly never used them. PCE and TCE are detected in upgradient SLW and GW monitoring wells, which define the western edge of the Kysor plume and possibly an Ingraham source. These contaminants migrated to the Four Winns property, accounting for (at least in part) the detections of PCE and TCE by ASI in Four Winns wells. The fact that the concentrations increase at the center of the site may reflect: (1) direct contributions of TCE and PCE to the soils by Four Winns, or (2) migration of the center of this plume from an upgradient source that has since been removed.

The extent to which the Ingraham property contributed to the contamination is uncertain. The property is adjacent to and upgradient of the Four Winns plant (see Figure 1-2). Mr. Ingraham stored barrels and drums containing solvents in a large depression in his yard before 1981; however, the barrels and drums were removed after 1981. Due to the already extensive contamination in the shallow and intermediate aquifers from the Kysor plume, it is not possible with available data to determine the extent of contamination released from the Ingraham property. However, the increase in contamination levels in GW-10S and GW-10D, as compared to upgradient wells SLW-11, SLW-12, GW-9S, and GW-9D, indicates that VOCs were present on the property (see Figures 5-5 and 5-6). The MDNR plans to investigate the shallow soils on the Ingraham property with a series of soil borings to be drilled during the summer of 1988. These borings,

followed by chemical analysis of soil samples, should assist in determining if significant contaminants still remain on the Ingraham property.

5.5.3 Four Star and Joynts/Four Winns (Sixth Avenue) Sources

The former Four Star Corporation on Sixth Avenue was a source of VOC contamination in the Cadillac industrial park. The FS and DNR series of monitoring wells were installed for the two studies (i.e., the 1983 Gosling-Czubak Associates [GCA] study for Four Star and the August 1986 study by MDNR, respectively). Appendix E contains specifications for these wells. The FS wells and the DNR wells are typically screened from 10 to 20 feet and from 10 to 35 feet below the water table, respectively.

Groundwater chemical analyses in 1983 detected approximately 37 ppm in FS 1-10, located 200 feet north of the Four Star building (GCA, 1983). Based on the relatively shallow FS and DNR wells, concentrations of TCE and PCE appear to fall below 0.10 ppm within approximately 600 feet downgradient of the building. This fairly rapid decrease in concentration may be due, in part, to the merging of the shallow and intermediate aquifers just north of FS 1-10. However, as shown by the detection of VOCs at depth in Northernair wells MW-10 (0.25 ppm at 140 feet) and MW-14 (0.78 ppm at 115 feet), the FS and DNR wells that do not go below 65 feet may not have intercepted the contaminant plume, which appears to be moving downward through the aquifer. (Appendix K includes the VOC field-screening data obtained during the drilling of MW-9 through MW-18.)

The detection of TCE in MW-10 and MW-14 northwest of Four Star (see Figure 4-1) may also be due (at least in part) to Kysor-related contamination. Because of the similarity of compounds present and the large areal extent of the Kysor plume, it is probably impossible to assess the degree to which each facility (i.e., Kysor and Four Star) contributed to the contamination in this area.

The plume in the North Park Subdivision, composed primarily of TCE, is relatively deep. Since it is believed the same compounds were released from Four Star and the Joynts/Four Winns facilities, it is again impractical to establish if the Joynts facility is the sole contributor to this plume. Geologic and hydrologic conditions are such that both facilities could potentially be responsible. SLW-4, SLW-13, and SLW-20 contain up to 0.6 ppm of TCE at depths of 130 to 150 feet. Downward vertical gradients and the presence of gravels with relatively high hydraulic conductivity would allow contaminants to reach these depths.

5.5.4 Rexair Source

One of the most significant findings of the Cadillac Area RI is the presence of a concentrated plume of TCE upgradient of the city of Cadillac municipal wellfield. The probable source of contamination was identified as Rexair on Seventh Street. This assessment was made based on the following factors:

- o the configuration and concentration of the TCE plume
- o the location of the plume with respect to groundwater flow direction in this area

- o the lack of any other known sources upgradient of the plume and Rexair
- o the absence of VOCs in two upgradient monitoring wells (i.e., SLW-53 and GW-2)
- o a site visit in 1985 by MDNR, which determined that TCE had been used at the site until approximately 1984

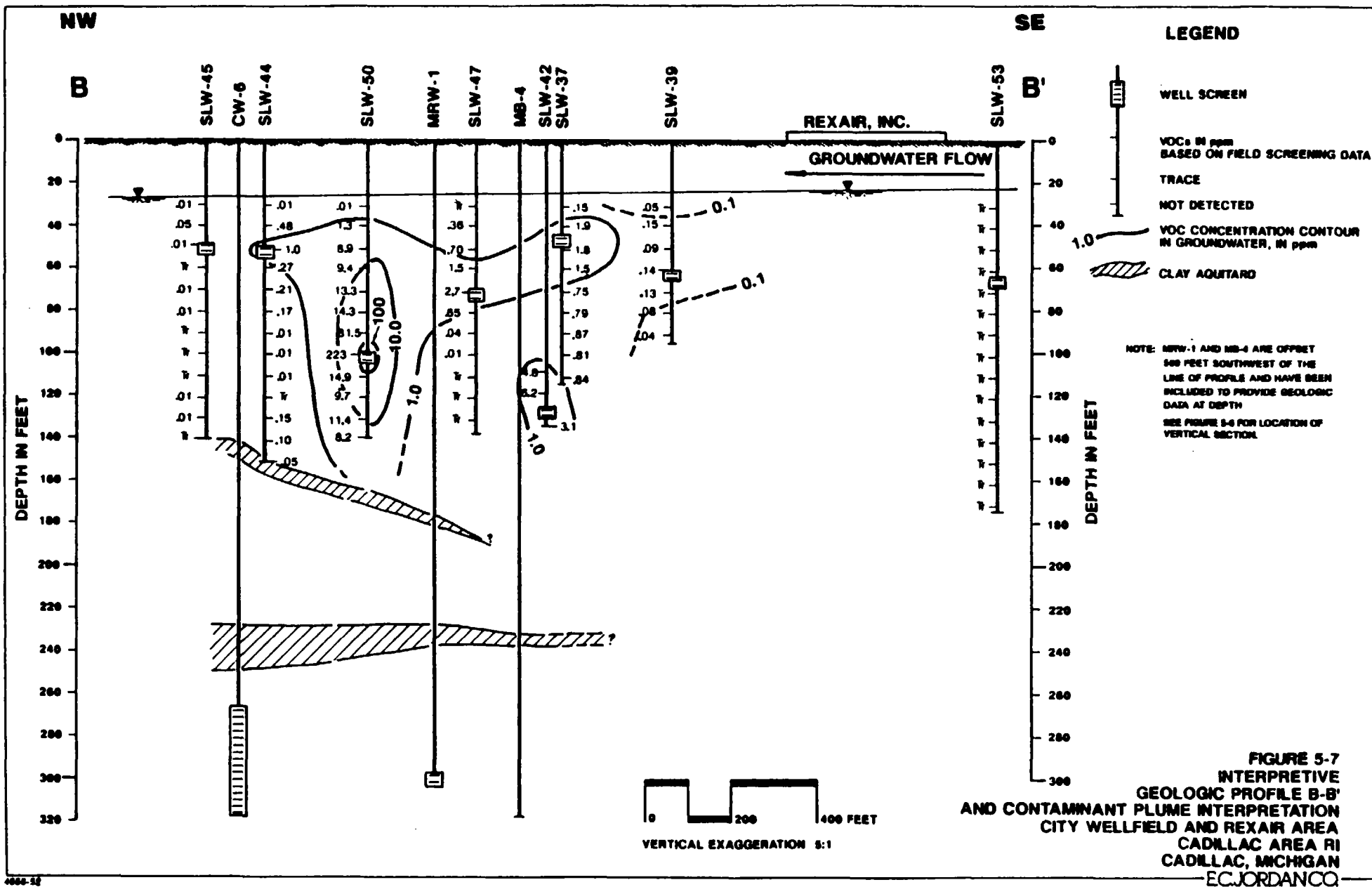
Rexair began operations at this site in 1969. The original building, built about 1959, was first used by the Cadillac Boat Company and subsequently by a garment manufacturer. The use of chemicals, if any, by these former occupants is unknown.

The plume (defined arbitrarily by the 0.1-ppm contour) is approximately 1,400 feet long in the northwest direction of groundwater flow; the width is presently a maximum of approximately 700 feet (see Figure 5-6). TCE contamination and the potential for impact to the city wellfield was the topic of a Technical Memorandum issued by Jordan to MDNR in August 1987.

Only the unconfined intermediate aquifer, the lower aquitard, and the lower aquifer exist in this area of the site. The plume center is located in the intermediate aquifer, approximately 350 feet east of the city wellfield, in which SLW-41 and SLW-50 contain 153 and 100 ppm of TCE, respectively. Significantly, this contamination extends from 100 to 140 feet in depth, only 700 feet downgradient from its source. In fact, the plume may be deeper; however, a dense gravel and silt formation prevented further auger advancement (see Section 4.3), which implies that the contamination is rapidly descending in the aquifer. Figure 5-7 is a geologic profile illustrating the relationship between the contaminant plume, lower aquitard, and city wellfield. This downward migration of the TCE plume through the intermediate aquifer may be the result of a combination of the following four factors:

- o the denser-than-water nature of TCE allowing it to sink through the aquifer
- o the downward vertical gradients that exist within the intermediate aquifer
- o the influence of the pumping of the lower aquifer by the city on the overlying intermediate aquifer
- o the relative thinness of the lower aquitard in this area, which would have a limited ability to restrict downward movement of contaminants toward the lower aquifer

Both the vertical and horizontal configurations of the plume suggest that the pumping of the lower aquifer is locally influencing flow in the intermediate aquifer. The depth of the plume at a relatively short distance from its source probably reflects both city wellfield pumping and the ability of concentrated TCE to sink because of its density and the relatively high hydraulic conductivity of the site sandy soils. Also, the concentration contours defining the



horizontal extent of the plume are skewed slightly toward the wellfield near CW-6 and SLW-44 (see Figure 5-6).

As shown in Figure 5-7 at SLW-50, a lobe of contamination appears to be drawn down from the shallower depths and some of the plume already overlies the thicker portions of the lower aquitard. The depth and continued contaminant migration may be influenced depending on which city well pumps are in operation. Pumpage on the west side of the field, where the aquitard is thicker and the pumps farthest away from the plume, probably influences the migration less than pumpage on the east side. There are probably seasonal variations related to the recreational nature of the area, with increases in water demand due to the transient population. Data are not yet available that definitely show contamination to a depth below the lower aquitard. However, the shape of the plume, the thinness of the aquitard on the east side of the wellfield, and the pump test data all suggest probable future impacts from the present contamination.

One round of chemical data was obtained from the deep wells MRW-1, MRW-2, and MRW-3; the groundwater samples showed about 4 parts per billion (ppb) of TCE, which is below the specified detection limit, but is still reported as being present. Review of the sampling procedures for that day shows sampling of a highly contaminated well (SLW-50) just prior to sampling the MRW wells. (This order of sampling was necessitated due to an ongoing pump test using the MRW wells.) Also, pump decontamination blank (MW-115) collected after sampling SLW-50 and before the MRW wells still had detectable levels of TCE; therefore, it is Jordan's belief that the TCE detections in the MRW wells reflect this sampling order.

Rexair Inc. has retained EDI Engineering & Science (EDI) to further investigate soil and groundwater contamination and identify the source location. A soil gas study has been described in a report "Technical Information Review and Soil Gas Study" issued by EDI in February 1988 describing their findings. The conclusions of the EDI report state that TCE is a primary contaminant in soils located on the eastern side of the Rexair building. Lesser amounts of trans-1,2-DCE and DCE were also detected at this location. A separate source of toluene contamination was identified on the western side of the Rexair building.

Based on the recommendations in the EDI soil gas report, Rexair's investigations of soil and groundwater contamination continued with the installation of monitoring wells in March 1988. One series of three wells screened at different depths has been near monitoring well SLW-50 and sampled to assess groundwater quality at depth. Although some groundwater quality data are available, a finalized report explaining well installation and sampling procedures has not yet been issued making discussion of the results inappropriate at this time.

5.5.5 Northernnaire Source

The fifth identifiable source of groundwater contamination within the industrial park is the Northernnaire facility, located on the northwestern corner of Sixth Street and Seventh Avenue. This site, which released hexavalent chromium and cadmium to the soils and hexavalent chromium to the groundwater, received

emergency response action under Superfund, and later was the focus of a detailed RI by MDNR and Jordan. A focused FS was completed for the contamination detected in soils. However, it was intended that remedial alternatives for groundwater remediation be assessed in the FS for the Cadillac Area RI. The primary rationale for this decision was that it was not likely the hexavalent chromium plume could be effectively remediated without consideration of the coincidental presence of VOCs within the aquifer.

Groundwater samples from selected monitoring wells installed for the Northernnaire study (i.e., the MW series) were obtained during the Cadillac Area RI. These data, combined with previous data sets, were used to interpret the hexavalent chromium plume movement and distribution.

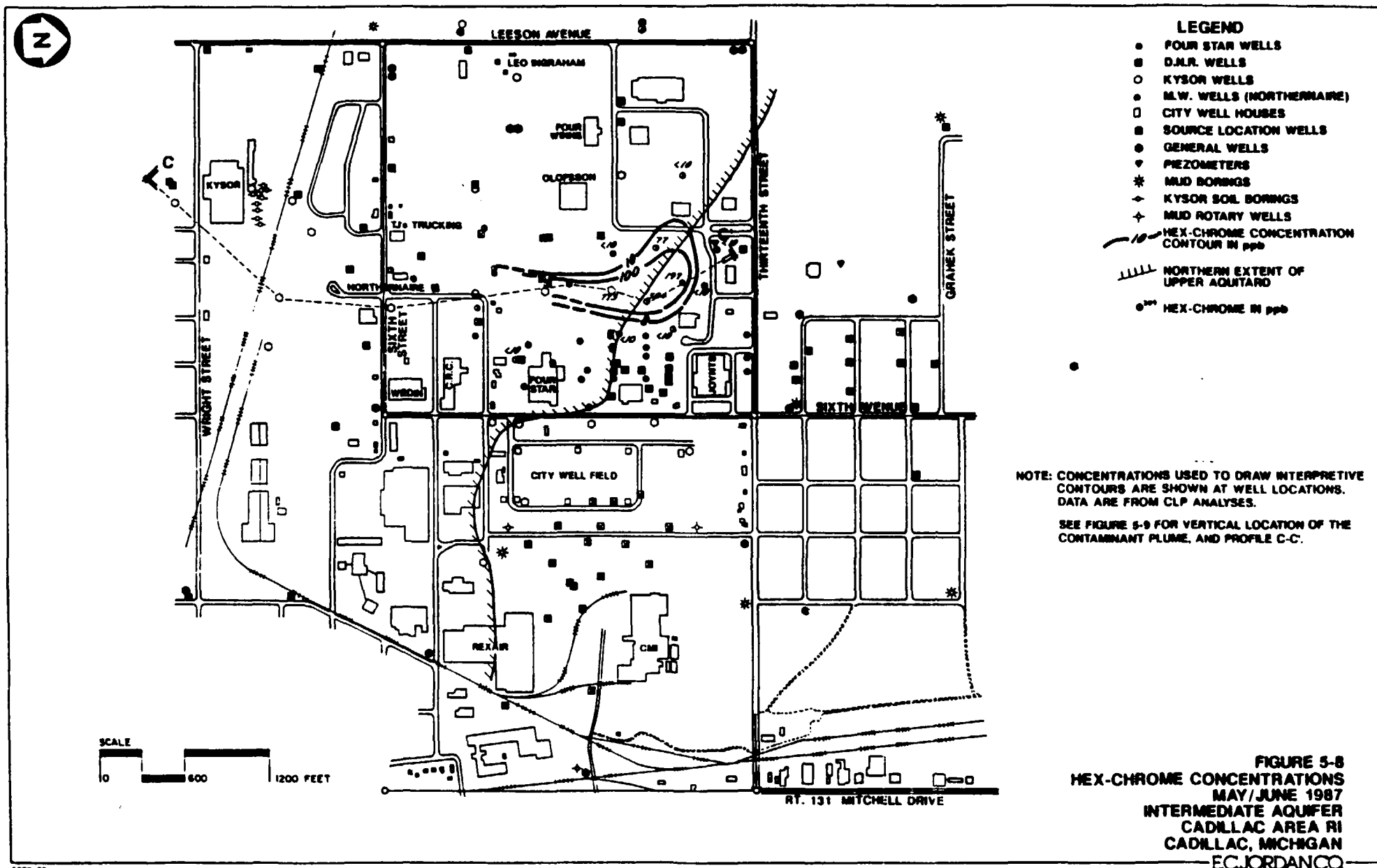
The most significant hexavalent-chromium contamination exists in the intermediate aquifer near the center of the industrial park. Concentrations from the June 1987 sampling approach 400 ppb in MW-7 and MW-12, and 800 ppb in K-15. Based on vertical distribution data obtained during monitoring well installation (see Appendix B), the plume is relatively thin (5 to 15 feet) and situated just above the lower aquitard. Figures 5-8 and 5-9 illustrate the plume configuration in horizontal and vertical perspectives. The hexavalent chromium reached the intermediate aquifer from the shallow aquifer probably in a similar manner as the VOCs; that is, by downward vertical gradients across the upper aquitard, and possibly in areas where the upper aquitard is very thin. The shape of the plume reflects the northwestward direction of groundwater flow.

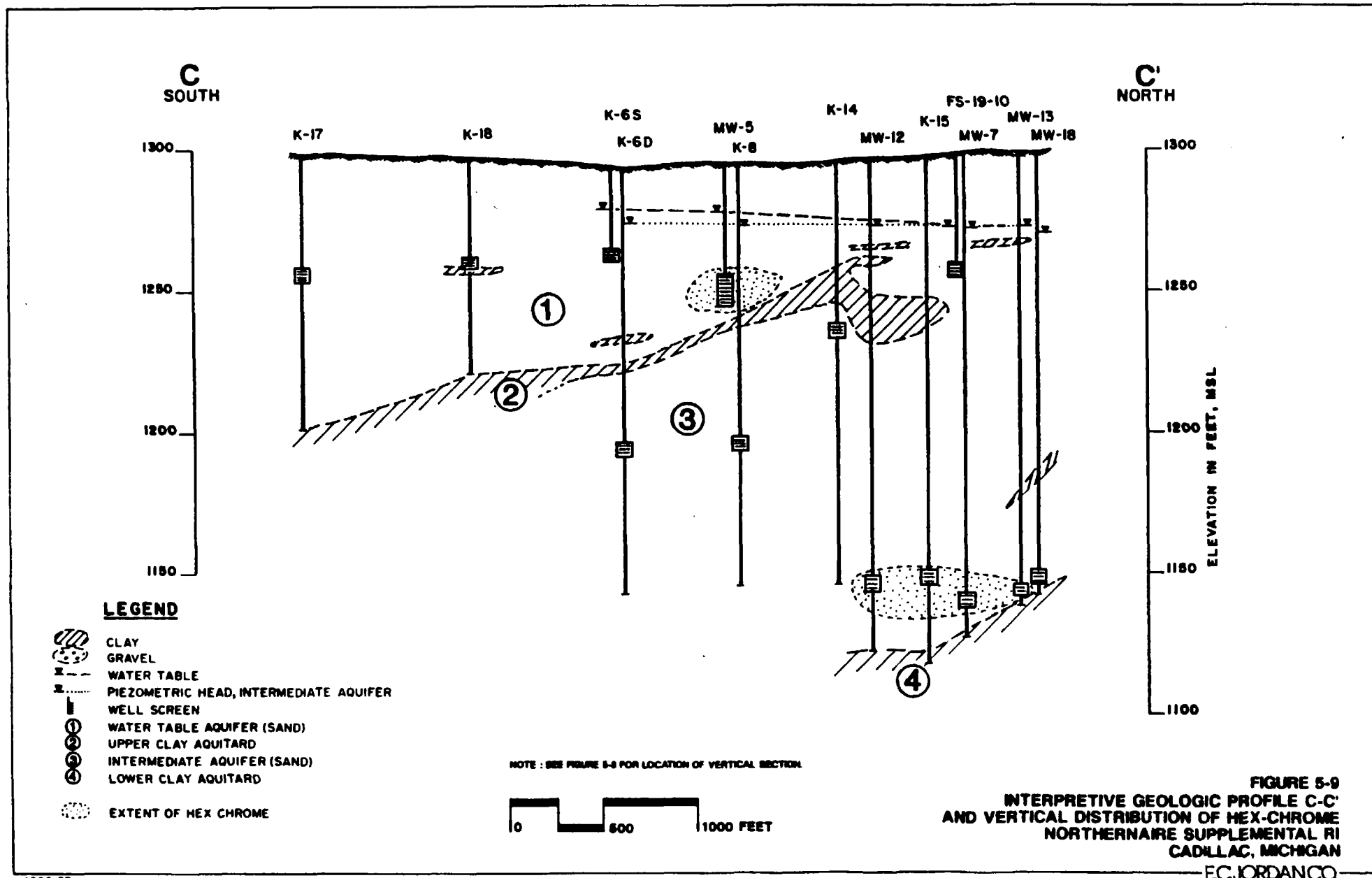
Concentrations of hexavalent chromium in individual wells showed minor fluctuations with time. Comparison of 1986 and 1987 hexavalent chromium groundwater analyses showed a general decrease in concentrations in most, but not all, wells. While K-15 showed a decline in hexavalent chromium from 2.2 ppm (February 1986) to 0.773 ppm (June 1987), a more downgradient well at the leading edge of the plume (MW-13) rose from 0.04 to 0.197 ppm during the same period; this may reflect the movement of the plume with groundwater flow. Also, seasonal variations may cause some vertical movement of the plume within the aquifer relative to the well screen. Appendix P contains CLP hexavalent chromium results.

In February 1986, hexavalent chromium was detected in the shallow aquifer in MW-5 only. However, in 1984 when the monitoring well was sampled, no hexavalent chromium was detected. In the February 1986 analysis, almost 1.6 ppm were detected; in May 1986, this level had dropped to approximately 0.5 ppm. Such large fluctuations in hexavalent chromium concentrations suggest that a small but concentrated slug was released from the Northernnaire facility and traveled northeastward in the direction of groundwater flow to MW-5.

5.5.6 Other Metals in Groundwater

Although chromium was identified as the primary metal of concern in groundwater originating at Northernnaire, several other metals were identified which had elevated concentrations in groundwater. Data for metals in groundwater were generated in the original and supplemental RIs for Northernnaire and in selected wells for the Cadillac Area RI. The principal metals with elevated concentrations, with maximum values indicated in parentheses, are as follows: calcium





(107 ppm), magnesium (19.4 ppm), potassium (8.1 ppm), and sodium (74.9 ppm). The Northernnaire report presented contoured concentration data, which strongly suggested that these metals were probably due to road-salting activities. In the shallow aquifer, concentrations were greatest near Sixth Street, decreasing in the direction of groundwater flow due to dispersion and dilution by infiltrating precipitation. The intermediate aquifer showed a similar relationship, with the greater concentrations near Sixth Avenue, decreasing toward the west-northwest. Recent data obtained for this RI confirm levels of contaminants identified previously.

While not a metal, cyanides were a focus of the Northernnaire study due to their presence in relatively large concentrations in plating baths and other wastes. No cyanides were detected in Northernnaire site soils. However, low levels of cyanide were detected in K-15 (22 ppb) and FS-19-10 (5 ppb) in 1985. The Northernnaire RI report noted that these levels were below the drinking water criterion of 50 ppb and did not further carry this contaminant into the health-based RA. The Cadillac Area RI provided two subsequent analyses for cyanide in K-15, both of which were reported as less than the detection limit of 10 ppb. This contaminant will be carried into the RA based on a presumption that the recent lower concentrations are due to sampling a different portion of the plume, also indicated by decreases in concentrations of hexavalent chromium in K-15. The maximum value of 22 ppb of cyanide in groundwater will be used for the risk calculation.

6.0 CONTAMINATION ASSESSMENT

The acquisition and compilation of geologic, hydrologic, historic, and chemical data at the Cadillac industrial park provided the means for interpreting hydrogeologic relationships and assessing contamination distribution in soil and groundwater. The preceding sections detailed methodologies used to collect these data and how it was used for site interpretation. It is believed that sufficient data currently exist to assess contaminant conditions, including environmental fate and transport. This introduction summarizes the major findings and conclusions derived from available data. Following this summary, this section presents general discussions of the contaminant assessments (Sections 6.1 and 6.2), and contaminant fate and transport (Section 6.3) as a framework for the baseline RA.

1. Five stratigraphic units that form a sequence of three aquifers and two confining clay aquitards were identified. The aquifers generally consist of fine to medium sands, but contain local variations in gravel and silt content. The clay aquitards are often silty and vary widely in thickness over relatively short distances (see Section 4.3).
2. Groundwater flow directions in the shallow aquifer has a broad, radial pattern, apparently emanating from an area of groundwater recharge (possibly Lake Cadillac) south of the Kysor facility and outside the industrial park boundaries. The predominant direction of shallow groundwater flow in the study area is northeastward. Along the southern (Wright Street) and western (Leeson Street) park boundaries, flow direction is more eastward and northward, respectively. This groundwater flow pattern has not fluctuated significantly over the past several years (see Section 5.3.1 and Figure 5-2). Groundwater flow rates average about 200 ft/yr.
3. Groundwater flow direction in the intermediate aquifer is in a north-northwestward direction (see Section 5.3.1 and Figure 5-3). Groundwater flow rates in this aquifer average about 80 feet per year.
4. Groundwater flow direction in the lower aquifer east of the City well field is approximately due west toward the wellfield. This flow pattern is believed to be directly influenced by the city pumping of the lower aquifer (see Sections 5.3.1 and 5.3.3 and Figure 5-4). The regional flow direction in the lower aquifer is not known.
5. In general, downward vertical gradients exist within and between each aquifer. These gradients, combined with areas where the clay aquitards are thin, diminish the effectiveness of the aquitard to restrict groundwater flow. These factors provide a mechanism for downward contaminant migration from upper to lower aquifers (see Section 5.3.2).
6. At least five contaminant sources (see Sections 5.5 and 6.3) resulted in contaminant plumes in the shallow and intermediate aquifers below the industrial park. The primary contaminants include VOCs and hexavalent chromium. VOC contamination is widespread in the shallow and intermediate aquifers west (downgradient) of the city wellfield. Contributors include

Kysor, Four Winns, Four Star (former), and Joynts (former). (Four Star and Joynts are considered a single source because they are close to each other, contributed similar compounds, and their plumes are inseparable downgradient of Joynts.) East and upgradient of the city wellfield, a concentrated TCE plume exists which is believed to have been released from the Rexair property. The remaining contamination source is the Northernaire facility, which released hexavalent chromium that was detected as a small "slug" in the shallow aquifer and a larger plume in the intermediate aquifer. The location and movement of the hexavalent-chromium plumes in the upper and intermediate aquifers do not suggest an immediate impact. However, seepage through the lower aquitard may allow migration into the lower aquifer and subsequent capture by the wellfield pumps.

6.1 SOILS CONTAMINATION

The Cadillac Area RI defined an area of soils behind the Kysor facility as a current continuing source of VOCs to groundwater contamination. The contaminated area is approximately 250 by 60 feet in size, and 25 feet deep. Soils in this area contain up to 750 ppm of VOCs, chiefly from a depth of 6 feet to 20 or 25 feet; the upper zone is clean backfill. The VOCs are dissolved by infiltrating precipitation and transported to the shallow aquifer. This continuing source of contaminants is probably responsible for the high VOC levels in groundwater detected downgradient of Kysor in SLW-25 (115 ppm in June 1987), despite the fact that solvent dumping has reportedly not occurred in the past seven years and groundwater movement is rapid through the area.

During drilling of the SLW borings, split-spoon soil samples of the unsaturated zone were analyzed for VOCs to (1) possibly locate other dumping areas near plant sites, and (2) determine if the soils contained contamination by the upward migration of gases released from underlying contaminated groundwater. The VOC analyses showed essentially no contamination in the SLW boring soils. This probably reflects the general absence of residual VOC contaminants in soils, as well as the very porous, silt- and clay-free nature of the unsaturated soils.

The location of residual VOC contaminants in soils (i.e., by spills, dumping, or drains), if they exist, could not be identified at the Rexair facility based on the soil analyses. However, it is possible that past releases could have been covered by new building construction at the facility or may have been to dry wells located within the original building. Subsequent investigations of soil gas by EDI for Rexair have identified a possible TCE source along the eastern side of the Rexair facility (Technical Information Review and Soil Gas Study, February 1988). At the time of issuing this RI, investigations are continuing at the Rexair site.

The Four Winns site on Frisbie Street was investigated by ASI but soil sampling and analysis was not extensive and thus some uncertainty remains as to the degree to which the soils below Four Winns are contaminated and may serve as a continuing source of contamination to groundwater. Also, soil samples at Leo Ingraham's property were only collected in the unsaturated zone where the

monitoring wells were installed. While it was known that drums of waste solvent had been stored at the Ingraham property, no incidents of dumping had been reported. Wastes were removed, together with visually contaminated soils, shortly after the removal of contaminated soils at Kysor. Based on water quality results discussed in the RI for this northwest corner of the industrial park, contaminant distribution characteristics suggest that the Ingraham property may have provided a contaminant source to groundwater. With these uncertainties concerning sources still remaining, the MDNR is planning a shallow soils investigation at Ingraham's similar to the Kysor shallow soil investigation, and possibly a further investigation of soils on the Four Winns property.

6.2 GROUNDWATER CONTAMINATION

The most serious groundwater contamination problem at the Cadillac industrial park is widespread VOC contamination in the shallow and intermediate aquifers. The possibility that some VOC contaminants may reach the lower aquifer of the city water supply is the primary potential health risk requiring detailed assessment. Hexavalent chromium contamination in the intermediate aquifer may also reach and contaminate the lower aquifer if leakage through the lower aquitard is significant and the flow in the lower aquifer at the point of leakage is toward the wellfield (e.g., if the radius of influence of the wellfield extends beneath the hexavalent chromium plume).

Prior reports and other information gathered for this report indicate that several industrial facilities at the Cadillac industrial park disposed of concentrated pure solvent products directly to the soils. Relatively high concentrations of solvents reported in several monitoring wells suggest that pure solvents have penetrated the upper and intermediate aquifers and may still exist in the subsurface.

Solvents such as TCE are denser than water (specific gravity = 1.46) and will tend to sink through an aquifer by the forces of gravity, if the soils are permeable enough and there is sufficient spill volume to force an initial entry into the aquifer. Referred to as dense, non-aqueous phase liquids (DNAPL) these solvents will sink until they reach an impermeable barrier such as a clay (or some finer grained material) aquitard/aquiclude. They may then form a pool or top of the barrier or follow the slope of the barrier downhill (Feenstra S., and Cherry, J.A; Subsurface Contamination by Dense Non-Aqueous Phase Liquid [DNAPL] Chemicals; presented at, the International Groundwater Symposium, International Association of Hydrogeologists, Halifax, Nova Scotia, May 1-4, 1988). Gravitational forces are the primary driving mechanism of the DNAPL whereas hydraulic forces have little effect. These conditions make the prediction of DNAPL movement along an aquitard or aquiclude surface difficult. In turn, depressions in an irregular barrier surface will intercept flows and cause pooling. Inadequate knowledge of the detailed surface topography of such barriers makes predictions of flow or locations of pools impossible.

Clean groundwater flowing through sinking "residuals" of a DNAPL spill will solubilize the contaminant to a concentration determined by the contaminant's solubility in water. It is believed that if groundwater contains a contaminant

at 5 to 10% of its solubility, then a separate, non-aqueous phase liquid probably exists in the soils and groundwater upgradient of the monitoring well (Feenstra and Cherry, 1988).

In two locations at the industrial park, groundwater contaminant concentrations approached the 10% solubility level. These include 1) downgradient of Kysor, and 2) downgradient of Rexair. Solvents are known to have been dumped behind Kysor and a source at Rexair is being investigated at the time this RI was issued. The field work for the Cadillac RI focused on defining the solubilized portion of the contamination in groundwater to provide information to determine its environmental fate and transport. No attempt was made to identify DNAPL residuals or pools since there is no cost-effective method available for such identification. DNAPL location in the subsurface would involve very close monitoring well grid spacings and essentially continuous vertical sampling through the aquifer.

No fewer than five industrial facilities are responsible for the VOC contamination, defined on plan views in Figures 5-5 and 5-6. The contamination is extensive both horizontally and vertically in the shallow and intermediate aquifers; TCE is the primary contaminant. Other contaminants commonly found but in lesser amounts include TCA, PCE, DCE, and toluene. Table 5-5 includes a complete list of compounds detected.

The contamination in the shallow and intermediate aquifers west of Sixth Avenue is so areally extensive that it appears the plumes from Kysor, Four Star, and Four Winns have merged. The Northernnaire plume is also in this area, but is made distinctive by the hexavalent chromium. The plume from the principal VOC contributor (Kysor) extends the entire length of the upper aquitard in the shallow aquifer. Concentrations in SLW-25 suggest that solvent residuals or plumes may exist behind Kysor. Highly contaminated soils in KB3 and KB7 also suggest significant residual contamination. To the east and at the northern limit of the upper aquitard, the plume merges with a fairly concentrated (37 ppm) VOC plume emanating from the Four Star facility. Likewise to the west, the Kysor plume includes the VOC releases at Four Winns, and possibly the Ingraham property.

VOC contamination in the intermediate aquifer west of Sixth Avenue is equally as widespread as in the shallow aquifer, but not as concentrated. SLW-10D has the highest concentration of VOCs (12 ppm). In order not to create a risk of cross contamination between the aquifers by penetrating the upper aquitard with wells between the Kysor and Northernnaire facilities where shallow aquifer contamination is greatest, VOC data for the intermediate aquifer are not available between these facilities. However, the contamination did not appear to extend south of Kysor, because the vertical contaminant profile in SLW-21D was clean.

In the North Park Subdivision, two smaller and less concentrated plumes were detected at vertically discrete depths. Maximum detected concentrations reached only 0.6 ppm. These plumes probably represent the leading edges of the Four Star and Joynts/Four Winns plumes and possibly the Kysor plume. The VOCs migrated downward, due to vertical gradients, and became diluted with dispersion. As shown in Figure 5-6, VOCs were not detected in several of the FS and

DNR wells. These wells may not be deep enough to intercept the plume, as suggested by the contamination detected at greater depths by vertical screened auger sampling of MW-10 and MW-14.

A discrete VOC plume is present east of Sixth Avenue; it is believed that the probable source is the Rexair facility. A potential receptor, the lower aquifer (and its consumers) at the city wellfield, was identified. The plume reaches a concentration of 153 ppm of TCE at a depth of 140 feet in SLW-41. The plume may extend even deeper, but a very dense gravel and silt formation prevented further auger penetration of the intermediate aquifer in this area. As discussed in Section 5.3.3, indications are that the city wellfield pumping in the lower aquifer is affecting groundwater and contaminant migration in the overlying, intermediate aquifer. Concentrations in SLW-41 and SLW-50 approach 10% of TCEs solubility (1100 ppm) suggesting the presence of TCE pools or residuals. This plume and a possible source location has received additional investigation by consultants contracted by Rexair. Data and interpretations from these studies should be forthcoming in reports by the consultants.

The hexavalent chromium plume originating from the Northernaire facility has not changed significantly from one year ago. However, concentrations in K-15 have decreased, while increasing in a more downgradient well (MW-13), suggesting northward plume migration. Concentrations remain below minimum detection levels (10 ppb) in adjacent wells at the plume's perimeter. Figures 5-8 and 5-9 illustrate the plume configuration in the intermediate aquifer.

To summarize, the assessment of the groundwater quality data and contaminant distribution provides for the identification of up to seven discrete plumes, including the following:

1. A concentrated (up to 115 ppm) and areally extensive VOC plume in the shallow aquifer attributable to a minimum of three sources: Kysor (south), Four Star (east), and Four Winns (west).
2. A less concentrated (12 ppm) but equally extensive VOC plume in the intermediate aquifer, which is due to the migration of Kysor and Four Star contaminants through the upper aquitard, over the northern edge of the aquitard, and possibly down ungrouted boreholes.
3. A small VOC plume (less than 0.1 ppm) in the North Park Subdivision at a depth of approximately 55 feet.
4. A second, small but more concentrated VOC plume (0.6 ppm) further north in the subdivision, which is present at a depth of 130 to 140 feet. These two plumes probably represent the leading edges of contaminant migration released by Four Star and Joynts/Four Winns, and possibly Kysor.
5. A highly concentrated VOC plume (153 ppm of primarily TCE) located between the Rexair facility and the city wellfield, which extends to a depth of at least 140 feet.

6. A small but fairly concentrated (1.5 ppm) hexavalent chromium plume in the shallow aquifer that was first detected in 1986 after previous groundwater sampling events showed no contamination.
7. A hexavalent chromium plume (up to about 0.8 ppm) in the intermediate aquifer just above the lower aquitard in the north-central area of the industrial park. Both hexavalent chromium plumes are attributable to operations at the Northernair facility.

6.3 ENVIRONMENTAL FATE AND IMPACT

The relative positions of the contaminant plumes, with respect to the city wellfield, provide a convenient basis for discussing the environmental fate and impact of the contamination. Six plumes west of Sixth Avenue, two in the shallow aquifer, and four in the intermediate aquifer are horizontally downgradient of the wellfield. The remaining plume, east of Sixth Avenue, is the only plume detected upgradient. This interpretation is based on flow directions within the upper two aquifers. However, if significant leakage occurs across the lower aquitard, contaminants may be introduced to the lower aquifer. Whether these contaminants might reach the wellfield depends on flow directions and the radius of influence of the pumping wells within the lower aquifer. The resultant risk to water supply consumers also depends on leakage rates relative to the total water pumped, because they determine the concentration that would appear in the water pumped. Low seepage rates with high pumping rates might result in very low or no detectable concentrations. Higher seepage rates might result in concentrations with a significant risk factor.

The environmental impact of a plume will depend primarily on its migration path and fate based on existing hydrogeologic conditions, chemical interactions, and the location of potential receptors. In the late 1970s, residences within the industrial park and the North Park Subdivision had private water well supplies tapping the intermediate aquifer. With the discovery of widespread contamination, however, these residences were provided with city water. Therefore, the city water supply, drawn from the lower aquifer below the industrial park, and its consumers are the most sensitive possible immediate receptors of contamination in the industrial park area. Fate and transport considerations for the remnant VOC contamination in soils behind Kysor are discussed in Section 6.3.2.

A number of mechanisms may be operative at a hazardous waste site which naturally decrease the contaminant concentrations found in the various media. These mechanisms generally constitute either physical or chemical processes. The principal contaminants present are the VOCs, mostly chlorinated hydrocarbons, and chromium, present in both trivalent and hexavalent forms. The primary site concerns are with VOCs in both soils (at Kysor) and groundwater, and hexavalent chromium in the groundwater.

In soils, the primary driving force for VOC migration is solubilization by percolating infiltration, which carries the contaminants to the groundwater. Some volatilization probably occurs upward through the unsaturated zone. Except in a few instances, there appears to be little degradation of the chlorinated solvents (low concentrations, if any, of DCE and vinyl chloride,

which are common degradation compounds of PCE and TCE), in either the soils or groundwater. The typical sandy soils do not provide active chemical surfaces for much reaction to occur. The sandy soils also do not provide much opportunity for the more soluble hexavalent chromium to react and be reduced to the trivalent state, which might then precipitate out. The contaminants of concern generally appear to be quite conservative in the groundwater environment. The most effective mechanisms for attenuating the contaminant concentrations appear to be hydrologically controlled. Dispersion within aquifers, and seepage and mixing between aquifers (across aquitards), appears to be decreasing concentrations. This is a natural consequence of the horizontal and vertical spread of the contaminant plumes as they move downgradient from the source. Some retardation of solutes may occur as the plumes seep through the aquitards and adsorb to the clay. The clay may provide sites for limited chemical degradation or reaction to occur. As a result, a contaminant plume in a lower aquifer is expected to be more dilute than in the overlying aquifer.

The contaminants are anticipated to migrate at approximately the same rate and direction as the groundwater flow. Horizontally, the flow rates are approximately 200 and 80 ft/yr in the shallow and intermediate aquifers, respectively. Some longitudinal dispersion may occur, decreasing the time of arrival and increasing the time of final impact from a time calculated based on average velocities.

6.3.1 Rexair Property Plume

Based on groundwater flow directions, the plume near Rexair in the intermediate aquifer has the greatest potential of reaching the lower aquifer. The plume is upgradient of the wellfield, and the leading edge is currently just beyond City Well No. 6 (see Figure 5-7). The center of the plume is approximately 300 feet east (upgradient) of the wellfield, at a depth of at least 140 feet. With the top of the city supply wells screened at a depth of approximately 265 feet, only 125 feet of sand and gravel aquifer(s), including a clay thickness ranging from 3 feet (MB-4) to 30 feet (CW-6), separates the plume from the water supply. With the ability of the detected contaminants to sink due to their density combined with downward vertical gradients existing in the intermediate aquifer itself and the creation of downward potentials across the lower aquitard by municipal pumping, mechanisms exist for contamination to reach the lower aquifer. Due to the proximity of the plume to the wellfield and lower aquifer, little opportunity exists for the plume to become further diluted through dispersion, should it reach the lower aquifer. This could adversely affect the quality of the lower aquifer groundwater as potable water supply. Some dilution could occur at the pumping well, however, because the intercepted plume would represent a small fraction of the total water pumped from the cone of influence. This dilution probably would not be sufficient to preclude either treatment of the water or possibly the establishment of another potable water source.

6.3.2 Shallow Aquifer VOC Plumes

Groundwater from the shallow aquifer is not currently used for any private or public supplies, but the potential does exist for the shallow aquifer to be tapped inadvertently for domestic use. VOC contamination in the shallow

aquifer is southwest of the city wellfield, and migrating in a northeast direction. However, the shallow aquifer merges with the northwestward flowing groundwater of the intermediate aquifer before reaching the city wellfield area. The primary identifiable "receptor," therefore, is the intermediate aquifer.

VOCs in the shallow aquifer are widespread and relatively concentrated in certain areas. The primary contributor is the Kysor facility, with significant additions of VOCs to the aquifer by Four Star and Four Winns located to the northeast and northwest, respectively. Based on borings behind (i.e., north) Kysor, significant contamination remains in the soils; therefore, the plume will continue to receive contaminants with essentially every precipitation and meltwater event. The transfer of contaminants from the soils to the groundwater means that the residual source in soils will slowly be depleted over the years. However, it appears that the soils will continue to significantly contaminate the underlying groundwater for many years. Due to the depth of the contaminants and low potentials for volatilization, hazards related to the contaminants in soils appear to be confined to transport to the groundwater. If the area were re-excavated, there would be associated contact and inhalation risks. Upon reaching the northern limit of the upper aquitard, the plume merges with the intermediate aquifer and becomes diluted. This includes the VOC plumes released from the former Four Star and Joynts facilities, which were initially very close to the northern limit of the upper aquitard.

6.3.3 Intermediate Aquifer VOC Plumes (West of Sixth Avenue)

The largest VOC plume in the intermediate aquifer west of Sixth Avenue is situated below the shallow aquifer plume, and just below the upper aquitard. The lower plume is likely a result of the shallow aquifer contamination migrating through the upper aquitard by various mechanisms (see Section 5.5.1). Contaminant concentrations in the intermediate aquifer are less than in the shallow aquifer, partly because the source of contamination (i.e., the VOCs in the shallow aquifer) seeps slowly through the aquitard, and because the intermediate aquifer is much larger, allowing for greater dilution.

Contamination in the intermediate aquifer west of the city wellfield is not an immediate threat to humans, inasmuch as private residences no longer use wells that tap the aquifer for drinking water supplies. Also, the city wellfield is horizontally upgradient of the intermediate aquifer groundwater flow direction. However, since this aquifer could be tapped for use, the health-based RA must assume the possibility of wells withdrawing contaminated water from either the shallow or intermediate aquifers. However, hydrogeologic conditions could exist, which would permit the contamination to reach the lower aquifer. These include a thin lower aquitard, strong downward vertical potentials, and/or a large cone of influence produced by lower aquifer pumping.

Determination of these parameters was largely beyond the original scope of the Cadillac Area RI. The Work Plan excluded explorations into the lower aquitard primarily based on MDNR's concern for the potential of creating conduits for contaminant transport between the intermediate and lower aquifers. (For the plume near Rexair, situated east of the city wellfield, the initial penetration of the lower aquitard was not by design, but due to its thinness. With the

potential severe impact of a very thin aquitard and the existing extreme overlying contamination, further investigation by drilling deep boreholes and wells away from areas of known contamination was a necessity.) The pump test, provided limited, additional information regarding the potential for inter-aquifer contaminant transport.

Assuming that no significant VOC seepage is occurring through the lower aquitard, the fate of the intermediate aquifer VOC plumes west of the city wellfield will be to continue migrating in a north-northwest direction. Because the shallow aquifer plume still has an active source of VOC recharge in the form of contaminated soils, it is expected that contamination of the intermediate aquifer will be indirectly recharged by this shallow contamination.

The remaining plumes in the intermediate aquifer include the two relatively small and unconcentrated plumes below the North Park Subdivision. These, in fact, may be a northerly continuation of the larger VOC plume resulting from Four Star, Joynts/Four Winns, and possibly Kysor contaminant releases (see Section 5.5.3). Residences in the North Park Subdivision are presently supplied with city water, and the plume is horizontally downgradient of the wellfield. No other identifiable receptors exist within a considerable distance.

6.3.4 Hexavalent Chromium Plumes

Hexavalent-chromium contamination was first detected in the shallow aquifer in MW-5 in 1986 after previous groundwater analyses contained no hexavalent chromium. The location of the chromium-impacted well (MW-5), directly downgradient of Northernair, indicates more positively that Northernair was the source. The sporadic detection of hexavalent chromium suggests that releases may have occurred in short, concentrated events. The fate of hexavalent chromium in the shallow aquifer will be to migrate in a northeastward direction, eventually reaching the intermediate aquifer by seepage through the upper aquitard or by passing the aquitard's northern limit. No human receptors of the hexavalent chromium in the intermediate aquifer presently exist. Since the source of contamination is no longer present, and remaining soils contamination is to be remediated, the plume should continue to be diluted by dispersion with no future source inputs (Focused FS; Jordan, July 1985).

The hexavalent chromium plume in the intermediate aquifer is well-defined by a series of MW monitoring wells north of the Northernair facility. The Northernair Site Supplemental RI, issued by Jordan in January 1987, describes the hexavalent-chromium plume in detail. Section 5.5.5 of this report compares 1986 and 1987 hexavalent-chromium groundwater data.

As discussed previously, seepage of the contaminated groundwater through the lower aquitard is possible. Subsequent dilution in the lower aquifer would further lower concentrations. While it is not possible to state with complete assurance that drinking water criteria (0.05 ppm) would be met if the hexavalent chromium reached the wellfield (captured by the cone of influence of the pumping), it does seem probable, given: (1) the less than 1 ppm concentration in the intermediate aquifer, (2) the low probable seepage rates, and (3) the high rate of pumping.

7.0 BASELINE RISK ASSESSMENT

This section uses the information presented earlier in this document to assess the potential public health and environmental risks posed by the Cadillac Industrial Park area groundwater contamination. The principle objective of conducting a baseline risk assessment (RA) of the groundwater in the study area is to estimate the magnitude and probability of actual or potential harm to public health or the environment from exposure to chemicals in the groundwater in the absence of remediation. The RA will be used to determine if remedial actions are necessary to reduce risks to acceptable levels, in order to protect potentially affected populations or environments.

7.1 INTRODUCTION

This RA is the baseline or no-action risk evaluation of the study area groundwater contamination, in which current and future potential risk to human health and the environment are assessed. Rather than look at individual contaminant plumes from specific source areas, most of this section focuses on the plumes in the various aquifers that could impact actual or potential groundwater receptors. This approach is more relevant from a public health and/or environmental viewpoint. The groundwater plumes that are evaluated are designated as:

- o Shallow plume (which contains all volatile organic chemicals (VOCs) that are currently in the upper aquifer and which have originated from various source areas);
- o Intermediate plume (which contains all VOCs that are currently in the intermediate aquifer west of Sixth Avenue at the site), with the exception of those that were found in the intermingled Northernnaire plume);
- o Northernnaire plume (which contains hexavalent chromium, cyanide, and the intermingling VOCs which were detected in the Northernnaire wells);
- o Rexair plume (a VOC plume east of Sixth Avenue at the site).

These major plumes are further subdivided into discreet plumes in the fate and transport section.

The baseline risk assessment is organized as follows:

Section 7.2 is a discussion of the selection of chemicals of concern (or indicator chemicals). A screening process is performed in order to focus the assessment on those compounds which pose the greatest threat to human health and/or the environment. The screening for each plume is based on the intrinsic toxicological properties, concentration, distribution and frequency, mobility, and persistence of each chemical.

Section 7.3, the Toxicity Assessment, consists of three subsections:

- o Brief toxicity profiles which describe potential adverse effects for each chemical of concern;
- o A dose-response assessment which identifies the pertinent available standards and criteria for the protection of human health. These criteria are subsequently used in the risk characterization to evaluate potential adverse effects of chemical concentrations measured in the groundwater.
- o A comparison of groundwater contaminant concentrations to chemical-specific applicable or relevant and appropriate requirements (ARARs) to ensure that public health or environmental requirements are being met.

Section 7.4 consists of a transport and fate analysis and public health and environmental exposure assessments. The transport and fate analysis utilizes various modeling techniques to predict chemical concentrations at downgradient receptor wells in the future. The public health exposure assessment identifies human populations that are potentially exposed, and characterizes possible routes of exposure. For the Cadillac site, ingestion of groundwater is assessed, and quantitative exposure dose levels are estimated. Present potential risk is assessed by utilizing the average and maximum chemical concentrations detected in each plume and assuming that someone could be drinking that water. Future potential risk is estimated by using the chemical concentrations predicted through time and distance from the source in the transport and fate analysis. Two scenarios are developed for each plume to represent both the most probable exposure conditions and conservative (or realistic worst-case) conditions. The environmental exposure assessment consists of an identification of the environments at risk of exposure and the pathways through which they could be exposed.

Section 7.5 characterizes the present and future potential risk to both human health and the environment posed by the Cadillac site groundwater contamination. The public health risk characterization compares the exposure levels calculated in Section 7.4 to the criteria identified in the toxicity assessment to quantify the magnitude of potential risk posed to human receptors. Carcinogenic and noncarcinogenic risk levels are estimated, and the potential for risk from multi-toxic exposure is discussed. Summary tables of risk estimates are also provided. The environmental risk characterization discusses the potential for risk to environments and nonhuman receptors. Both the public health and environmental risk characterizations are then briefly summarized in Section 7.6, and potential adverse effects that could be caused by contaminated groundwater at the Cadillac site are highlighted.

7.2 SELECTION OF CHEMICALS OF CONCERN

The selection of chemicals of concern is a screening process which is performed to reduce the field of chemicals that are assessed in a baseline risk assessment to a manageable number. In determining the chemicals of concern, Jordan

utilized the guidance provided in the Superfund Public Health Evaluation Manual (U.S. EPA, 1986). This guidance states that the selection should be based on the inherent toxicity, measured concentrations in various media at the site, distribution and frequency of detection, and physical and chemical parameters which are related to the environmental mobility and persistence of each chemical.

Selecting chemicals of concern for the Cadillac site investigation for human receptors is a relatively simple process for several reasons. While many sites have multiple chemicals present in many environmental media, groundwater is the only medium of concern at Cadillac. In addition, there were only a total of sixteen chemicals (including chromium and cyanide, but not including the additional metals which were detected but which are probably due to road salting activities; see Section 5.5.6) detected in the four groundwater plumes defined in Section 7.1. Differences in chemical mobility and persistence are usually a primary consideration in the selection process. However, most of the chemicals that are present in Cadillac groundwater plumes are volatile organic compounds, and therefore exhibit similar behavior in groundwater, making differences in mobility and persistence slight. This similarity of properties reduces the number of selection criteria that must be considered.

Chemicals of concern for the Cadillac site were chosen based on consideration of the relative importance of the inherent toxicity, frequency of detection in and distribution among the various plumes, and measured concentrations of each chemical detected. This was accomplished by first identifying the chemicals which were detected in each of the four plumes of concern, and then performing a statistical analysis of chemical concentrations within each plume. Data collected on June and August 1987 were used for this analysis, which can be found in Appendix Q. Pertinent data from the statistical analysis were then summarized and are presented in Table 7-1, along with information pertaining to the toxicity and persistence of each chemical. Table 7-1 can be summarized as follows:

- o Chemicals are divided by chemical class (volatile organic vs. inorganic) and classified as "noncarcinogenic" or "carcinogenic".
- o The frequency of detection of each chemical in each plume is expressed as a fraction (e.g. 1,1,1-trichloroethane = 2/43, which indicates that 1,1,1 trichloroethane was found above detection limits in two out of a total of 43 samples).
- o The maximum concentration of each chemical in each plume is also noted in parentheses, beneath each frequency.
- o RfDs, U.S EPA verified risk reference doses, are used to indicate the relative toxicity of each noncarcinogen, while CAG (Carcinogen Assessment Group) cancer potency values serve the same purpose for carcinogens.
- o Surface water half-lives are included in the "comments" column to ensure that no extremely persistent chemicals are overlooked.

TABLE 7-1

CADILLAC AREA GROUNDWATER STUDY: DATA FOR SELECTION OF CHEMICALS OF CONCERN (p. 1)

----- FREQUENCY AND DISTRIBUTION -----

COMPOUND	DELAIR (ppb)	UPPER AQUIFER (ppb)	INTER. AQUIFER (ppb)	NORTHERNAIRE PLUME (ppb)	COMMENTS
VOLATILE COMPOUNDS - NONCARCINOGENS					
1,1,1-TRICHLOROETHANE	2/43 (17)	26/45 (32000)	4/21 (1200)	4/22 (440)	Rfd = 0.086 SW half-life = 0.14-7 days
1,1-DICHLOROETHANE	--	3/45 (420)	--	--	Rfd = 0.12 SW half-life = 1-5 days
ACETONE	--	6/45 (920)	3/21 (420)	--	Rfd = 0.1 SW half-life = NA
ETHYLBENZENE	--	3/45 (670)	--	--	Rfd = 0.1 SW half-life = 1.5-7.5 days
TOLUENE	6/43 (42)	8/45 (3200)	1/21 (1)	--	Rfd = 0.3 SW half-life = 0.17 days
TRANS-1,2-DICHLOROETHENE	3/43 (48)	--	2/21 (4600)	2/22 (40)	-- SW half-life = 1-6 days
XYLENES	--	2/45 (1700)	--	--	Rfd = 2 SW half-life = 1.5-9 days
VOLATILE COMPOUNDS - CARCINOGENS					
BENZENE (GROUP A)	--	1/45 (20)	--	--	CAG VALUE = 0.029 SW half-life = 1-6 days
CHLOROFORM (GROUP B2)	3/43 (6)	2/45 (15)	1/21 (4)	--	CAG VALUE = 0.081 SW half-life = 0.3-30 days
1,1-DICHLOROETHYLENE (GROUP C)	--	17/45 (1000)	3/21 (99)	2/22 (26)	CAG VALUE = 0.58 SW half-life = 1-6 days

TABLE 7-1

CADILLAC AREA GROUNDWATER STUDY: DATA FOR SELECTION OF CHEMICALS OF CONCERN (p. 2)

----- FREQUENCY AND DISTRIBUTION -----

COMPOUND	REXAIR (ppb)	UPPER AQUIFER (ppb)	INTER. AQUIFER (ppb)	NORTHERNAIRE PLUME (ppb)	COMMENTS
1,2-DICHLOROETHANE (GROUP B2)	: : -- :	: : 3/45 : (2000)	: : -- :	: : -- :	: : CAG VALUE = 0.091 : SW half-life = 0.17 days :
METHYLENE CHLORIDE (GROUP B2)	: : 5/43 : (780)	: : 5/45 : (690)	: : 2/21 : (240)	: : -- :	: : CAG VALUE = 0.0075 : SW half-life = 1.2-5.8 days :
TETRACHLOROETHYLENE (GROUP B2)	: : -- :	: : 10/45 : (720)	: : 1/21 : (17)	: : 1/22 : (3)	: : CAG VALUE = 0.051 : SW half-life = 1-30 days :
TRICHLOROETHYLENE (GROUP B2)	: : 31/43 : (153000)	: : 30/45 : (78000)	: : 15/21 : (7500)	: : 18/22 : (2500)	: : CAG VALUE = 0.011 : SW half-life = 1-90 days :
INORGANIC COMPOUNDS - CARCINOGENS					
CHROMIUM (HEXAVALENT) (GROUP A - INHALATION ONLY):	: : -- :	: : -- :	: : -- :	: : 10/24 : (773)	: : CAG VALUE=NOT APPLICABLE FOR : ORAL EXPOSURE. : RFD = 0.005 : SW half-life = 3 days :
INORGANIC COMPOUNDS - NONCARCINOGENS					
CYANIDE	: : -- :	: : -- :	: : -- :	: : -- :	: : RFD = 0.029 : SW half-life = 0.33-0.8 days :

Based on the analysis of the available data, only four chemicals were eliminated from further consideration in the baseline risk assessment. These chemicals, and the justification for their elimination, are as follows:

- o 1,1-DICHLOROETHANE (1,1-DCA). This noncarcinogenic chemical was detected in only the shallow aquifer, in three of forty-five samples, at a maximum concentration of 420 ppb. It is adequately represented by 1,1,1-trichloroethane (1,1,1-TCA), a similar chlorinated aliphatic volatile organic chemical, which was detected more frequently (26 of 45 samples in the same aquifer) and at higher concentrations (maximum concentration 32000 ppb). 1,1,1-TCA is also more toxic.
- o ETHYLBENZENE. Ethylbenzene was detected in the shallow aquifer only, in three of forty-five samples, at a maximum concentration of 670 ppb. It is adequately represented by toluene, which is also an aromatic volatile organic chemical. Ethylbenzene is slightly more toxic than toluene; however, the maximum concentration of toluene (3200 ppb) is almost five times more than the maximum concentration of ethylbenzene. Furthermore, toluene was detected more frequently (eight of forty-five samples in the same aquifer) and at higher concentrations.
- o XYLENES. As with 1,1-DCA and ethylbenzene, xylenes were only detected in the shallow aquifer, in two of forty-five samples, maximum concentration 1700 ppb. Xylenes are also adequately represented by toluene, which was detected more frequently, at higher concentrations, and is over 60 times more toxic.
- o BENZENE. Benzene is a Group A, known human carcinogen. This would ordinarily be a strong incentive to include it in the baseline risk analysis, although it's inclusion is not technically required. However, benzene was only detected once (in the shallow aquifer), and at a low concentration (20 ppb). This concentration is higher than its MCL (maximum contaminant level, drinking water criteria) of 5 ppb. However, due to the fact that benzene was only detected once, and there is no history of use or disposal of this chemical by the industries around Cadillac, benzene will not be considered further in the analysis of potential risk.

Cyanide was not detected in the June and August 1987 sampling events. However, as stated in Section 5.5.6, this contaminant was of concern in the Northernnaire study because of its presence in plating baths. Low levels of cyanide were detected in Northernnaire wells in 1985 (maximum concentration = 22 ppb), and it can be assumed that the recent lower concentrations (non-detects) may be due to the fact that we are now sampling a different portion of the Northernnaire plume. Therefore, the maximum value of cyanide detected in 1985 will be evaluated in order to assess the potential for that chemical to pose a risk to public health.

Table 7-2 lists the chemicals of concern that have been identified for the Cadillac area groundwater study baseline risk assessment for public health receptors.

TABLE 7-2
CHEMICALS OF CONCERN
CADILLAC AREA GROUNDWATER STUDY
BASELINE RISK ASSESSMENT

VOLATILE ORGANIC COMPOUNDS

Noncarcinogens

1,1,1-Trichloroethane
Acetone
Toluene
Trans-1,2-dichlorethene

Carcinogens

Chloroform
1,1-Dichloroethene
1,2-Dichloroethane
Methylene Chloride
Tetrachloroethene
Trichloroethene

INORGANIC COMPOUNDS

Chromium (hexavalent)
Cyanide

Aquatic and terrestrial organisms are potentially exposed to chemical contamination in surface waters, sediments and surface soils. No surface water bodies which might be receptors of contamination exist at the Cadillac site; therefore, exposure is only possible via surface soil contamination. Surface soil contamination had been identified behind the Kysor facility in the past, but has since been remediated by excavation and backfilling. This chemical contamination is thus not available to terrestrial organisms. Because of the negligible or non-existent potential for exposures to environmental receptors, no chemicals of concern for environmental receptors have been identified.

7.3 TOXICITY ASSESSMENT

This section contains qualitative and quantitative indices of the potential hazard of each of the contaminants of concern, i.e., their inherent ability to cause harm to human health. (No environmental contaminants of concern have been identified for this site; therefore, this section is concerned solely with potential hazard to human health.) It consists of three subsections:

- o Brief toxicity profiles which describe potential adverse effects for each chemical of concern;
- o Identification and definition of pertinent available standards and criteria for the protection of human health;
- o Comparison of groundwater contaminant concentrations to chemical-specific applicable or relevant and appropriate requirements (ARARs) to ensure that requirements are being met.

7.3.1 Toxicity Profiles for Indicator Chemicals

Toxicological profiles for the chemicals of concern (indicator chemicals) are presented below. These profiles contain a brief summary of the major toxicological properties and the adverse human health effects of each chemical. Toxicological profiles for environmental receptors are not necessary as no environmental contaminants of concern have been identified for this site. They are qualitative overviews and are not meant to be comprehensive health evaluations. References are provided for each chemical.

7.3.1.1 Acetone. Acetone is a highly volatile and flammable organic solvent which is not persistent in the environment. It has an air half-life of 14 hours (Graedel, 1978) and a soil half-life of 10 hours to 9 days (U.S.EPA, 1981a) and is expected to leach from soil into groundwater due to its weak soil sorption and high water solubility (U.S.EPA, 1981b).

Acetone can be absorbed both through inhalation and ingestion but is rapidly excreted and does not bioaccumulate. Toxicity information is minimal, and subchronic exposure information is based only on animal studies. When rats drank water with 2.5% acetone contamination (about 25,000 mg/l), decreased body weight was reported in rats ingesting high doses (Bruckner and Peterson, 1981).

Human chronic toxicity studies are based on occupational exposures. One study of workers inhaling about 20 ppm acetone reported mucus membrane irritation (Parmessiani & Sassi, 1954); however, another 17-year occupational study reported no significant toxicity at inhalation exposures less than 2,150 ppm (Oglesby et al., 1949).

Acetone interacts with other chemicals in the body in that it potentiates the toxicity of those chemicals. In particular, it potentiates the hepatotoxicity of 1,1,1-trichloroethane (Plaa et al., 1975). Information on teratogenicity is not available, but acetone is known to cross the placenta (Dowty et al., 1975). Occupational exposure standards have been established, and an oral reference dose has been set at 0.1 mg/kg/day (U.S.EPA, 1986).

Information on the carcinogenicity of acetone is not available, and the U.S.EPA has not classified it as to carcinogenicity (Group D).

7.3.1.2 Toluene. Toluene (methylbenzene), an organic solvent formed during petroleum and coal tar distillation, is used in the manufacture of other chemicals and is found as a component of gasoline. In some media, toluene has short environmental half-lives. The air half-life is 1.3 days (Singh et al., 1981) and the water half-life is 4.1 hours (Macay and Yeun, 1983). It has a moderately low potential for adsorption and certain portions of spills may migrate into groundwater (Wilson et al., 1981).

While toluene is a relatively common water contaminant, available studies have not indicated that it is highly toxic. Human studies have shown rapid absorption through the respiratory tract (Astrand et al., 1972). Gastrointestinal absorption information is limited to animal studies and is reported as relatively rapid (Pyykko et al., 1977). There is no available information on the oral exposure toxicity in humans, but very limited animal oral exposure studies indicate central nervous system (CNS) inhibition (Kimura et al., 1971).

Numerous human occupational studies of inhalation exposure to toluene have been done, and both acute and chronic exposures to varying air concentration of toluene have reported CNS toxicity (von Oettingen et al., 1942a,b; Carpenter et al., 1944; Wilson, 1943; Munchinger, 1963; Hanninen et al., 1976).

Toluene is subject to abuse as a "recreational drug", and studies of chronic toluene abusers and occupational studies of chronically exposed workers have reported liver (Greenburg et al., 1942; Grabski, 1961) and renal function effects (Kroeger et al., 1980; Moss et al., 1980).

Pregnant animals exposed to toluene by ingestion and inhalation had decreased fetal weights (Hudak and Ungavry, 1978; Nawrot and Staples, 1979). Adequate data to evaluate the teratogenicity, mutagenicity or carcinogenicity are not available. Occupational standards have been set. U.S.EPA has not classified toluene as to carcinogenicity (Group D). The oral risk reference dose is 0.03 mg/kg/day (U.S.EPA, 1986) based on a study of inhalation exposure study in rats.

7.3.1.3 1,1,1-Trichloroethane. 1,1,1-Trichloroethane (1,1,1-TCA) is a halogenated hydrocarbon which is used primarily as a metal degreaser and cleaner,

but also has other industrial uses. It is more persistent in the atmosphere than in water, and information on the soil half-life of 1,1,1-TCA is variable. Air half-lives have been estimated ranging from one to eight years (Singh et al., 1981; Makide and Rowland, 1981; U.S.EPA, 1984a). Water half-life has been estimated at greater than six months (U.S.EPA, 1985c). While this compound has a fairly short soil half-life, it has a low soil adsorption coefficient and can leach into groundwater.

1,1,1-TCA is rapidly absorbed in the body from both the respiratory (U.S.EPA, 1985c) and gastrointestinal tracts (Stewart et al., 1969), and gastrointestinal tract absorption appears to be complete (Stewart et al., 1969).

The prime target organs of this compound are the central nervous system (CNS) and liver. Information based on human responses is limited and focuses on inhalation. Short-term, high level inhalation of 1,1,1-TCA resulted in CNS depression (U.S.EPA, 1984a). Stewart et al. (1975) exposed 20 people to 500 ppm, 7.5 hours per day for three weeks and no body chemistry effects were detected. Fatigue, irritation and headache were noted. Liver fatty vacuolation has been reported (Caplan et al., 1976).

Animal studies have reported CNS effects at high oral exposure levels (Bruckner et al., 1985), cardiac arrhythmias (Carlson, 1981) and liver chemistry changes at high inhalation doses (McNutt et al., 1975). Limited animal studies indicate 1,1,1-TCA is not teratogenic.

Increased incidence of liver cancer has been found in laboratory animals (NTP, 1982E) orally exposed to high doses, and 1,1,1-TCA is under review by the U.S.EPA for evidence of human carcinogenic potential. The U.S.EPA Office of Drinking Water (1987) has set the maximum contaminant level goal (MCLG) at 200 ug/l in drinking water. An oral reference dose (RfD) has been set at 0.086 mg/kg/day (U.S.EPA, 1986).

7.3.1.4 Methylene Chloride. Methylene chloride (dichloromethane), a halogenated hydrocarbon, is used commercially as an industrial solvent, and as a component of paint removers and aerosol products.

This is a relatively volatile compound with an atmospheric half-life of 53 to 127 days (Singh et al., 1981; Makide & Rowland, 1981). The half-life of methylene chloride in water has been estimated from 1-40 days depending on the organic content of the water (U.S.EPA, 1984a). Soil half-life information is not available, however, methylene chloride is expected to evaporate quickly from surface soil. If this compound enters the subsurface soil, biodegradation may be slow (Wilson et al., 1983), and, due to a relatively high solubility, it is expected to leach into the groundwater.

Information on the behavior of this compound in the human body is limited. Several studies indicate that gastrointestinal absorption may be fairly rapid and complete (Roberts and Marshall, 1976; Llewellyn, 1966, Stewart and Nake, 1976; Friedlander et al., 1987). Human and animal studies of absorption of methylene chloride through inhalation indicate 34% to 70% absorption (U.S.EPA, 1984a).

Inhalation exposures of animals at the subchronic level has resulted in liver and kidney effects, depressed weight gain, elevated carboxyhemoglobin levels and at high doses, increased mortality (U.S.EPA, 1985c). Chronic animal inhalation studies have resulted in increased incidence of mammary (Burek et al., 1984), pituitary gland and lung tumors, as well as leukemia and mesothelioma (Dow, 1982). Chronic inhalation exposure of humans to this compound resulted in CNS effects and anorexia (U.S.EPA, 1985c).

Chronic oral exposure of animals to methylene chloride in two studies by the National Coffee Association in 1982 found increased incidence of liver cancer in both rats and mice. This compound has been placed in Carcinogenesis Assessment Group B2 (probable human carcinogen based on adequate animal study data). The cancer potency factor is $0.0075 \text{ mg/kg/day}^{-1}$ (U.S.EPA, 1986). Methylene chloride has also been found to be mutagenic in numerous types of organisms (U.S.EPA, 1985c). There is no available evidence of teratogenicity.

U.S.EPA (1986) has set an oral reference dose for methylene chloride at 0.06 mg/kg/day based on a chronic study of rats exposed to the compound in drinking water.

7.3.1.5 Trichloroethene (TCE). Trichloroethene (TCE) is a synthetic compound that is used mainly as a metal degreaser. Its half life in air and surface water is short, four days or less, but it moves quickly through soil to groundwater where it may persist for months to years (U.S.EPA, 1985d). Under certain conditions, TCE in groundwater may degrade to dichloroethylene and vinyl chloride (Parsons, 1984; Vogel, 1985).

TCE can be absorbed into the body through the respiratory and gastrointestinal tract (Daniel, 1963; U.S.EPA, 1980). TCE is metabolized mainly in the liver, lungs and kidneys, and these organs are where the major effects have been reported. Evidence of carcinogenicity have been found in animals studies in both sexes following both oral and inhalation exposure. While TCE is reported to be a common contaminant of water supplies (Ziglio et al., 1983), there is no available data on oral toxicity in humans. Animal studies have reported increased liver and kidney weights (Tucker et al., 1982) liver carcinomas (NCI, 1976; NTP, 1982; Herren-Treund et al., 1986, 1987; Bell, 1978), renal adenocarcinomas (NTP, 1982a, 1986a; Maltoni et al., 1986), leukemia (Maltoni et al., 1986), and lung tumors (Fukada et al., 1983; Maltoni et al., 1986).

TCE is classified as a probable human carcinogen (CAG B2). The cancer potency factor is $0.011 \text{ mg/kg/day}^{-1}$ (U.S.EPA, 1987). Very limited information is available on reproductive effects or teratogenicity; one animal study reported decreased maternal and fetal weight gain and delayed fetal development (Schwetz et al., 1975). TCE toxicity is enhanced with exposure to carbon tetrachloride (Pessayre et al., 1972) or ingestion of alcohol (Gessner and Cabana, 1970).

7.3.1.6 Chloroform. Chloroform is a volatile chlorinated hydrocarbon that is used a refrigerant, industrial solvent, pesticide, fire extinguisher, and component in pharmaceuticals and toiletries. It has an 80 day air half-life (U.S.EPA, 1986), and 30-day-or-less surface water half-life (Zoeteman et al., 1980). Chloroform has a greater-than 27 week soil half-life (Wilson et al., 1983) and will leach into the groundwater.

The compound can enter the body by both inhalation and ingestion. It was used as a human anesthetic until the early 1900's, and its effects on the central nervous system (CNS) are well known. Chloroform exposures in animal studies have resulted in liver and kidney effects (Torkelson et al., 1976; Palmer et al., 1979; Pohl, 1979).

While chloroform is being evaluated by U.S.EPA for evidence of its human carcinogenic potential, animal studies have reported several types of tumors after oral exposure. Liver cancer was reported in male and female mice (NCI, 1976), renal cancer and adenoma and thyroid tumors in rats (Palmer et al., 1979) and hepatomas in mice (Eschenbrenner and Miller, 1945). Case-control and epidemiological studies of humans orally exposed to chloroform support an association with increased incidence of bladder, colon and rectal cancer (Alavanja et al., 1978; Cantor et al., 1978; Brenniman et al., 1978; Hogan et al., 1979; Struba, 1979; Gottlieb et al., 1981; Young et al., 1981).

Teratogenic effects have been observed in animals. Deformed fetuses were reported in rats and mice exposed to chloroform during gestation (Schwetz et al., 1974; Murray et al., 1979). Chloroform is fetotoxic at levels which produce severe maternal toxicity (Thompson et al., 1974).

An oral reference dose (RfD) for humans has been set at 0.01 mg/kg/day. This RfD is based on a study of dogs ingesting chloroform (Heywood et al., 1979).

7.3.1.7 Tetrachloroethene (PCE). Tetrachloroethene (or perchloroethene, PCE) is a volatile synthetic compound which is used as a metal degreaser, in the dry cleaning industry, and as a chemical intermediate. Its half-life is about 30 to 40 days in air and a month or less in soil (Zoeteman et al., 1980). PCE is very mobile in soil and leaches quickly into groundwater where it may persist for months to years if volatilization does not occur. Under certain conditions, PCE in groundwater may degrade to trichloroethylene and then to dichloroethylene and vinyl chloride (Parsons, 1984; Vogel, 1985).

This compound can be absorbed into the body by inhalation, ingestion, or dermal exposure and is stored in body fat (Monster, 1979). The principal toxic effects in both humans and animals are CNS depression, and liver and kidney effects (U.S.EPA, 1985a).

The carcinogenic potential of PCE is currently under review by the U.S.EPA. This compound was classified as a possible human carcinogen by EPA in 1985, and it has been reclassified as a Group B2 (probable human carcinogen). The initial cancer potency factor was 0.051 mg/kg/day, and this value was used to compute the cancer risk in the absence of other data. Based on a 1977 National Cancer Institute oral animal study, there is limited evidence of increased incidence of hepatocellular cancer following PCE ingestion. A recent National Toxicology Program mouse study found evidence of liver cancer in PCE-exposed mice. A subsequent NIOSH study of dry cleaning employees did not find deaths due to liver cancer but did find an excess of urinary cancers. However, this result may be confounded due to other exposures.

The proposed maximum contaminant level goal (MCLG) for drinking water has been set at zero (U.S.EPA, 1986).

7.3.1.8 Cyanides. Cyanides are sometimes naturally occurring substances present in a number of foods and plants, are produced by certain bacteria, fungi, and algae, are a dietary requirement in minute amounts (vitamin B₁₂). Cyanides are also a component of cigarette smoke. However, most of the cyanides found in the environment are released as a waste product of industrial processes.

Cyanides represent a group of compounds which may exist as gases or solids with varying physical and chemical properties and subsequently different environmental fates. Of particular interest at this site are cyanides used in the metal treatment and electroplating industries. Total cyanides were measured at two wells at this site, and, since cyanide salts are used in the metal plating process, it is suspected that these are cyanide salts.

Absorption of cyanide varies with route of exposure and presumably type of cyanide compound. Humans inhaling hydrogen cyanide absorbed 56%-77% (Landahl and Herrmann, 1950). Gastrointestinal absorption in humans depends on the pH of the gut and on the lipid solubility and pKa of the specific cyanide compound. Rapid lethal effects from cyanide ingestion indicate that GI absorption does occur (Gosselin et al., 1976). Cyanides are moderately lipid-soluble and are able to be absorbed dermally. Inhalation and dermal exposures usually occur simultaneously from a vapor form.

CNS effects, respiratory effects, cardiac effects, and thyroid disorders have been reported in workers occupationally exposed to cyanide. Similar effects have been reported in animal studies. Fatalities from high doses of cyanide are well documented by suicides and homicides (U.S.EPA, 1987).

Evidence of teratogenicity (Willhite, 1982) and developmental effects (Doherty et al., 1983) has been reported in hamsters and thyroid effects in rats exposed to thiocyanate during gestation (Kruetler et al., 1963). There is no evidence reported of cyanide reproduction toxicity, genotoxicity or carcinogenesis and it is in Carcinogenesis Assessment Group D (not evaluated).

Oral reference doses (RfD) have been established for a number of the cyanide compounds, and all are based on a two-year study of rats eating food fumigated with hydrogen cyanide. The reference dose (RfD) for free cyanide is 0.029 mg/kg/day, and the RfDs for cyanide compounds range from a high of 0.1 mg/kg/day for silver cyanide to 0.04 mg/kg/day for sodium cyanide, calcium cyanide and cyanogen (U.S.EPA, 1986). Water quality criteria for free cyanide for protection of human health is 200 ug/L (U.S.EPA, 1984c).

7.3.1.9 Hexavalent Chromium. Hexavalent chromium (Chromium VI) is an industrial metal used in the chrome plating and metal finishing industries and is used as a rust inhibitor. It can enter the environment either during the discharge of industrial waste waters, or as mists emitted during the manufacturing process into the atmosphere, adsorbed to particles in the air and subsequently deposited in the soil as a result of precipitation or atmospheric fallout. Chromium can be absorbed into the body via inhalation, ingestion, or dermal exposure. Oral absorption in humans has been reported at 11% and less; oral absorption in animals has been reported at 2.3% and less (U.S.EPA, 1987).

Hexavalent chromium can exist in several different ionic forms. However, all of these forms are treated the same for the Carcinogenicity Assessment Group classification. Hexavalent chromium has been classified as a CAG Group A (known human carcinogen) compound by inhalation exposure only. The inhalation classification is based on numerous human occupational studies of chromate production facilities and studies of the chrome pigment industry (U.S.EPA, 1987). All these studies found an association between occupational inhalation exposure to chromium and lung cancer. However, chromium has not been shown to be carcinogenic following oral exposure.

Industrially-used chromium can be hexavalent or trivalent (III), and workers are nearly always exposed to both forms (Casarett and Doull, 1986). Most of the studies used to determine the carcinogenicity classification did not attempt to determine whether the etiologic agent was hexavalent or trivalent chromium. When taken into the body, hexavalent chromium is reduced to trivalent chromium at the intracellular level by the cytochrome-P 450 system in the presence of NADPH and can then bind to macromolecules such as RNA (Gruber and Jennette, 1978). The toxicity of hexavalent chromium may be related to intermediates in the reduction process which subsequently bind to cell macromolecules (Casarett and Doull, 1986).

Besides lung cancer, human inhalation exposures have been associated with liver and kidney damage, internal hemorrhage, and respiratory effects (U.S.EPA, 1987). Dermal exposure has resulted in contact dermatitis and ulceration (Denton et al., 1954). Unfortunately, oral exposure data is limited to a very few animal studies, and no adverse effects were detected in those studies. Hexavalent chromium has been found to be mutagenic in numerous bacteria (U.S.EPA, 1987), but data are not available on reproductive toxicity or teratogenicity.

A reference dose (RfD) of 0.005 mg/kg/day was established based on a one-year-long drinking water study in rats (U.S.EPA, 1986). The MCL for chromium is 0.05 mg/l, and a MCLG for drinking water of 0.12 mg/L has been proposed (U.S.EPA, 1986).

7.3.1.10 Trans-1,2-dichloroethylene. Trans-1,2-dichloroethylene (trans-1,2-DCE), or 1,2-dichloroethene is a halogenated aliphatic hydrocarbon. It is a synthetic compound used as an intermediate in the manufacture of other chlorinated solvents. Both the cis- and trans- isomers of 1,2-dichloroethylene are expected to behave similarly in the environment. Based on information on other structurally related chlorinated compounds, trans-1,2-dichloroethylene is expected to biodegrade slowly under favorable conditions in subsurface soil (Tabak et al., 1981). It does not sorb strongly to soil, and is expected to leach into groundwater where it is relatively stable. However, in some groundwaters, the 1,2-DCEs have been shown to degrade biologically to vinyl chloride, a potent carcinogen (U.S.EPA, 1985b). Trans-1,2-DCE is not expected to bioaccumulate in plants or animals.

No oral or inhalation studies have been conducted on the absorption of trans-1,2-dichloroethylene. As a low molecular weight lipid soluble compound, however, it is expected to be absorbed quickly and readily by all routes (oral, inhalation and dermal). Based on studies of trichloroethylene, (Daniel, 1963

and Monster et al., 1976) it is estimated that 100% of ingested DCE and 35-50% of inhaled DCE may be absorbed. Although no studies of distribution have been conducted on trans-1,2-DCE, if it behaves similarly to 1,1-DCE, the highest concentrations would be expected to be found in the liver and kidney following exposure.

In studies in humans, at high concentrations the DCEs possess anesthetic and narcotic properties (Irish, 1963). The trans-isomer is about twice as potent as the cis-isomer in depressing the CNS (Albrecht, 1927).

Animal inhalation studies have reported slight liver degeneration, decreases in the red blood cell count and pulmonary effects. The severity of these effects increased with dose and length of exposure (Freundt et al., 1977).

No studies have been conducted regarding carcinogenic effects or teratogenic and reproductive effects of trans-1,2-DCE (Greim, et al., 1975). Trans-1,2-DCE was not mutagenic when assayed in several in vitro and in vivo systems. A MCLG of 0.07 mg/L has been proposed (U.S.EPA, 1986).

7.3.1.11 1,1-Dichloroethylene. 1,1-Dichloroethylene (1,1-DCE), also known as vinylidene chloride and dichloroethene, is a synthetic chemical. It is used as a co-monomer in the production of a number of polymers. Polymers of 1,1-DCE are used as food wrap. 1,1-DCE is chemically stable in water, mobile in soil, and expected to migrate with ground water. It is not believed to bioaccumulate in plants or animals (U.S.EPA, 1983).

When 1,1-DCE is administered orally to rats and mice, it is rapidly and completely absorbed. It is initially seen in the liver and kidneys and then distributed throughout other soft tissues. The highest concentrations are found in the liver (Jones and Hathway, 1978a, McKenna et al., 1978b). The acute toxicity of 1,1-DCE is probably from a toxic metabolite rather than the parent compound. The rate of excretion is relative rapid.

In short-term exposure studies of 1,1-DCE in animals, the liver is the most sensitive endpoint (Reynolds et al., 1975; Chieco et al., 1982). In addition, kidney lesions have been seen at doses similar to those affecting the liver (Prendergast et al., 1967). The liver was also the target organ in longer-term studies of 1,1-DCE in animals (Rampy et al., 1977; Quast, et al., 1983).

Studies in humans on workers exposed to 1,1-DCE via inhalation in combination with other compounds have reported liver function abnormalities, headaches, vision problems, weakness, fatigue and neurological sensory disturbances (NIOSH, 1979). At high concentrations (>4000 ppm), 1,1-DCE causes rapid onset of CNS depression with subsequent unconsciousness (Irish, 1963).

No teratogenic effects were reported in studies of 1,1-DCE in rats and rabbits. No studies of the effect of 1,1-DCE on reproductive capacity have been conducted. The International Agency for Research on Cancer (IARC) has concluded that 1,1-DCE is mutagenic based on Ames testing in the presence of a metabolic activation system. Animals studies have shown 1,1-DCE to be carcinogenic in some inhalation studies (Maltoni et al., 1977, 1980) and in one oral study of drinking water (Maltoni et al., 1985). The U.S.EPA Carcinogen Assessment

Group has classified 1,1-DCE as a Group C (possible human carcinogen). The cancer potency factor for 1,1-dichloroethylene is 5.8×10^{-1} (mg/kg/day)⁻¹, and the reference dose (RfD) is 9.0×10^{-5} mg/kg/day.

7.3.1.12 1,2-Dichloroethane. 1,2-Dichloroethane (1,2-DCA) is a synthetic compound, and is used by metal working industries as a solvent and metal cleaner. This volatile compound has a half-life of several months in the atmosphere, and can persist in surface water for months to years (U.S.EPA, 1985b).

1,2-Dichloroethane can be absorbed by humans through the lungs (Spencer et al., 1951; Urusova, 1953), gastrointestinal tract (Alumot et al., 1976) and skin (Urusova, 1953). Reports of accidental human ingestion indicate very high absorption of this compound (NIOSH, 1976), and animal studies indicate rapid and high oral absorption (Reitz et al., 1982). Higher oral absorption was found when rats drank water contaminated with this compound versus contaminated corn oil (Spreafico et al., 1978, 1979, 1980). Information on inhalation absorption rates is limited but absorption by this route is expected to be rapid and complete (Urusova, 1953; Reitz et al., 1980, 1982).

Acute oral exposures in humans have resulted in headache, dizziness, weakness, dilated pupils, vomiting of blood and bile, epigastric pain, diarrhea, heart pains, pulmonary edema, and unconsciousness (Wirtschafter and Schwartz, 1939; McNally and Fostedt, 1941). Animal studies of acute oral toxicity have reported central nervous system (CNS) effects, cardiovascular collapse, respiratory effects, and increased liver fat (Alumot et al., 1976; Spencer et al., 1951; Irish, 1963).

Chronic lower dose exposure studies in rats have reported increased forestomach carcinomas and hemangiosarcomas of the circulatory system in male rats and mammary gland adenocarcinomas in female rats (NCI, 1978). Mice had increased respiratory adenomas in both sexes, and females had increased mammary adenocarcinomas and uterine polyps or carcinomas (NCI, 1978). Dermal exposure of mice to this compound resulted in benign lung tumors (van Duuren et al., 1979).

Based on these animal studies, the U.S.EPA has classified 1,2-dichloroethane as CAG B2 (probable human carcinogen based on animal data). This compound has been found to be mutagenic in bacteria (Brem et al., 1974; Rannug Bieje, 1979). No data indicating reproductive effects or teratogenicity are available. The CAG potency factor for 1,2-DCA is 9.1×10^{-2} , and the drinking water MCL is 0.005 mg/l.

7.3.2 Dose Response Evaluation

This section contains the quantitative indices of toxicity that will be used to estimate the risks associated with the contaminants of concern. Various regulatory agencies have developed standards, guidelines and criteria to protect public health and/or the environment from adverse effects from chemical exposure. The National Contingency Plan (NCP) identifies these health based standards/guidelines/criteria and categorizes them, along with other technology-based values, as either "potential ARARs" (applicable or relevant

and appropriate requirements) or "other criteria or guidelines to be considered". Those health-based values which are relevant to the assessment of potential risk at Cadillac have been identified for the twelve chemicals of concern and are presented in Table 7-3.

To compare the estimated body doses calculated in the Exposure Assessment section to an applicable standard or guideline, it is necessary to convert a standard or guideline to body dose units (ug/kg/day). For example, to convert units of mg/L into ug/kg/day, the following formula was used:

$$\text{mg/L} \times \frac{1000 \text{ ug}}{\text{mg}} \times \frac{2\text{L}}{\text{day}} \times \frac{1}{70 \text{ kg}} = \text{Equivalent Daily Dose (ug/kg/day)}$$

The standard water consumption level for an adult is assumed to be 2 liters per day, and the average adult weight is assumed to be 70 kg. For a child, the standard water consumption level is assumed to be 1 liter per day for a 35 kg child, resulting in the same equivalent daily dose as for an adult.

The following subsections describe the standards or guidelines which are used to evaluate carcinogens and noncarcinogens in the Cadillac groundwater.

7.3.2.1 Evaluation of Carcinogens. The unit cancer risk criteria or carcinogenic potency factor [developed for approximately 58 chemicals by the U.S.EPA Carcinogen Assessment Group (CAG)] was used to evaluate carcinogens. If toxicological evidence indicates that a chemical may be a potential carcinogen, mathematical models are used to calculate the estimated excess cancer risk associated with exposure to the chemical. CAG calculated the unit risks using a non-threshold linearized multi-stage model for low-dose extrapolation. This model leads to a plausible upper limit (upper 95 percent confidence limit) of carcinogenic risk. It is based on a 70 kilogram (150 lb) person who drinks 2 liters of water per day over a 70-year lifetime. The risk value obtained represents increased carcinogenic risk over a person's lifetime from a single chemical. The unit cancer risks are expressed in units of (mg/kg/day)⁻¹.

7.3.2.2 Evaluation of Noncarcinogens. For noncarcinogens, to ensure that the most applicable standard or guideline was used for a particular route of exposure, a hierarchy was developed and followed to select the most appropriate value. The medium of concern at Cadillac is groundwater, and ingestion is the only route of exposure which is quantitatively assessed. For oral exposure to noncarcinogens, the following values were used in the order listed below:

(1) U.S.EPA Reference Dose (RfD)

The reference dose (RfD) is defined as the estimated amount of toxicant in mg/kg body weight (bw)/day for a 70 kg person which is not anticipated to result in adverse effects after lifetime chronic exposure. The degree of uncertainty is one order of magnitude or more.

Using relevant toxicologic data for toxic chemicals (noncarcinogens), the U.S.EPA has established no observed adverse effect levels

TABLE 7-3
DOSE-RESPONSE AND CHEMICAL-SPECIFIC ADAR TABLE
HEALTH ADVISORY (ng/l)

COMPOUND	HEALTH ADVISORY (ng/l)										MDC (ng/l)		
	RfD (ng/kg/day)	UNIT CANCER RISK (ng/kg/day)-1	MCL (ng/l)	MCLB (ng/l)	1-DAY (10 kg)	10-DAY (10 kg)	LONGER TERM (10 kg)	(70 kg)	LIFETIME (70 kg)	Water & Fish Ingestion	Fish Ingestion Only	Drinking Water Ingestion Only	
	ORAL	ORAL											
ACETONE	0.1	---	---	---	---	---	---	---	---	---	---	---	
CHRONIUM	3E-3 (VI)	NA (VI)	0.05	0.12	1.4	1.4	0.24	0.04	0.17	0.05	---	0.05	
CYANIDE	2.90E-2	---	---	---	0.22	0.22	0.22	0.75	0.75	0.200	---	0.200	
1,2-DICHLOROETHANE	---	9.10E-2	0.005	0	0.74	0.74	0.74	2.60	NA	9.4E-4	0.243	---	
1,1-DICHLOROETHYLENE	9.00E-3	9.00E-1	0.007	0.007	2.0	1.0	1.0	3.5	0.007	3.3E-3	1.05E-3	3.3E-3	
TRANS-1,2-DICHLOROETHYLENE	---	---	---	0.07	2.72	1.0	1.0	3.5	0.35	---	---	---	
METHYLENE CHLORIDE (DICHLOROETHANE)	6E-2	7.3E-3	---	---	13.3	1.9	---	---	---	1.9E-4	0.0157	1.9E-4	
TETRACHLOROETHYLENE	1E-2	3.10E-20	---	0	---	34.0	1.94	6.0	---	0E-4	0.05E-3	0E-4	
TOLUENE	3E-1	---	---	2.0	10.0	6.0	---	---	10.0	14.3	424	15	
1,1,1-TRICHLOROETHANE	0.6E-2	---	0.2	0.2	140.0	35.0	35.0	125.0	1.0	10.4	1030	---	
TRICHLOROETHYLENE	---	1.1E-2	0.005	0	---	---	---	---	0.26	2.7E-3	0.0007	2.0E-3	
CHLOROFORM	1E-2	0.1E-2	---	---	---	---	---	---	---	1.9E-4	0.0157	1.9E-4	

Footnote: 1Chloroform and tetrachloroethylene are presently being reevaluated by CAS (IRIS).

(NOAELs); no observed effect levels (NOELs); and lowest observed adverse effect levels (LOAELs). By using these values and incorporating uncertainty factors, a reference dose (RfD) is calculated. The toxic endpoint chosen for calculating RfDs is the most sensitive effect seen in a test animal. RfDs have also been developed for the noncarcinogenic effects of carcinogens. These values do not provide information on the protection from their carcinogenic effects. RfDs are developed by the U.S.EPA Environmental Criteria and Assessment Office (ECAO) Cincinnati, Ohio.

(2) U.S.EPA Maximum Contaminant Level Goal (MCLG) and Maximum Contaminant Level (MCL)

Pursuant to section 1412 of the Safe Drinking Water Act, U.S.EPA has promulgated drinking water goals (MCLGs) and drinking water standards (MCLs) for certain organic and inorganic substances.

Maximum Contaminant Level Goals (MCLGs) are non-enforceable health goals and are based on toxicological endpoints. They are set at a level at which no known or anticipated adverse effects on the health of persons occur and which allows an adequate margin of uncertainty. For contaminants for which no safe threshold is known to exist (i.e., for carcinogens), the MCLG is set at zero as a health goal.

Maximum contaminant levels (MCLs) specify the maximum permissible level of a contaminant in water which is used as a public water system. MCLs are enforceable regulatory requirements, which must be met when evaluating the availability and cost of treatment techniques. MCLs are based on MCLGs, taking costs and other factors into consideration.

(3) U.S.EPA Health Advisories

The U.S.EPA Office of Drinking Water has developed Health Advisories (HA) to describe contaminant levels in drinking water at which adverse effects would not be anticipated to occur. Health Advisory numbers are developed from data describing noncarcinogenic endpoints of toxicity. They do not incorporate potential carcinogenic risks from exposure.

Health Advisories are based upon the identification of adverse health effects associated with the most sensitive and meaningful non-carcinogenic endpoint of toxicity (Health Advisory, Office of Drinking Water, 30 September 1985). The general formula for developing HA values is as follows:

$$\frac{(\text{NOEL or LOEL}) (\text{Body Weight})}{(\text{Uncertainty Factor}) (\text{Assumed Consumption of Water})} = \text{Health Advisory}$$

HA values are derived for 1-day, 10-day, longer-term and lifetime exposure when applicable information is available, and are based on a child weighing 10 kg and drinking 1 liter of water per day or an

adult weighing 70 kg and drinking 2 liters of water per day. (Longer-term HA values are developed for adults and children, life-term HA values are developed for adults).

(4) U.S.EPA Ambient Water Quality Criteria (AWQC)

Under Section 304(a)(1) of the Clean Water Act, the U.S.EPA is required to develop ambient water quality criteria to protect against adverse health effects from lifetime ingestion of drinking water and aquatic organisms. These criteria are to reflect the latest scientific knowledge (at the time of promulgation) on the identifiable effects of pollutants in surface water and groundwater on public health and welfare, aquatic life, and recreation. AWQCs are not legally enforceable. However, the Superfund Amendments and Reauthorization Act (SARA) requires that remedial actions must at least attain AWQC when they are relevant and appropriate to a site.

Two human health criteria have been derived: 1) a criterion which considers that people will drink 2 liters of water daily and eat a daily average of 6.5 grams of fish, and 2) a criterion which considers eating fish only and not drinking water.

Where human health criteria are set for carcinogens or suspected carcinogens, the value corresponds to carcinogenic risks of one in one million for people who eat an average of 6.5 grams of fish daily for 70 years.

7.3.3 Chemical-Specific ARARs Comparison

The potential ARARs which currently exist for the assessment of public health are the final promulgated Maximum Contaminant Levels (MCLs), Maximum Contaminant Level Goals (MCLGs), Ambient Water Quality Criteria (AWQC), National Ambient Air Quality Standards (NAAQs), and any duly promulgated and consistently applied State standards. NAAQs are not applicable at Cadillac as inhalation will not be quantitatively assessed. Contaminant concentrations which have been detected to date in Cadillac groundwater are compared to MCLs, MCLGs, and AWQC in this section.

Table 7-4 is a comparison of current (June and August 1987) contaminant concentrations in the four major plumes to the relevant health-based chemical-specific ARARs identified above. The value of the ARAR is noted, and if a contaminant concentration exceeds that value in a particular plume it is marked with an "X". As can be seen, each plume contains several chemicals at concentrations above ARARs. It should also be noted that benzene also exceeded its MCL (0.005 mg/l) in the upper aquifer the one time it was detected (see Section 7.2). Trichloroethylene (TCE) exceeds all three relevant ARARs in all four plumes. Contaminant levels in all four plumes currently exceed applicable health-based requirements.

TABLE 7-4
CADILLAC AREA GROUNDWATER STUDY: HEALTH-BASED CHEMICAL SPECIFIC ARARS COMPARISON

COMPOUND	ARAR (NG/L)			EXCEEDANCE OF ARARS (MAXIMUM VALUE DETECTED)			
	NCL	NCLB	AMOC*	REXAIR	UPPER AQUIFER	INTER. AQUIFER	NORTHERNAIRE PLINE
1,1,1-TRICHLOROETHANE	: 0.2 :	: :	: :	: :	: X :	: X :	: X :
	: :	: 0.2 :	: :	: :	: X :	: X :	: X :
	: :	: :	: 18.4 :	: :	: X :	: :	: :
ACETONE	: --- :	: :	: :	: :	: :	: :	: :
	: :	: --- :	: :	: :	: :	: :	: :
	: :	: :	: --- :	: :	: :	: :	: :
TOLUENE	: --- :	: :	: :	: :	: :	: :	: :
	: :	: 2.0 :	: :	: :	: X :	: :	: :
	: :	: :	: 14.3 :	: :	: :	: :	: :
TRANS-1,2-DICHLOROETHENE	: --- :	: :	: :	: :	: :	: :	: :
	: :	: 0.07 :	: :	: :	: :	: X :	: :
	: :	: :	: --- :	: :	: :	: :	: :
CHLOROFORM	: --- :	: :	: :	: :	: :	: :	: :
	: :	: --- :	: :	: :	: :	: :	: :
	: :	: :	: 0.00019 :	: X :	: X :	: X :	: :
1,1-DICHLOROETHYLENE	: 0.007 :	: :	: :	: :	: X :	: X :	: X :
	: :	: 0.007 :	: :	: :	: X :	: X :	: X :
	: :	: :	: 10.000033 :	: :	: X :	: X :	: X :
1,2-DICHLOROETHANE	: 0.005 :	: :	: :	: :	: X :	: :	: :
	: :	: 0 :	: :	: :	: X :	: :	: :
	: :	: :	: 0.00094 :	: :	: X :	: :	: :
METHYLENE CHLORIDE	: --- :	: :	: :	: :	: :	: :	: :
	: :	: --- :	: :	: :	: :	: :	: :
	: :	: :	: 0.00019 :	: X :	: X :	: X :	: :
TETRACHLOROETHYLENE	: --- :	: :	: :	: :	: :	: :	: :
	: :	: 0 :	: :	: :	: X :	: X :	: X :
	: :	: :	: 0.0008 :	: :	: X :	: X :	: X :
TRICHLOROETHYLENE	: 0.005 :	: :	: :	: X :	: X :	: X :	: X :
	: :	: 0 :	: :	: X :	: X :	: X :	: X :
	: :	: :	: 0.0028 :	: X :	: X :	: X :	: X :
CHROMIUM (HEXAVALENT)	: 0.05 :	: :	: :	: :	: :	: :	: X :
	: :	: 0.12 :	: :	: :	: :	: :	: X :
	: :	: :	: 0.05 :	: :	: :	: :	: X :
CYANIDE	: --- :	: :	: :	: :	: :	: :	: :
	: :	: --- :	: :	: :	: :	: :	: :
	: :	: :	: 0.2 :	: :	: :	: :	: :

* AMOC are adjusted for drinking water exposure only (U.S.EPA, 1986) or calculated for water + fish ingestion if drinking water exposure value does not exist.

7.4 EXPOSURE ASSESSMENT

The purpose of this exposure assessment is to identify the populations and environments which are at risk of exposure to chemicals in the Cadillac area contaminated groundwater and the major pathways through which they could come into contact with the chemicals. It has been divided into three sections: (1) Fate and Transport Assessment; (2) Public Health Exposure Assessment; and (3) Environmental Exposure Assessment.

The fate and transport section describes the various contaminant plumes that are assessed in this exposure assessment. Various groundwater modeling techniques are then utilized to predict future chemical concentrations at downgradient receptor wells. This section summarizes information that is presented in much greater detail in Appendix R, Fate and Transport Calculations, and focuses the analysis on the elements that are necessary for the assessment of exposure and risk.

In the public health exposure assessment, the human populations that may live in or use the area around the site are identified. Then the potential routes of exposure to the contaminated groundwater are evaluated to identify the most significant routes. Although there are several routes through which people could be exposed to groundwater (such as bathing and showering), the route of major concern is lifetime ingestion. Quantitative estimates of the exposure dose levels are developed for this route, both at present contaminant levels in on-site wells and at modeled future downgradient concentrations. Hypothetical scenarios of exposure encompassing a range of exposure conditions are developed for each plume in order to attempt to model both conservative and the most probable dose levels that people could now or eventually be exposed to.

The environmental exposure assessment first identifies the environments and environmental receptors in the Cadillac area that could potentially be exposed to chemicals from the site. Then the potential for significant environmental exposure at the site is qualitatively discussed.

7.4.1 Fate and Transport

This section summarizes the fate and transport of contaminants (VOCs and hexavalent chromium) in the various groundwater plumes present in the study area, and an evaluation of the presence of volatile organic compounds remaining in soils at the Kysor facility. During the course of earlier investigations, elevated levels of inorganics such as calcium, sodium, potassium, and magnesium were also found in the shallow and intermediate aquifers. These contaminants are believed to have been caused by salting of roads for ice and dust control. Although they affect water quality, they are not believed to be associated with any manufacturing facility.

The first subsection of this fate and transport assessment presents some general aspects of the contaminant plumes identified at the site. The remaining subsection considers each identifiable plume and zones of higher concentration within larger plumes (e.g., the chromium plume within a larger and more diffuse volatile plume). For each plume, the estimates of concentrations calculated in Appendix S are summarized as a function of time at selected

receptor locations at the boundary of the industrial park (along the plume pathway) and, in two instances, at the municipal well field.

7.4.1.1 General Considerations. Several major factors control the migration of contaminants in the environment. In this study, the principal focus has been on groundwater, although in one area (at Kysor) some consideration has also been given to contaminants in soils. The major factors controlling migration include:

- o the hydrogeology (including the unsaturated zone),
- o chemical specific factors (e.g., solubility, adsorption potential, volatility, cosolvation, colloidal transport, and density),
- o size, mode, date, and frequency of contaminant release, and
- o mobilizing factors (e.g., precipitation, rusting of drums).

The geology and hydrogeology of the site have been described in the RI conducted for the site as well as in prior assessment reports. These reports have been supplemented by the more recent pump test performed by Keck for MDNR, and the groundwater flow model for the site prepared by MDNR. Chemical specific factors are available from the literature for the groundwater contaminants. The size, mode, and frequency of contaminant release may be known in some cases, but generally must be inferred from the chemical distributions in the groundwater and soils. Mobilizing factors such as leaching by precipitation, or repeated spill events, may be apparent or may be deduced from the distribution of the contaminants in the groundwater, e.g., a continuous plume or a series of pulses.

The availability of data and the current state of the art for contaminant transport modeling rarely warrant the use of sophisticated transport models. Within the range of accuracy attainable through numerical analyses and necessary for risk assessment evaluation, the application of appropriate analytical expressions describing solute transport is more common. Variability of parameters within the system are treated by assuming an average or conservative worst case situation which permits the use of constants for the controlling parameters in the equations.

Contaminants in the groundwater in the Cadillac Industrial Park appear to be present as the result of single or multiple slug source inputs, such as at Northernnaire and Kysor, respectively, or as an areal source, such as from the shallow to the intermediate aquifers in an area near Northernnaire. The equations used to model these two types of contaminant sources, the instantaneous slug point source equation and the vertical and horizontal spread (VHS) model (reference: 50 FR 48897), can be found in Appendix R.

Evaluation of the potential for impact of residual volatile organic contaminants in soil was done using EPA's organic leaching model (OLM) for impacts on groundwater while estimating losses of volatiles to the atmosphere by a model described in the EPA guidance "Superfund Exposure Assessment Manual" (OSWER directive 92855-1, Sept. 22, 1987). The OLM model (51 FR 41088) and the volatilization model are also described in greater detail in Appendix R.

7.4.1.2 Plume Analyses. In the following sections, the Appendix R plume analyses are summarized by identifying the appropriate receptor locations, and summarizing the estimated time of impact to receptor wells and concentrations in each plume as it passes by the receptor location. Locations of hypothetical receptor wells for each plume can be found on Figure 7-1. In two instances, the Northernnaire hexavalent chromium plume and the Rexair volatile organic plume, estimates of the potential for impact on the municipal water supply are also made. For the purposes of this section only a brief summary of the parameters necessary to complete the exposure assessment are included. Appendix R contains a thorough and detailed explanation of how these parameters were estimated.

The following major plumes are addressed: (1) a VOC plume in the shallow aquifer; (2) the Rexair plume in the intermediate aquifer; and (3) the general VOC plume in the intermediate aquifer. The lesser plumes associated with Four Star and Northernnaire are considered as part of the intermediate aquifer plume. The Four Winns plume merges with the shallow Kysor plume and also possibly with some small contribution from the Ingraham property. While some groundwater data are available downgradient of Ingraham, and some remediation of soils on the Ingraham property is reported, a complete evaluation of the Ingraham property is not possible due to the lack of soils data. This report recommends soils testing at the Ingraham property, but does not presently consider the suspected Ingraham plume further since it appears that potential concentrations due to this plume are less in that vicinity than in the intermingling Kysor and Four Winns plumes.

The Shallow Aquifer Plumes

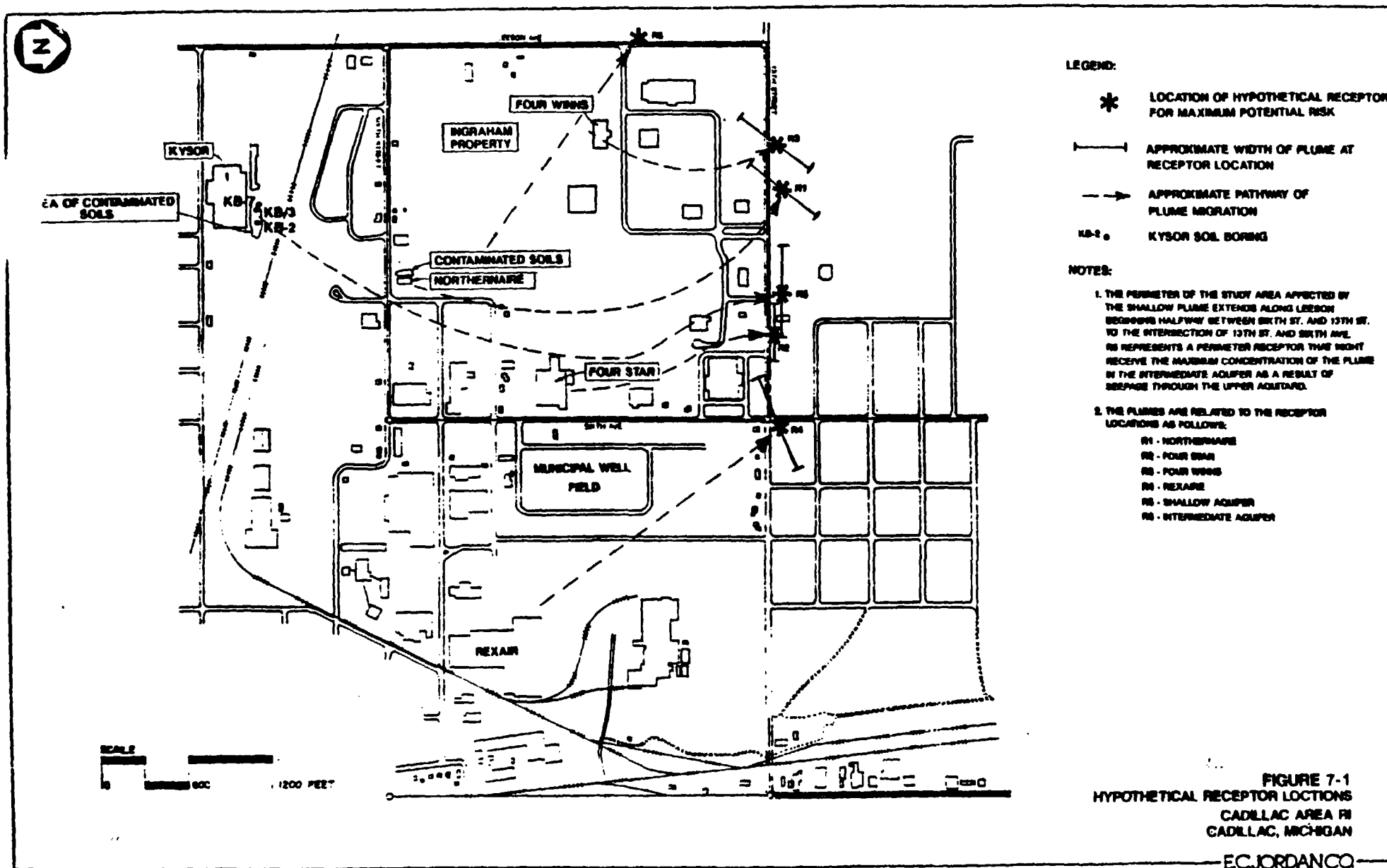
This section considers the following sources of shallow aquifer contamination:

- o The residual contaminants in soils at Kysor;
- o the general shallow Kysor plume; and
- o the Four Winns plume.

Evaluation of Soils at Kysor -- Based on MDNR and the Cadillac RI soil and groundwater data for this assessment, the following observations have been made:

1. Two areas of contaminated soils remain, one near boring KB-3 and KB-7, and the other near KB-2.
2. The area near KB-3 and KB-7 is estimated to contain total volatiles at a concentration of about 400 mg/kg, yielding a solvent mass in soils of about 900 pounds or 410 kg.
3. The area near KB-2 contains an estimated 3 pounds of total solvents.
4. The distribution of contaminants is about 67% xylenes, 10% each of toluene, TCE and ethylbenzene, and about 2 to 3% of 1,1,1-TCA.

A model was run to assess the rate of loss of the solvent to both leaching and volatilization. The EPA organic leaching model (OLM) and a volatilization



model from EPA's site assessment guidance manual were applied for both TCE and xylene. These models were run to assess the relative potential for transport by each mechanism, and as a check with observed concentrations in groundwater. The modeling suggests that, due to the depth of the contamination, the loss rate of solvents would be only about 2% per year, mostly by volatilization (more than 99% of the loss). The OLM predicted xylene concentrations of about 600 to 700 ppb in the leachate, while in SLW-25 (the most contaminated monitoring well), the xylene concentration was about 1,000 ppb. While some dilution through dispersion would occur between the source and SLW-25, the numbers are reasonably close.

The volatilization model was run for a 5 year period and indicated only slight reductions in the soil concentration, and a total loss from the soil of about 10% of the originally estimated mass. This has implications for a potential long-term impact on groundwater due to the presence of the remaining solvents in soils. The contribution of soils in the Kysor area to the groundwater contamination will be addressed qualitatively in the remainder of this risk assessment.

Shallow Plume (Originating in the Vicinity of Kysor) -- The slug point source equation was used to assess the shallow plume originating in the vicinity of Kysor. Contaminant concentrations were modeled to a theoretical receptor location on the projected center line of the plume at 13th Street (see Figure 7-1). The plume is assumed to stay in the shallow aquifer until it approaches 13th Street, where, locally, the upper aquitard terminates, and then to migrate according to the flow in the intermediate aquifer. It is predicted that the plume would arrive at the receptor well at approximately the year 2000, and to leave the vicinity of the well approximately 15 years later. For the distribution of contaminants currently found in the shallow plume, the predicted concentration for the contaminants of concern at 13th St. would be:

Chemical	Maximum Concentration (ppm)	10-Year Average (2000-2010) (ppm)	15-Year Average (2000-2115) (ppm)
1,1-DCE	0.011	0.005	0.003
1,1,1-TCA	1.066	0.487	0.335
TCE	2.600	1.190	0.818
PCE	0.012	0.006	0.004
Toluene	0.053	0.025	0.017

When considering the maximum concentrations predicted above, and in each of the subsequent plume analyses, it is important to note that these values represent point concentrations. A well would actually draw from an interval and thus the resultant concentration would be less than the maximum value.

Four Winns Plume -- The Four Winns plume was evaluated in the same manner as the general shallow plume, but with a hypothetical receptor well location further west on 13th Street (see Figure 7-1). It is estimated that the plume would reach the receptor well sometime in 1988, and pass the vicinity of the

well approximately 7 years later. The predicted concentrations for the individual contaminants of concern arriving at 13th Street are:

Chemical	Maximum Concentration (ppm)	8-Year Average (1987-1995) (ppm)
Acetone	1.792	0.417
1,1,1-TCA	2.139	0.493
TCE	0.248	0.057
PCE	0.089	0.020
T-1,2-DCE	0.005	0.002
1,2-DCA	0.260	0.060

The Rexair Plume

The Rexair plume is located in the intermediate aquifer, upgradient of the municipal wellfield, and was evaluated using the instantaneous slug source equation. Two receptor locations were considered: (1) the hypothetical supply well on 13th Street, and (2) the municipal well aquifer.

It was estimated that some of the plume would escape being drawn down through the lower aquitard and would continue on toward 13th Street. It was estimated that the plume would reach the 13th Street receptor well about 1992, and would pass the well about 15 years later. For the distribution of contaminants currently found in the Rexair plume, the resultant concentration for the individual contaminants of concern arriving at 13th St. would be:

Chemical	Maximum Concentration (ppm)	10-Year Average (1992-2002) (ppm)	15-Year Average (1992-2007) (ppm)
TCE	53.000	22.000	15.130
1,1,1-TCA	0.006	0.003	0.002
Toluene	0.015	0.007	0.005
T-1,2-DCE	0.017	0.008	0.005
Chloroform	0.002	0.0009	0.0006
MeCl	0.270	0.113	0.077

In the municipal supply well TCE concentrations could eventually be about 100 ppb. Seepage of the plume through the aquitard would take about two years. This estimate is based on limited data and could vary on actual thickness and quality of the aquitard in places, and the pumping stress which creates the vertical gradient. The date when the plume might pass through is unknown as it also depends on when the plume actually comes into contact with the aquitard. Furthermore, investigation is expected in defining this plume by Rexair's consultant. Concentrations of other chemicals in the plume were not calculated.

ed, as they are already so low in comparison to TCE that they would be negligible by the time that they descend through the intermediate aquifer, pass through the aquitard, and migrate to the municipal supply well.

Intermediate Aquifer Plumes

In this section, the contaminants present in the intermediate aquifer west of 6th Avenue are considered. First, the generalized plume, largely due to release at Kysor, is considered. Then, concentrated areas represented by Four Star and Northernair releases are presented.

Generalized Intermediate Aquifer Plume (Kysor and Others) --The Vertical and Horizontal Spread (VHS) model was used to predict future concentrations in the intermediate aquifer plume, since the zone of introduction is not a point slug source but a constant source over a large area. The receptor for this plume is at Leeson Avenue near the intersection of Frisbee Street (see Figure 7-1). This is the point where effects may be noted earliest and at highest concentration. The plume will reach the hypothetical receptor well in about 1992, and with no source remediation will decrease at a rate of approximately 2% per year. Based on current concentrations in the intermediate aquifer, the following concentrations at the receptor well were estimated:

Chemical	Maximum Concentration (ppm)	Average Concentration (ppm)
-----	-----	-----
TCE	2.140	1.200
T-1,2-DCE	1.200	0.665
MeCl	0.070	0.038

Some seepage of the VOC plume through the lower aquitard into the municipal water aquifer is possible. Part of this VOC plume also intermingles with the chromium plume. Resultant concentrations of VOCs in the municipal well would be in the same range as that calculated for the chromium (1 ppb). Due to the numerous uncertainties in the model and the low resultant concentrations, the potential for impact on the water supply is considered insignificant and not assessed further here.

Four Star Plume -- The contribution of the Four Star facility to the intermediate aquifer plume cannot be completely separated from that of the generalized plume. However, it is focused on more closely in this section to the extent possible.

The slug source equation was used to evaluate this plume, as its shape suggests that this equation would be applicable. The plume will reach the hypothetical receptor well on 13th Street in approximately 1990, and pass the well approximately 15 years later. For the distribution of contaminants currently found in the Four Star plume, the predicted concentrations for the contaminants of concern arriving at 13th Street are:

Chemical	Maximum Concentration (ppm)	10-Year Average (1990-2000) (ppm)	15-Year Average (1990-2005) (ppm)
TCE	6.730	2.450	1.650
PCE	1.740	0.585	0.437
1,1,1-TCA	0.132	0.049	0.034

Northernaire Chromium Plume -- The hexavalent chromium plume was defined within the intermediate aquifer in the central portion of the industrial park during the supplemental remedial investigation conducted for the Northernaire CERCLA site. While a shallow plume to the northeast of the facility was also detected, it is believed that this detection, in one well, is a minor plume of low concentration which is being quickly removed by groundwater flow from the shallow aquifer.

The analysis of the Northernaire chromium plume was performed using the slug source equation. It was estimated that the plume would reach the hypothetical receptor well on 13th Street approximately in 1994, and pass the well approximately 20 years later. The resulting maximum and average hexavalent chromium concentrations would be:

Chemical	Maximum Concentration (ppm)	8-Year Average (1994-2002) (ppm)	20-Year Average (1994-2014) (ppm)
Chromium	0.600	0.294	0.122

The potential for the chromium plume to impact the municipal water supply was also evaluated. A simple mass balance approach was used with the rate of chromium potentially received by a municipal supply well calculated by assuming a uniform seepage of the plume through the lower aquitard. The analysis suggested that the plume could pass through the aquifer in about three years assuming no retardation or degradation of the hexavalent chromium. Some reduction in concentration could take place due to the organic content of the lower aquitard. However, to be conservative, no such transformation was assumed. The analysis predicted a concentration of 1 microgram per liter of hexavalent chromium in the water supply well. This is well below the 50 microgram per liter MCL (drinking water standard). Therefore, the impact of the chromium plume on the municipal wellfield is not assessed further in this risk assessment.

7.4.2 Public Health Exposure Assessment

The public health exposure assessment consists of two subsections: (1) Exposed Population Analysis; and (2) Exposure Pathway Assessment.

7.4.2.1 Exposed Population Analysis. This section discusses the general characteristics of the human populations in the vicinity of the Cadillac area

groundwater study and the activities which would promote contact with chemicals from the study area.

The town of Cadillac had a population of 10,199 in the 1980 U.S. Census (Table 7-5). The current population of Cadillac is estimated to be 11,000-12,000 with a 40% seasonal increase between May and September (Maher, 1988). Most of the seasonal increase is retirees returning for the summer.

The year-round population of Cadillac is predicted to increase due to anticipated economic growth. The lake front areas were recently developed, the downtown is currently undergoing renovation, and the Cadillac Chamber of Commerce is actively recruiting industry for a new industrial park. It has been estimated the population of Cadillac will increase 20% over the next 25 years as a result of this growth (Cadillac Chamber of Commerce).

The site under investigation includes the Cadillac industrial park which is located along the northern boundary of the city. Several private residences, including a trailer park, are located in the industrial park. Another residential area, the North Park subdivision, is located adjacent to the northern boundary of the industrial park and within the study area. Prior to 1986 most of these residences used drinking water from private wells which tapped the intermediate aquifer below the site.

Cadillac is the county seat of Wexford county. Much of the land immediately surrounding Cadillac is used for Christmas tree farming. Within the county, land is used primarily for agriculture, outdoor recreation and tourism.

The city of Cadillac currently uses the deep aquifer under the site as a source of drinking water. With the anticipated population increase, an even greater demand will be placed on the deep aquifer with an increased number of persons potentially exposed.

7.4.2.2 Exposure Pathway Assessment. Four components are necessary to define a complete exposure pathway: (1) a source of chemical release to the environment; (2) an environmental transport medium and transport mechanism (e.g. solubilization and mobilization in groundwater); (3) an exposed population and point of exposure; and (4) a route of exposure for the receptor at the exposure point. The first component was defined earlier in the remedial investigation, and several sources were identified. The second component was discussed briefly in Section 7.4.1, the fate and transport analysis, and more thoroughly in Appendix R. Component three was discussed in Section 7.4.2.1. The fourth component of the exposure pathway is discussed below. By combining information on all four components, the extent of exposure can be estimated. For the Cadillac study area, the only route of exposure which is quantitatively assessed is ingestion of contaminated groundwater.

Exposure dose levels were calculated for hypothetical present and future potential ingestion of groundwater. Two exposure scenarios were developed for each plume to represent both the most probable exposure conditions and the upper bound conservative exposure conditions (referred to as a realistic worst case exposure). Exposure assumptions (such as weight of exposed individuals, amount ingested, etc.) were obtained using U.S. EPA guidance and the current

TABLE 7-5
POPULATION OF CADILLAC
U.S. CENSUS 1980

<u>Age Group</u>	<u>Male</u>	<u>Female</u>	<u>Total</u>	<u>Percent</u>
< 5 years	409	354	763	7%
5-14 years	798	773	1571	15%
15-64 years	2908	3381	6289	62%
≥ 65 years	<u>569</u>	<u>1007</u>	<u>1576</u>	15%
TOTAL	4684	5515	10,199	

From Table 32, 1980 U.S. Census General Population Characteristics -- Michigan

literature. Each of the parameters were varied to provide a range of hypothetical exposure conditions in order to reduce the uncertainty of the subsequent risk estimates. While these scenarios may not necessarily predict actual exposures to the groundwater, they provide a range of possible doses. In the following subsections, groundwater exposure is first discussed in a general manner, and then specific exposure pathways for the various plumes are evaluated for the relevant chemicals of concern.

Groundwater Exposure Pathways

There are several pathways through which people could come into contact with contaminated groundwater. Since there are no surface water discharge points, the relevant pathways are all related to use of the groundwater for domestic household purposes. Currently, there are no domestic supply wells in use that are intercepting the contaminated groundwater plumes in the study area. However, there are no restrictions on the use of the contaminated aquifers as drinking water supplies, and, in fact, the North Park subdivision has used wells screened in the intermediate aquifer as a drinking water supply in the past. Furthermore, the municipal supply wells for the town of Cadillac are within the study area and, although they are screened in a deeper aquifer, they may be impacted by contaminants from the Rexair plume in the intermediate aquifer at some time in the future. This exposure assessment addresses the present and future potential for impact to hypothetical groundwater receptors, both within the current boundaries of the plumes and immediately downgradient at the boundary of the industrial park, as well as the potential for the Rexair plume to impact the municipal supply wells.

Exposure dose levels associated with the ingestion of contaminated groundwater from each plume are estimated utilizing the following equation:

$$\text{Body Dose Level (mg/kg/day)} = C \times A \times T \times \frac{1}{\text{BW}} \times \frac{F}{365 \text{ days}} \times \frac{Y}{70 \text{ years}}$$

Where:

C = Concentration in groundwater (mg/l)

A = Amount of water ingested (l/day)

T = Toxicokinetic Factor (100% gastrointestinal absorption assumed)

BW = Body Weight (kg)

F = Frequency of Contact (days; 365 per year assumed)

Y = Exposure in a lifetime (years)

For noncarcinogens, the last term in the equation drops out, since the dose does not have to be averaged over a lifetime.

Two hypothetical exposed populations are evaluated: children (age 6 to 15) who ingest one liter of water per day for 10 years, and adults who ingest two liters of water per day for their entire lifetime. This was done for several reasons. First, by separating out childhood exposures, those chemicals which pose a risk even when less-than-lifetime ingestion is assumed are highlighted. Second, children are, as a group, more sensitive to chemical exposure than adults. Exposure may occur for a shorter duration, but it may actually have a

greater effect because of greater sensitivity, or because of exposure to a cancer-causing agent with a long latency period.

There are several other routes through which receptors could come into contact with site-derived chemicals through household use, including dermal contact and inhalation while bathing and showering. While these routes may add to the overall risk from use of the contaminated groundwater, they are not as significant as the ingestion of drinking water. Therefore, they are addressed only qualitatively as an additional source of risk in Section 7.5 (Risk Characterization). The following paragraphs explain the approach taken for the present and future assessment of potential exposure to the groundwater plumes defined in the Fate and Transport section, i.e. the shallow aquifer plume, the Rexair plume, and the intermediate aquifer plume, as well as the smaller plumes that are associated with each aquifer. The exposure parameters, calculations and resulting body doses associated with each of these plumes can be found in Appendix S.

Shallow Aquifer Plume

Evaluation of present exposure to the shallow aquifer plume utilizes the average (most probable) and maximum (worst case) contaminant concentrations currently found in on-site wells in the large generalized plume. Although no one is currently known to be using the groundwater in the shallow aquifer as a drinking water source, this route is assessed in order to establish baseline conditions.

The assessment of the future potential exposure of the shallow aquifer as a drinking water source utilizes the modeling efforts summarized in Section 7.4.1. The shallow plume originating in the vicinity of Kysor is assessed separately from the Four Winns plume; both are evaluated at hypothetical receptor wells on 13th Street (the nearest probable location of receptor wells and the location with the greatest probable concentrations). Concentrations would decrease further from the source due to dilution in the aquifer.

The shallow Kysor plume will impact the receptor well for approximately 15 years (2000-2015). Future potential risk to children (35 kg children, age 6-15, who ingest 1 liter of water per day) is assessed for the most probable case by using the average concentration over the 10 years that the contaminant concentrations are highest at the receptor well. Worst case exposure assumes that water containing the maximum contaminant concentrations is ingested over the same 10 year period. Adult exposure (70 kg adults who ingest 2 liters of water per day for a lifetime) is assessed by using the average concentration over the entire 15 years that the plume is at the receptor well. Worst case exposure is assessed by using the maximum concentration and assuming that water is ingested for a lifetime.

The Four Winns plume will impact the hypothetical receptor well from approximately 1987 through 1995. Since the plume will be at the receptor well for about 8 years, most probable future exposure to both children and adults is assessed by using the average contaminant concentrations over that period. Worst case exposure is assessed by using the maximum concentration predicted at the receptor well and assuming that children would ingest the groundwater over

the 10 years that they are 6-15 years old, and adults would ingest the groundwater over their entire lifetime.

The Rexair Plume

The Rexair plume is assessed in a similar manner to the shallow plume. Present exposure is assessed using the average and maximum contaminant concentrations now in the plume in order to establish baseline conditions. Assessment of future potential exposure is performed at two different receptor locations: (1) a hypothetical supply well on 13th Street; and (2) the municipal supply wells.

The plume is predicted to affect the 13th Street receptor well from about 1992 through 2007. As with the shallow Kysor plume, future potential exposure to children (35 kg children, age 6 to 15, who ingest 1 liter of water per day) is assessed by using the highest 10 year average concentration at the receptor well for the most probable case. Worst case exposure is assessed by assuming that water which contains the maximum contaminant concentrations is ingested over the same 10 year period. Adult exposure (70 kg adults who ingest 2 liters of water per day for a lifetime) is assessed by using the average concentration over the entire 15 years that the plume is at the receptor well. Worst case exposure is assessed by using the maximum concentration and assuming that water that is contaminated at that level will be ingested for a lifetime. Lifetime ingestion is a very conservative assumption which probably overestimates the actual risk, since the plume will only be at the well for about 15 years. However, this duration was used to assess the worst case exposure possible, so that the upper bound of risk could be estimated.

Seepage of the Rexair plume through the aquitard into the municipal well aquifer would take in excess of two years, and the maximum TCE concentration would eventually be about 100 ppb. Future potential risk to individuals who obtain their drinking water from the municipal well supply is assessed by assuming that water contaminated with TCE at this level would be ingested for a lifetime.

Intermediate Aquifer Plumes

As with the shallow and Rexair plumes, present exposure to the contaminants in the intermediate aquifer is assessed using the average and maximum contaminant concentrations detected at the present time in on-site monitoring wells in order to establish baseline conditions. The large generalized plume is assessed separately for the chromium, cyanide, and intermingled VOCs in the Northernnaire plume, as the Northernnaire plume is found much deeper in the intermediate aquifer. Future potential exposure is assessed separately for the generalized intermediate aquifer plume, the Four Star plume, and the Northernnaire chromium plume.

The generalized intermediate aquifer plume comes from a relatively constant source (seepage from the shallow aquifer) rather than a point slug source. Therefore it is estimated that the plume would reach a hypothetical receptor well on Leeson Avenue near Frisbee Street in about 1992, and with no source remediation would decrease at a rate of about 2% per year (see Section 7.4.1.2). Future potential exposure is estimated using the maximum contaminant

concentrations predicted to assess worst case conditions, and average contaminant concentrations for most probable conditions (see Appendix S). Duration of exposure is 10 years for children and lifetime for adults.

The Four Star plume will reach a hypothetical receptor well on 13th Street in 1990, and pass the well in 2005. Future potential exposure to receptors using this well is assessed in a similar manner to the shallow plume and the Rexair plume, i.e. future potential exposure to children is assessed by using the average concentration over the 10 years that the contaminant concentrations are highest at the receptor well for the most probable case. Worst case exposure is assessed by assuming that water which contains the maximum contaminant concentrations is ingested over the same 10 year period. Adult exposure is assessed by using the average concentration over the entire 15 years that the plume is at the receptor well. Worst case exposure is assessed by using the maximum concentration and assuming that water that is contaminated at that level will be ingested for a lifetime.

The Northernnaire chromium plume is presumed to reach the 13th Street hypothetical receptor well in 1994, and to pass the well approximately 20 years later. As with the Four Star plume, most probable future potential risk to children is assessed by using the average chromium concentration over the 10 years that the concentration is highest at the receptor well. Worst case exposure is assessed by assuming that water which contains the maximum predicted chromium concentration is ingested over the same 10 year period. Adult exposure is assessed by using the predicted average concentration over the 20 years that the plume is at the receptor well. Worst case exposure is assessed by using the maximum chromium concentration and assuming that water that is contaminated at that level will be ingested for a lifetime.

7.4.3 Environmental Exposure Assessment

E.C. Jordan field biologists performed a site investigation in February, 1988 to identify potential ecological receptors of contamination at the Cadillac site. No aquatic ecosystems which might be receptors of contamination appear to exist at the site. However, terrestrial systems have been identified as potential ecological receptors of contamination. This exposure assessment contains a description of ecological receptors present at the site and an evaluation of potential exposure pathways.

7.4.3.1 Identification of Ecological Receptors. The primary ecological receptors at the Cadillac Industrial Park area are terrestrial plant communities and associated wildlife. The various environments and species potentially found at the site are described below.

The composition of terrestrial plant communities at the site reflects past logging activities, planting of ornamental species, reforestation efforts, and the dry, sandy soils present at the site (see Table 7-6). Most of the site is open, short-growth field, with grasses (Gramineae) and spotted knapweed (Centaurea maculosa) being the dominant plant species. Small trees and shrubs are scattered throughout many of the field areas. These include sweetfern (Comptonia peregrina), hawthorn (Crataegus sp.), witch hazel (Hamamelis virginiana), staghorn sumac (Rhus typhina), cherry (Prunus sp.), hack

Table 7-6--Plant Species Observed at the Cadillac Site

Scientific Name	Common Name	Abundance and Distribution
<u>Acer rubrum</u>	Red maple	Common tree in residential areas
<u>Acer saccharinum</u>	Silver maple	Occasional tree in residential areas
<u>Asclepias syriaca</u>	Common milkweed	Occasional herb in field areas
<u>Betula populifolia</u>	Gray birch	Occasional tree in residential areas
<u>Camelina microcarpa</u>	Smallseed falseflax	Common herb in field areas
<u>Centaurea maculosa</u>	Spotted knapweed	Dominant herb in field areas
<u>Comptonia peregrina</u>	Sweetfern	Occasional small shrub scattered in fields
<u>Cornus</u> sp.	Dogwood	Occasional tree in residential areas
<u>Crataegus</u> sp.	Hawthorn	Occasional small tree scattered in fields
Gramineae	Grasses	Common to dominant herbs in field areas
<u>Hamamelis virginiana</u>	Witch hazel	Occasional small tree scattered in fields; also dominant in small groves in some areas
<u>Oenothera biennis</u>	Evening primrose	Occasional herb in field areas
<u>Picea abies</u>	Norway spruce	Occasional tree in residential areas
<u>Pinus resinosa</u>	Red pine	Dominant tree in wooded (reforested) areas; occasional in fields
<u>Pinus strobus</u>	White pine	Common tree in residential areas; dominant tree before logging (Everett Bole, personal communication)
<u>Populus deltoides</u>	Eastern cottonwood	Occasional tree in residential areas
<u>Prunus</u> sp.	Cherry	Occasional small tree scattered in fields
<u>Pyrus melanocarpa</u>	Hack chokeberry	Occasional small tree scattered in fields
<u>Quercus rubra</u>	Red oak	Common tree in residential areas
<u>Rhus typhina</u>	Staghorn sumac	Occasional small tree scattered in fields
<u>Solidago</u> sp.	Goldenrod	Occasional herb in field areas
<u>Thuja orientalis</u>	Oriental arborvitae	Occasional tree in residential areas
<u>Verbascum thapsus</u>	Mullein	Occasional herb in field areas

chokeberry (Pyrus melanocarpa), and an occasional red pine sapling (Pinus resinosa).

Red pine has also been planted in certain areas as part of a reforestation/logging program, creating small sections of coniferous woodland. Because these stands of pine have not been properly managed, they have grown into a dense assemblage of tall, uniformly thin individuals.

Several species have been planted in or have colonized residential areas. These include white pine (Pinus strobus), Norway spruce (Picea abies), red oak (Quercus rubra), silver maple (Acer saccharum), oriental arborvitae (Thuja orientalis), gray birch (Betula populifolia), and red maple (Acer rubrum). Based on the composition of plant communities at the site, Jordan biologists performed a wildlife habitat assessment.

The purpose of the wildlife habitat assessment was to identify organisms which may inhabit, feed, or pass through the site. Given the presence of short-growth field with scattered trees or shrubs, coniferous woodlots, residential area plantings, and regional presence of wetlands, numerous species of mammals and birds are expected at the Cadillac site. Reptiles and amphibians are not expected because they generally require water or moist situations for breeding, which are not present at the site itself. The site lies outside the expected home range of reptiles and amphibians which inhabit nearby wetlands and rivers.

Numerous species of mammals and birds may exploit habitat resources at the Cadillac site, despite its relatively barren and industrial appearance (see Tables 7-7 and 7-8). Several species can utilize man-made structures. These include the chimney swift (Chaetura pelagica), the pigeon (Columba livea), the big brown bat (Eptesicus fuscus), the Norway rat (Rattus norvegicus), and the eastern chipmunk (Tamias striatus). Some birds may also make use of vegetation in residential areas, including the eastern bluebird (Sialia sialis), the common grackle (Quiscalus quiscula), the American robin (Turdus migratorius), and the mourning dove (Zenaida macroura). Song birds, such as the eastern meadowlark (Sturnella magna) and the American goldfinch (Carduelis tristis), may benefit from scattered trees and shrubs in field areas, which they use as singing perches.

The coniferous woodlots and short-growth fields offer habitat to small mammals such as the snowshoe hare (Lepus americanus) and the white-footed mouse (Peromyscus leucopus). These species may in turn serve as a food source for predators such as the red fox (Vulpes vulpes) and red-tailed hawk (Buteo jamaicensis).

Jordan contacted the Michigan Natural Features Inventory regarding the possible presence of rare, threatened, or endangered species of plants or animals or critical habitats at the Cadillac site. They have no data indicating that such species or habitats do in fact exist at the site.

7.4.3.2 Evaluation of Potential Exposure Pathways. Pathways through which terrestrial organisms may be exposed to chemicals include, in general, direct contact with surface soils, direct contact with wetland sediments, direct contact with or ingestion of wetland surface waters, and ingestion of biota

Table 7-7--Mammals Expected at the Cadillac Site

Scientific Name	Common Name	Species Habitat Present at the Cadillac Site
<u>Eptesicus fuscus</u>	Big brown bat	Abandoned man-made structures
<u>Erethizon dorsatum</u>	Porcupine	Pine forest (red pine)
<u>Lepus americanus</u>	Snowshoe hare	Areas planted with conifers (red pine)
<u>Marmota monax</u>	Woodchuck	Sandy soils (burrowing), woodlots
<u>Mephitis mephitis</u>	Striped skunk	Small woodlots, suburban areas
<u>Mustela frenata</u>	Long-tailed weasel	Fields, small woodlots
<u>Myotis lucifugus</u>	Little brown bat	Abandoned man-made structures
<u>Peromyscus leucopus</u>	White-footed mouse	Woodland edges, occasionally grass/weed fields
<u>Peromyscus maniculatus bairdii</u>	Deer mouse	Open, disturbed areas
<u>Rattus norvegicus</u>	Norway rat	Residential, business, and manufacturing areas
<u>Sciurus carolinensis</u>	Gray squirrel	Red pine wooded areas
<u>Sorex cinereus</u>	Masked shrew	Woodlands, fields
<u>Spermophilus tridecemlineatus</u>	Thirteen-lined ground squirrel	Dry, sandy soils and short grass/weed fields
<u>Sylvilagus floridanus</u>	Eastern cottontail	Interspersed wooded and cleared areas
<u>Tamias striatus</u>	Eastern chipmunk	Human habitations, brush
<u>Tamiasciurus carolinensis</u>	Red squirrel	Coniferous trees (red pine)
<u>Taxidea taxus</u>	Badger	Open areas, sandy soils
<u>Urocyon cinereoargenteus</u>	Gray fox	Wooded and open areas with small mammals
<u>Vulpes vulpes</u>	Red fox	Wooded and open areas with small mammals
<u>Zapus hudsonius</u>	Meadow jumping mouse	Open areas, woodland edges

Source: Rollin H. Baker, Michigan Mammals, Michigan State University Press, 1983.

Table 1-8--Birds Expected at the Cadillac Site

Scientific Name	Common Name	Species Habitat Present at the Cadillac Site
<u>Buteo jamaicensis</u>	Red-tailed hawk	Woodland (red pine) interspersed with fields
<u>Buteo lagopus</u>	Rough-legged hawk	Open areas
<u>Carduelis tristis</u>	American goldfinch	Open weedy fields with scattered small trees
<u>Chaetura pelagica</u>	Chimney swift	Buildings with chimneys
<u>Charadrius vociferus</u>	Killdeer	Bare areas, sparse vegetation, waste areas
<u>Columba livia</u>	Pigeon, rock dove	Human habitations
<u>Corvus brachyrhynchos</u>	American crow	Interior and edges of coniferous woodlots (red pine)
<u>Dendroica petechia</u>	Yellow warbler	Scattered small trees, roadsides
<u>Eremophila alpestris</u>	Horned lark	Open areas with short grasses and bare, exposed soil
<u>Falco sparverius</u>	American kestrel	Forest edges near open ground
<u>Hirundo rustica</u>	Barn swallow	Little-used buildings, rural areas
<u>Melospiza melodia</u>	Song sparrow	Forest edges, roadsides, suburbs
<u>Molothrus ater</u>	Brown-headed cowbird	Forest (red pine) edges
<u>Otus asio</u>	Common screech owl	Small woodlots (red pine)
<u>Poocetes gramineus</u>	Vesper sparrow	Open areas with short vegetation and perches
<u>Quiscalus quiscula</u>	Common grackle	Suburban areas
<u>Sialia sialis</u>	Eastern bluebird	Fields, woodland edges, residential areas
<u>Spizella arborea</u>	American tree sparrow	Weedy pastures
<u>Spizella pusilla</u>	Field sparrow	Open areas with low shrubs or trees
<u>Sturnella magna</u>	Eastern meadowlark	Open areas with widely scattered shrubs
<u>Troglodytes aedon</u>	House wren	Human dwellings near wooded areas
<u>Turdus migratorius</u>	American robin	Woodland (red pine) edges, residential areas
<u>Tyrannus tyrannus</u>	Eastern kingbird	Clearings, fields, forest edges
<u>Zenaidura macroura</u>	Mourning dove	Open, bare ground, suburbs, woodland edges

Sources: DeGraff, R.M., and D.D. Rudis, New England Wildlife: Habitat, Natural History, and Distribution, Northeast Forest Experiment Station, Technical Report NE-108, 1986.
 Peterson, R.T., A Field Guide to the Birds East of the Rockies, Houghton Mifflin Company, 1980.

containing chemicals bioconcentrated in their tissues (food chain exposure). Home range, feeding behavior, and habitat requirements of terrestrial organisms, as well as chemical concentrations in environmental media, are factors affecting the magnitude and probability of exposure.

Analytical data available for the Cadillac site indicate that exposures to terrestrial flora and fauna will be insignificant to non-existent because no surficial chemical contamination currently exists. (Note: There are contaminated soils at Northernair which have not yet been removed. However, these soils have been dealt with in the Northernair Supplemental Remedial Investigation, and are due to be removed in the Spring or Summer of 1988.) Direct contact with wetland sediments and surface waters by terrestrial flora and fauna is not possible as no wetland areas or surface waters bodies were identified at the site. Surface soil contamination existed in the past behind the Kysor facility. However, the current contamination lies well below the surface and the upper zone consists of clean backfill. Therefore, exposures of terrestrial organisms to this contamination via direct contact or ingestion is unlikely.

7.5 RISK CHARACTERIZATION

This section characterizes the risk to humans and environments potentially exposed to chemicals identified in the contaminated groundwater in the Cadillac area. The type and extent of this risk depends upon the nature, duration and frequency of exposure to the contaminants, and on the characteristics of the exposed populations. This information, discussed in Section 7.4, combined with appropriate dose-response toxicity data presented in Section 7.3, is the basis for the developed risk characterization presented in this section.

The risk characterization has been divided into two sections: (1) Public Health Risk Characterization; and (2) Environmental Risk Characterization.

7.5.1 Public Health Risk Characterization

The results of the human health risk characterization are presented in this section. The risk associated with the ingestion of contaminated groundwater from each plume described in Section 7.4 is quantified, and then the significance of the calculated estimates of risk are discussed. Both baseline present and future potential risks are assessed. Carcinogenic and noncarcinogenic risks are determined as applicable to each plume.

Incremental carcinogenic risk estimates are determined by multiplying the body dose levels (calculated as described in Section 7.4) by cancer potency factors. To put these incremental risk levels into perspective, they should be evaluated against a reference (no-effect) level. However, for carcinogens, it is current EPA science policy that cancer induction is a non-threshold phenomenon. Therefore, any exposure poses some probability of causing cancer, and a "safe" (i.e. no-effect) level cannot be determined. EPA guidelines indicate that the target total individual carcinogenic risk resulting from exposures at a Superfund site may range anywhere from 10^{-4} (that is, a dose corresponding to one excess cancer case in a population of 10,000) to 10^{-7} (one excess cancer

case in a population of 10,000,000). Remedial alternatives being considered should be able to reduce total potential carcinogenic risks to individuals to levels within this range (U.S.EPA, 1986). Using this range as a baseline, this risk characterization refers to the calculated carcinogenic risk estimates as being "below the target range" (i.e., less than 10^{-7}), "within the target range" (10^{-7} to 10^{-4}) or "above the target range" (greater than 10^{-4}).

Noncarcinogenic risk estimates are determined by dividing body dose levels for each noncarcinogen by the relevant standard, criterion, or guideline. This results in a ratio. If this ratio is less than or equal to one, the associated body dose level is likely to be without significant lifetime risk to human populations. If the ratio is greater than one, the predicted body dose level could potentially result in adverse health effects, although this is by no means a certainty as the relevant standards or guidelines have conservative safety factors incorporated into their values, and are not exact numbers. However, for the purposes of this risk characterization, if the ratio is less than or equal to one, the risk will be termed "insignificant". If the ratio is greater than one, the risk will be termed "significant".

All risk estimates are presented in Appendix S. In addition, summary risk tables are provided in this section to identify the risks associated with potential ingestion of the contaminated groundwater plumes..

The methods for estimating risks described above consider the risks associated with exposure to single chemicals only, and do not consider the health risks resulting from exposure to a mixture of chemicals. It is also necessary to assess the effects of multiple toxicants on exposed populations. EPA guidelines for assessing effects of exposure to chemical mixtures are used (U.S.EPA, 1986). The EPA approach assumes additivity of effect for each compound, based on noncarcinogenic and carcinogenic endpoints. For noncarcinogens, the following formula is used:

$$HI = E_1/AL_1 + E_2/AL_2 + \dots + E_i/AL_i$$

Where: HI = Hazard Index

E_i = Exposure level to the ith toxicant

AL_i = Maximum acceptable level for the ith toxicant (where acceptable level indicates applicable criteria, standards or advisories derived by EPA).

For carcinogens, the following formula is used:

$$P = D_1B_1 + D_2B_2 + \dots + D_iB_i$$

Where: P = Summary incremental risk

D_i = Dose of the ith carcinogen

B_i = CAG potency value for the ith carcinogen

The hazard indices and summary cancer risks are presented in the summary risk tables and discussed in the text. Following the risk characterization for groundwater ingestion, the uncertainties and limitations of the risk characterization are briefly discussed.

7.5.1.1 Groundwater Ingestion Risk Characterization. The following subsections contain a description of the potential risks associated with ingestion of contaminated groundwater in the Cadillac area. All risk calculations can be found in Appendix S. Summary tables and discussion of the noncarcinogenic and carcinogenic risk estimates are presented below. Risks are characterized according to plume and by exposure scenario.

Shallow Aquifer Plume

Table 7-9 is a summary of the risk calculations for present exposure to the large generalized contaminant plume in the shallow aquifer. Nine chemicals of concern have been detected in the shallow plume. The noncarcinogenic risk associated with the ingestion of water contaminated with 1,1,1-trichloroethane (1,1,1-TCA) ranges from 1.03 to 10.6 (the more that a noncarcinogenic risk estimate exceeds 1, the greater the concern that adverse effects could occur). Acetone is the only chemical which was detected in the shallow plume which poses an insignificant risk in all exposure scenarios. However, acetone is known to potentiate the effects of 1,1,1-TCA, and therefore should not be considered innocuous. Summary hazard indices range from 1.17 to 13.9.

Virtually all of the predicted carcinogenic risks are within or exceed the EPA target range, with 1,1-dichloroethene (1,1-DCE), trichloroethene (TCE), and 1,2-dichloroethane (1,2-DCA) contributing the most to the total risk (approximately 96%). Summary incremental cancer risks exceed the EPA target range for all scenarios, ranging from 4×10^{-4} to 5×10^{-2} . Therefore, the shallow aquifer would pose very high risks to the general exposed population if it were to be used as a drinking water supply at the present time.

The large generalized shallow aquifer plume is assessed from separate source areas, i.e. Kysor and Four Winns, in order to assess future potential risk to exposed populations. Although these plumes are intermingled, a portion of the large plume originating from each source area was to a hypothetical downgradient receptor location (see Section 7.4.1).

Table 7-10 summarizes the risk calculations for the plume which originates in the Kysor area. It is predicted that this plume will impact the receptor well for 15 years, from 2000 to 2015. Five chemicals of concern were detected in this plume, two noncarcinogens and three carcinogens. Neither of the noncarcinogens pose a significant risk at predicted future concentrations in any of the exposure scenarios assessed. However, all three carcinogens pose risks within or above the EPA target risk range for all exposure scenarios. TCE appears to pose the highest risk [approximately five times higher than 1,1-DCE and 15 times higher than tetrachloroethene (PCE)]. Summary hazard indices range from 7×10^{-5} to 1×10^{-3} . This indicates that, if it were to be ingested, the portion of the shallow plume that originates in the Kysor area may still pose an excess risk in the future at a downgradient well, even when attenuation is considered.

There are seven chemicals of concern that have been detected in the portion of the shallow plume that originates in the Four Winns area. The Four Winns plume intermingles with the shallow plume which has sources upgradient of Four Winns. The risk calculations associated with future potential exposure to this

RISK CALCULATION SUMMARY
Cadillac
Shallow Piling (Present)

Noncarcinogenic Effects

Compounds	Most Probable Case		Realistic Worst Case		Most Probable Case		Realistic Worst Case	
	Child	Total Risk	Child	Total Risk	Adult	Total Risk	Adult	Total Risk
1,1,1-TRICHLOROETHANE	1.03	97.6%	10.63	94.9%	1.03	97.6%	10.63	94.9%
TOLUENE	0.01	1.2%	0.30	2.7%	0.01	1.2%	0.30	2.7%
ACETONE	0.01	1.2%	0.26	2.3%	0.01	1.2%	0.26	2.3%
SUMMARY HAZARD INDEX	1.06		11.20		1.06		11.20	

Carcinogenic Effects

Compounds	Most Probable Case		Realistic Worst Case		Most Probable Case		Realistic Worst Case	
	Child	Total Risk	Child	Total Risk	Adult	Total Risk	Adult	Total Risk
1,1-DICHLOROETHENE	2.72E-04	34.9%	2.37E-03	34.9%	1.56E-03	34.9%	1.66E-02	34.9%
TETRACHLOROETHENE	1.34E-05	3.3%	1.90E-04	2.7%	9.41E-05	3.3%	1.09E-03	2.2%
TRICHLOROETHENE	1.29E-04	31.9%	3.50E-03	31.6%	9.05E-04	31.9%	2.45E-02	31.6%
CHLOROFORM	1.32E-07	0.0%	4.96E-04	0.1%	9.26E-07	0.0%	3.47E-05	0.1%
1,2-DICHLOROETHANE	3.88E-05	9.6%	7.43E-04	10.9%	2.71E-04	9.6%	9.20E-03	10.9%
METHYLENE CHLORIDE	1.04E-06	0.3%	2.11E-05	0.3%	7.26E-06	0.3%	1.48E-04	0.3%
SUMMARY INCREMENTAL CANCER RISK		4.03E-04		6.79E-03		2.83E-03		4.75E-02

RISK CALCULATION SUMMARY
Cadillac: Shallen Kysar Place
Receptor: 13th Street

Noncarcinogenic Effects

Groundwater Ingestion

Carcinogenic Effects

Compounds	Most Probable Case		Realistic Worst Case		Most Probable Case		Realistic Worst Case	
	Child	Percent Total Risk	Child	Percent Total Risk	Adult	Percent Total Risk	Adult	Percent Total Risk
1,1-DICHLOROETHENE	1.10E-05	17.8%	2.60E-05	17.9%	1.21E-05	17.7%	1.82E-04	17.9%
TRICHLOROETHENE	9.34E-05	80.3%	1.17E-04	80.4%	9.51E-05	80.6%	8.17E-04	80.4%
TETRACHLOROETHENE	1.14E-06	1.7%	2.50E-06	1.7%	1.19E-06	1.7%	1.75E-05	1.7%
SUMMARY INCREMENTAL CANCER RISK	6.64E-05		1.43E-04		6.85E-05		1.02E-03	

combined plume are summarized in Table 7-11. The fate and transport evaluation of this plume predicts that it will impact the receptor well for approximately 7 years, from 1988 to 1995. As with the shallow Kysor plume, none of the individual noncarcinogens pose significant risks. However, both 1,1,1-TCA and acetone are present in this plume, and may have a synergistic effect that is not quantified in this analysis. Therefore, this plume does cause concern for noncarcinogenic adverse effects in the future. All four carcinogens pose risks within or above the target range at predicted future concentrations in all exposure scenarios. Summary incremental cancer risks range from 5×10^{-5} to 2×10^{-3} . Therefore, the portion of the shallow plume that originates in the Four Winns area may pose an elevated risk in the future if it were to be ingested.

The above estimates of future downgradient concentrations do not include the potential for residual soil contamination to act as a continual source to the shallow groundwater plumes. Soil contamination exists at both Kysor and Four Winns. (Adequate data are not available to assess the fate, transport, exposure, and risk from Four Winns soil contamination. However, it is assumed that soil contamination at Four Winns is sufficient to warrant remediation.) If these soils are not remediated, contaminants could continue to leach into the shallow aquifer and cause higher risks and/or exposure to contaminants for a longer period of time than is assumed above and in Section 7.4.1. The magnitude of these risks cannot be predicted; however, they should not be overlooked.

Rexair Plume

The plume which originates at Rexair is located in the intermediate aquifer, upgradient of the Cadillac municipal well field. Risk to hypothetical present receptors is estimated in Appendix S, and summarized in Table 7-12. Six chemicals of concern have been detected in the Rexair plume, three noncarcinogens and three carcinogens. None of the noncarcinogens in the plume would pose a significant risk to exposed individuals, even if it were to be ingested for a lifetime, and there is no evidence that suggests that these three chemicals would act synergistically. However, TCE risks exceed the target range in all four scenarios, and chloroform and methylene chloride pose cancer risks within the EPA target range for three of the four exposure scenarios assessed. Virtually all (greater than 99%) of the total risk is posed by TCE. Summary incremental cancer risks range from 5×10^{-4} to 5×10^{-2} . This indicates that very high risks to the exposed population would result if the water in the Rexair plume were used as a drinking water source at the present time.

Future potential risk to exposed populations is estimated at two receptor locations, a hypothetical well of 13th Street and the municipal well field.

Table 7-13 summarizes the calculated risks at the 13th Street well. The Rexair plume would impact this location from approximately 1992 to 2007. The same chemicals of concern would still be located in the plume in the future; therefore, the three noncarcinogens would pose a much lower risk than was calculated in the present exposure scenarios. However, TCE risks would exceed the target range for all four exposure scenarios and methylene chloride would still pose risks within the target range. In fact, average contaminant concentrations

RISK CALCULATION SUMMARY
Cadillac: Four Wins Place
Receptor: 13th Street (Future)

Noncarcinogenic Effects

Carcinogenic Effects

Compounds	Most Probable Case		Realistic Worst Case		Most Probable Case		Realistic Worst Case	
	Child Total Risk	Percent	Child Total Risk	Percent	Adult Total Risk	Percent	Adult Total Risk	Percent
1,1-DICHLOROETHENE	2.69E-09	33.5%	1.44E-04	33.4%	2.69E-09	33.5%	1.01E-03	33.4%
TRICHLOROETHENE	2.05E-06	4.1%	1.11E-05	4.1%	2.05E-06	4.1%	7.79E-09	4.1%
TETRACHLOROETHENE	3.33E-06	6.7%	1.89E-05	6.8%	3.33E-06	6.7%	1.30E-04	6.8%
1,2-DICHLOROETHANE	1.77E-05	39.7%	9.64E-05	39.7%	1.77E-05	39.7%	6.74E-04	39.7%
SUMMARY INCREMENTAL CANCER RISK:	4.96E-05		2.71E-04		4.96E-05		1.89E-03	

TABLE 7-12

Groundwater Ingestion

Noncarcinogenic Effects

RISK CALCULATION SUMMARY

Cadillac

Repair Ponds (Present)

Compounds	Most Probable Case Child	Percent Total Risk	Realistic Worst Case Child	Percent Total Risk	Most Probable Case Adult	Percent Total Risk	Realistic Worst Case Adult	Percent Total Risk
1,1,1-TRICHLOROETHANE	0.00	0.7%	0.01	0.8%	0.00	0.7%	0.01	0.8%
TOLUENE	0.00	0.3%	0.00	0.6%	0.00	0.3%	0.00	0.6%
TRANS-1,2-DICHLOROETHENE	0.03	98.0%	0.69	98.6%	0.03	98.0%	0.69	98.6%
SUMMARY HAZARD INDEX		0.03		0.70		0.03		0.70

Groundwater Ingestion

Carcinogenic Effects

Compounds	Most Probable Case Child	Percent Total Risk	Realistic Worst Case Child	Percent Total Risk	Most Probable Case Adult	Percent Total Risk	Realistic Worst Case Adult	Percent Total Risk
CHLOROFORM	9.92E-08	0.0%	1.90E-06	0.0%	6.94E-07	0.0%	1.39E-05	0.0%
METHYLENE CHLORIDE	5.91E-07	0.1%	2.39E-05	0.3%	4.14E-06	0.1%	1.67E-04	0.3%
TRICHLOROETHENE	4.97E-04	99.9%	6.87E-03	99.6%	3.40E-03	99.9%	4.81E-02	99.6%
SUMMARY INCREMENTAL CANCER RISK		4.90E-04		6.90E-03		3.49E-03		4.83E-02

TABLE 7-13

Groundwater Ingestion
Noncarcinogenic Effects

RISK CALCULATION SUMMARY
Cadillac: Repair Plant
Receptors: 13th Street

Compounds	Most Probable Case		Realistic Worst Case		Most Probable Case		Realistic Worst Case	
	Child	Percent Total Risk	Child	Percent Total Risk	Adult	Percent Total Risk	Adult	Percent Total Risk
1,1,1-TRICHLOROETHENE	0.00	0.9%	0.00	0.0%	0.00	0.9%	0.00	0.0%
TOLUENE	0.00	0.6%	0.00	0.6%	0.00	0.6%	0.00	0.6%
TRANS-1,2-DICHLOROETHENE	0.11	98.5%	0.24	98.6%	0.07	98.5%	0.24	98.6%
SUMMARY HAZARD INDEX		0.11		0.25		0.08		0.25

Groundwater Ingestion
Carcinogenic Effects

Compounds	Most Probable Case		Realistic Worst Case		Most Probable Case		Realistic Worst Case	
	Child	Percent Total Risk	Child	Percent Total Risk	Adult	Percent Total Risk	Adult	Percent Total Risk
CHLOROFORM	2.98E-07	0.0%	6.61E-07	0.0%	2.98E-07	0.0%	6.63E-06	0.0%
TRICHLOROETHENE	9.80E-04	99.6%	2.38E-03	99.6%	1.07E-03	99.6%	1.67E-02	99.6%
METHYLENE CHLORIDE	3.44E-06	0.3%	8.27E-06	0.3%	3.94E-06	0.3%	5.79E-05	0.3%
SUMMARY INCREMENTAL CANCER RISK		9.91E-04		2.39E-03		1.07E-03		1.67E-02

over the 10 year period when the concentrations will be highest at the receptor well (1992 to 2002) are predicted to be higher than current average concentrations. Again, TCE accounts for greater than 99% of the predicted risks. Summary incremental cancer risks range from 1×10^{-5} to 2×10^{-2} . Therefore, if groundwater from the Rexair plume were to be used as a drinking water source at a hypothetical downgradient receptor well in the future, it will pose high risks.

The future impact of the Rexair plume on the municipal supply well is assessed for TCE only (see Section 7.4.2). The plume is predicted to be drawn into the municipal well field approximately two years from 1988. The cancer risk associated with lifetime ingestion of TCE in the municipal well field is 3×10^{-5} (see Table 7-14). This indicates that there is cause for concern for the municipal well field. The predicted concentration (100 ppb) is also well above the MCL for TCE of 5 ppb.

Intermediate Aquifer Plume

The risk associated with present exposure to the plumes in the intermediate aquifer on the west side of the study area (downgradient of the municipal well field) is assessed separately for: (1) the large generalized VOC plume and (2) the hexavalent chromium plume (originating at Northernnaire) and intermingled VOCs and cyanide. The Northernnaire plume is found much deeper in the intermediate aquifer and, although the intermingled VOCs probably descended through the aquifer from the larger generalized plume, it is unlikely that a domestic supply well would be screened at a wide enough interval to intercept both plumes.

The risk calculations for hypothetical exposure to the large generalized intermediate plume are summarized in Table 7-15. Nine chemicals of concern have been detected in this plume, four noncarcinogens and five carcinogens. T-1,2-dichloroethene poses a significant risk in all four exposure scenarios assessed (hazard ratio = 5.99 to 65.7), and accounts for virtually all of the total noncarcinogenic risk. As in the shallow plume, acetone and 1,1,1-TCA commingle in this plume, increasing the concern for potential synergistic effects. Summary hazard indices range from 6.06 to 66.5. Summary incremental cancer risks range from 1×10^{-4} to 4×10^{-3} , with TCE and 1,1-DCE accounting for more than 97% of the total risk in all four exposure scenarios. Therefore, the intermediate aquifer would pose very high risks to the general exposed population if it were to be used as a drinking water supply at the present time.

Table 7-16 summarizes the risks associated with hypothetical ingestion of chromium, cyanide, and intermingled VOCs in the deeper portion of the intermediate aquifer (near Northernnaire). Seven chemicals of concern have been detected in this portion of the aquifer, including four noncarcinogens and three carcinogens. Cyanide is included in this analysis, although it was not detected in the two latest sampling rounds (see Section 7.2 for explanation). Using the maximum value of cyanide ever detected (1985), the hazard ratio is insignificant (0.02). Therefore cyanide will not be discussed further in the risk assessment. The only noncarcinogen which poses a potentially significant risk at present levels is chromium; the potential risk associated with this

DISK: 213
FILE: REX_NOM

TABLE 7-14

RISK CALCULATION SUMMARY

Cadillac: Repair Plus

Receptor: Municipal Hall (Future)

This table calculates estimated body doses and incremental cancer risks.

The equations to calculate body dose level and incremental cancer risks are:

Body Dose (ug/kg/day)	= Concentration (ug/l)	× Amount Ingested (l/day)	× Fraction Absorbed ×	1 ----- Body Weight (kg)	×	Frequency of Exposure ----- 365 days	× No. of Years of Exposure ----- 70 yr. Lifetime
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Incremental Cancer Risk = Body Dose x Carcinogenic Potency Factor

[illegible]

TABLE 7-15

Groundwater Ingestion

Noncarcinogenic Effects

RISK CALCULATION SUMMARY

Cadillac

Intermed. Site Plume (Present)

Compounds	Most Probable Case		Realistic Worst Case		Most Probable Case		Realistic Worst Case	
	Child	Percent Total Risk	Child	Percent Total Risk	Adult	Percent Total Risk	Adult	Percent Total Risk
1,1,1-TRICHLOROETHANE	0.06	1.0%	0.40	0.6%	0.06	1.0%	0.40	0.6%
TOLUENE	0.00	0.0%	0.04	0.1%	0.00	0.0%	0.04	0.1%
ACETONE	0.01	0.2%	0.00	0.0%	0.01	0.2%	0.00	0.0%
TRANS-1,2-DICHLOROETHENE	3.99	98.8%	69.71	99.3%	3.99	98.8%	69.71	99.3%
SUMMARY HAZARD INDEX		6.06		66.19		6.06		66.19

Groundwater Ingestion

Carcinogenic Effects

Compounds	Most Probable Case		Realistic Worst Case		Most Probable Case		Realistic Worst Case	
	Child	Percent Total Risk	Child	Percent Total Risk	Adult	Percent Total Risk	Adult	Percent Total Risk
1,1-DICHLOROETHENE	2.84E-05	21.8%	2.34E-04	40.2%	1.99E-04	21.8%	1.64E-03	40.2%
TETRACHLOROETHENE	1.67E-07	0.1%	3.94E-06	0.6%	1.17E-06	0.1%	2.40E-05	0.6%
TRICHLOROETHENE	1.01E-04	77.6%	3.37E-04	57.7%	7.00E-04	77.6%	2.36E-03	57.7%
CHLOROFORM	6.61E-08	0.1%	1.32E-06	0.2%	4.63E-07	0.1%	9.26E-06	0.2%
METHYLENE CHLORIDE	4.62E-07	0.4%	7.35E-06	1.3%	3.24E-06	0.4%	5.14E-05	1.3%
SUMMARY INCREMENTAL CANCER RISK		1.30E-04		5.83E-04		9.11E-04		4.00E-03

TABLE 7-16

Groundwater Ingestion

Noncarcinogenic Effects

RISK CALCULATION SUMMARY

Cadillac

Northernline Plume (Present)

Compounds	Most Probable Case		Realistic Worst Case		Most Probable Case		Realistic Worst Case	
	Child	Percent Total Risk	Child	Percent Total Risk	Adult	Percent Total Risk	Adult	Percent Total Risk
CHROMIUM	0.59	92.4%	4.42	89.7%	0.59	92.4%	4.42	89.7%
CYANIDE	0.00	0.0%	0.02	0.4%	0.00	0.0%	0.02	0.4%
1,1,1-TRICHLOROETHANE	0.01	2.3%	0.15	2.8%	0.01	2.3%	0.15	2.8%
TRANS-1,2-DICHLOROETHENE	0.03	5.3%	0.97	11.1%	0.03	5.3%	0.97	11.1%
SUMMARY HAZARD INDEX		0.64		5.16		0.64		5.16

Groundwater Ingestion

Carcinogenic Effects

Compounds	Most Probable Case		Realistic Worst Case		Most Probable Case		Realistic Worst Case	
	Child	Percent Total Risk	Child	Percent Total Risk	Adult	Percent Total Risk	Adult	Percent Total Risk
1,1-DICHLOROETHENE	6.02E-06	30.7%	6.16E-05	35.3%	2.82E-05	30.7%	4.31E-04	35.3%
TETRACHLOROETHENE	2.00E-08	0.2%	6.24E-07	0.4%	1.46E-07	0.2%	4.37E-06	0.4%
TRICHLOROETHENE	9.06E-06	69.1%	1.12E-04	64.4%	6.34E-05	69.1%	7.86E-04	64.4%
SUMMARY INCREMENTAL CANCER RISK		1.31E-05		1.74E-04		9.17E-05		1.22E-03

chemical ranges from 0.6 to 5. The summary incremental cancer risk for the three carcinogens ranges from 1×10^{-5} to 1×10^{-3} , and TCE and 1,1-DCE pose more than 99% of the total risk in all exposure scenarios. Therefore, this portion of the intermediate aquifer may pose very high risks to the general exposed population.

Future potential risk associated with use of the intermediate aquifer as a drinking water supply is assessed separately for the three identified source areas, i.e. Kysor, Four Star, and the Northernnaire hexavalent chromium plume. Although the plumes associated with Kysor and Four Star are intermingled and cannot be completely separated, the portion of the large plume originating from each source area was modeled to a hypothetical downgradient receptor location (see section 8.4.1).

The risk calculations for future potential exposure to the intermediate Kysor plume are summarized in Table 7-17. Unlike the other plumes assessed thus far in the risk characterization, this plume appears to originate from a relatively constant source rather than a point slug source. Three chemicals of concern were detected in this portion of the plume.

Trans-1,2-Dichloroethene was the only noncarcinogenic chemical detected, and at predicted future concentrations it would pose a significant risk in all four hypothetical exposure scenarios (hazard ratio = 9.5 to 17.1). Both carcinogens detected pose risks within the EPA target range in all exposure scenarios. Summary incremental cancer risks range from 6×10^{-5} to 7×10^{-4} , and TCE accounts for almost 98% of the total calculated carcinogenic risk. This indicates that chemicals in the portion of the intermediate plume which originates in the Kysor area may pose high risks to individuals in the future based on the assumed exposure scenarios.

The portion of the intermediate aquifer plume which originates in the vicinity of Four Star was modeled to a hypothetical receptor well on 13th Street. Table 7-18 summarizes the risk calculations performed for future potential exposure to the Four Star plume. Three chemicals of concern were detected in the plume, 1,1,1-TCA, PCE, and TCE. The noncarcinogenic risks associated with 1,1,1-TCA are insignificant for all four exposure scenarios. The summary incremental cancer risks associated with future potential PCE and TCE exposure range from 2×10^{-4} to 5×10^{-3} , with each chemical contributing approximately half to the total risk. Thus, chemicals originating in the Four Star portion of the intermediate aquifer plume could also pose a high risk in the future in the absence of remediation.

Table 7-19 summarizes the calculations of future potential risk associated with the hexavalent chromium plume which originated at Northernnaire. Hazard ratios associated with future potential ingestion of hexavalent chromium in the intermediate aquifer range from 0.7 to 3.4. Future use of the portion of the intermediate aquifer containing the Northernnaire plume as a drinking water source may pose a significant risk to exposed populations.

7.5.1.2 Uncertainties and Limitations. Regardless of the type of risk estimates calculated, it should be emphasized that all estimates of risks are based upon numerous assumptions, introducing many uncertainties into the risk

RISK CALCULATION SUMMARY
Cadillac: Inter. Kysor Plaza
Receptor: Leeson Ave. (Future)

Noncarcinogenic Effects

Compounds	Most Probable Case Child	Percent Total Risk	Realistic Worst Case Child	Percent Total Risk	Most Probable Case Adult	Percent Total Risk	Realistic Worst Case Adult	Percent Total Risk
TRANS-1,2-DICHLOROETHENE	9.30	100.0%	17.14	100.0%	9.30	100.0%	17.14	100.0%
SUMMARY HAZARD INDEX	9.30		17.14		9.30		17.14	

Carcinogenic Effects

Compounds	Most Probable Case		Realistic Worst Case		Most Probable Case		Realistic Worst Case	
	Child	Percent Total Risk	Child	Percent Total Risk	Adult	Percent Total Risk	Adult	Percent Total Risk
METHYLENE CHLORIDE	1.16E-06	2.1%	2.14E-06	2.2%	8.14E-06	2.1%	1.90E-05	2.2%
TRICHLOROETHENE	5.39E-05	97.9%	9.61E-05	97.8%	3.77E-04	97.9%	6.73E-04	97.8%
SUMMARY INCREMENTAL CANCER RISK:	5.50E-05		9.82E-05		3.85E-04		6.88E-04	

TABLE 7-18

Groundwater Ingestion

Noncarcinogenic Effects

RISK CALCULATION SUMMARY

Cadillac Four Star Plaza

Receptors: 13th Street

Compounds	Most Probable Case Child	Percent Total Risk	Realistic Worst Case Child	Percent Total Risk	Most Probable Case Adult	Percent Total Risk	Realistic Worst Case Adult	Percent Total Risk
1,1,1-TRICHLOROETHANE	0.02	100.0%	0.04	100.0%	0.01	100.0%	0.04	100.0%
SUMMARY HAZARD INDEX		0.02		0.04		0.01		0.04

Groundwater Ingestion

Carcinogenic Effects

Compounds	Most Probable Case Child	Percent Total Risk	Realistic Worst Case Child	Percent Total Risk	Most Probable Case Adult	Percent Total Risk	Realistic Worst Case Adult	Percent Total Risk
TETRACHLOROETHENE	1.22E-04	52.5%	3.62E-04	94.5%	1.36E-04	55.1%	2.54E-03	94.5%
TRICHLOROETHENE	1.10E-04	47.5%	3.02E-04	45.5%	1.11E-04	44.9%	2.12E-03	45.5%
SUMMARY INCREMENTAL CANCER RISK		2.32E-04		6.64E-04		2.48E-04		4.66E-03

DISK: 214
FILE:NORT_FUT

TABLE 7-19
RISK CALCULATION SUMMARY

Cadillac: Northernair Chromium Plume
Receptor: 13th Street (Future)

This table calculates estimated body doses and risk ratios.

The equations to calculate body dose level and risk ratio are:

$$\text{Body Dose} = \frac{\text{Groundwater Concentration (ug/l)} \times \text{Amount Ingested (l/day)} \times \text{Fraction Absorbed} \times 1}{\text{Body Weight (kg)}}$$

$$\text{Risk Ratio} = \frac{\text{Body Dose}}{\text{Standard or Guideline}}$$

Compound	Concentration (ug/l)	Amount Ingested (l/day)	Fraction Absorbed	Body Weight (kg)	Number of Exposures (Per Year)	Body Dose (ug/kg/day)	Standard/ Guideline (ug/kg/day)	Risk Ratio
CHROMIUM VI								
Most Probable								
Child	295	1	1	35	365	8.43	5	1.69
Adult	122	2	1	70	365	3.49	5	0.70
Realistic Worst								
Child	600	1	1	35	365	17.14	5	3.43
Adult	600	2	1	70	365	17.14	5	3.43

characterization process. These uncertainties are introduced in many areas, including (but not limited to):

- o Exposure Assumptions
- o Toxicokinetics
- o Fate and Transport Modeling
- o Additional Routes of Exposure
- o Use of Established Dose-Response Values
- o Possible Synergistic/Antagonistic Effects
- o Sensitive Subpopulations

These areas of uncertainty are discussed briefly below.

Groundwater ingestion assumptions have been standardized by EPA guidelines (i.e., a 70 kg adult drinks 2 liters of water per day for a 70 year lifetime, and a child drinks 1 liter of water per day over the exposure period). However, use of standardized exposure parameters may overestimate or underestimate the actual risk. For instance, EPA guidelines state that a gastrointestinal (GI) toxicokinetic factor of one should be used in the absence of known GI absorption rates. In effect, this means that it is assumed that human GI absorption of the chemical is the same as that of the test animal used to derive the pertinent standard or guideline. This is a good estimate for many chemicals, but it may overestimate or underestimate actual absorption for other chemicals by as much as an order of magnitude or more.

There are also limitations in the accuracy of fate and transport calculations. The assumption of exposure to point concentrations (maximum levels at receptor wells) may lead to overestimation of risk. The use of conservative parameters (such as lifetime ingestion and maximum contaminant levels) can also lead to an overestimation of risk. However, it should also be noted that ingestion is the only route of exposure that is assessed in this risk characterization. Other routes of exposure associated with domestic use of water (including showering, bathing, cooking, etc.) may also add significantly to the calculated risks, leading to an underestimation of actual risks. These factors may offset each other, but the extent to which this happens cannot be determined.

The use of established standards, criteria, guidelines, and unit cancer risk values based on animal studies introduces additional uncertainties into the risk estimates. This is due to such factors as extrapolating potential adverse human health effects from animal studies, extrapolating effects observed at high doses to low dose effects, and modeling dose-response effect.

The formulas used for assessing chemical mixtures have inherent limitations in that they do not take possible synergistic or antagonistic effects into account. When data on chemical interactions were available (e.g. 1,1,1-TCA and acetone) they are noted in the risk characterization. However, due to the number of chemicals detected and the possibility of unknown chemical interactions, it is possible that the actual total risk for each scenario is greater than that calculated.

In addition, within any population there are certain individuals who are intrinsically more sensitive to the hazardous chemicals found in the Cadillac

groundwater plumes. This includes infants, the elderly, pregnant women, and people with immunodeficiencies or specific organ pathology. It is not possible to determine the size of all of these groups. However, if individuals in any of the sensitive subpopulations noted above were to use the groundwater in the Cadillac area as a drinking water source, the calculated risk estimates would probably underestimate their actual risk.

If the uncertainties associated with a risk characterization are felt to outweigh its usefulness or meaning, the risk estimates should not be used and risk should be assessed qualitatively. However, by using a range of realistic exposure assumptions, some of the uncertainty that is inherent in the risk characterization process has been reduced. A realistic lower and upper bound for risk have been provided for each exposure scenario. It should be stressed that these numbers are not indices of absolute risk, but rather a range in which the actual risk should be contained. They should be used to determine the location and relative magnitude of risk from Cadillac groundwater, the relative magnitude of risk from plume to plume, and the magnitude of risk from Cadillac groundwater compared to groundwater in other areas.

This risk characterization serves as the baseline or present and future potential "no action" assessment for the Cadillac area groundwater study. The derived risk estimates are all based on hypothetical exposure scenarios, assuming exposures that do not presently exist. This was done in order to establish baseline conditions in each plume so that the effectiveness of various remedial alternatives can be compared against that baseline in the Cadillac Area Groundwater Feasibility Study. However, it should be emphasized that a risk does not exist unless there are two basic conditions: (1) the presence of a chemical at sufficient concentrations, and (2) an exposed population that is coming into contact with that chemical. Therefore, the risks estimated here are hypothetical. However, there are no restrictions on use of this water as a domestic supply, and therefore the calculated risks could apply if the water were to be used in the future.

Regardless of the assumption, uncertainties, and limitations that are associated with this and all risk characterizations, this risk characterization does, within these limitations, present an accurate representation of the present and future potential for risk to public health posed by the contaminated groundwater in the Cadillac area.

7.5.2 Ecological Risk Characterization

Ecological risk characterization is the process of predicting the probability of toxic effects to the ecosystem or ecosystem components (communities, populations, and individuals) resulting from exposure to the contaminants of concern. At the Cadillac site, no chemicals of concern in surface waters, surface soils, or sediments have been identified. Potential terrestrial receptor organisms were characterized (see Section 7.4.2), but there were no identifiable possible routes of exposure for these organisms to site contamination. Therefore, risks of adverse effects to these organisms due to exposure to chemical contaminants from the site are minimal to non-existent.

7.6 SUMMARY

Sections 7-2 through 7-5 of the baseline risk evaluation characterize the potential risk to human and environmental receptors posed by the groundwater in the Cadillac Industrial Park area. The groundwater plumes which are evaluated are designated as:

- o Shallow Plume
- o Intermediate Plume
- o Northernnaire Plume
- o Rexair Plume

Section 7-2 identifies the following chemicals of concern in the groundwater, based on the inherent toxicity, frequency of detection in and distribution among the various plumes, and concentrations of each chemical detected:

- 1,1,1-Trichloroethane
- Acetone
- Toluene
- Trans,-1,2-Dichloroethene
- Chloroform
- 1,1-Dichloroethene
- 1,2-Dichloroethane
- Methylene Chloride
- Tetrachloroethene
- Trichloroethene
- Hexavalent Chromium
- Cyanide

Section 7.3 contains information on the potential hazard of each of the chemicals listed above. First, the potential adverse effects of each chemical are briefly summarized in toxicity profiles. Next, the pertinent available standards and criteria for the evaluation of the chemicals are listed in Table 7-3, and briefly explained in the text. Finally, June and August 1987 contaminant concentrations in the four major plumes are compared to chemical-specific applicable or relevant and appropriate requirements (ARARs) in Table 7-4. Each of the chemicals of concern are potentially hazardous to human health. In addition, all four plumes currently do not meet applicable health-based requirements for these chemicals.

The exposure assessment (Section 7.4) identifies the populations and environments which are at risk of exposure to chemicals in the groundwater, and the major pathways through which they could come into contact with the chemicals. The first subsection, Fate and Transport, defines the plumes that are assessed. The major plumes that are listed above are further subdivided into areas of higher concentration, as follows:

- o Shallow Aquifer Plume
 - Residual Soils at Kysor
 - General Shallow Kysor plume
 - Four Winns plume

- o Rexair Plume
- o Intermediate Aquifer Plume
 - Generalized Intermediate Plume
 - Four Star Plume
- o Northernnaire Plume

Various modeling techniques are then utilized to follow contaminant migration in each of these plumes in order to predict chemical concentrations at downgradient receptor wells in the future .

The public health exposure assessment identifies the human populations that may live in or use the area around the site. There are currently no domestic supply wells that intercept the contaminated groundwater plumes in the study area. However, there is a potential for impact to these populations in the future, either through use of the shallow or intermediate aquifer as a domestic supply, or if the plumes impact the municipal supply wells. The potential for the groundwater to impact present and future human populations is assessed by calculating exposure dose levels associated with the ingestion of contaminated groundwater from each plume.

The environmental exposure assessment describes the terrestrial systems present at the site that are potential ecological receptors of contamination. There were no potential routes of exposure identified, as current soil contamination lies well below the surface. Therefore, exposures of terrestrial organisms to contamination via direct contact and ingestion is unlikely.

Section 7.5 characterizes the risk to humans and environments potentially exposed to chemicals identified in the contaminated groundwater in the Cadillac area. In the public health risk characterization, each plume is assessed using exposure dose levels calculated in Section 7.4 from present concentrations (from current on-site well data) and (modeled) future concentrations, and the appropriate guideline or criterion identified in Section 7.3. Virtually all of the plumes pose a risk to human health, both at current contaminant concentrations and at predicted future concentrations. The chemicals which contribute the most to the total risks in each plume are:

SHALLOW AQUIFER PLUME

Present Exposure

(General Plume Only):
 1,1,1-Trichloroethane
 1,1-Dichloroethene
 Trichloroethene
 1,2-Dichloroethane

Future Potential Exposure

(General Kysor Plume):
 Trichloroethylene
 1,1-Dichloroethene
 Tetrachloroethene

 (Four Winns Plume):
 1,1-Dichloroethene

1,2-Dichloroethene
Tetrachloroethene
Trichloroethene

REXAIRE PLUME

Present Exposure

Trichloroethene
Methylene Chloride

Future Potential Exposure

(13th Street Receptor Well):
Trichloroethene
Methylene Chloride
(Municipal Supply Well):
Trichloroethene

INTERMEDIATE PLUME

Present Exposure

(General Plume Only):
Trans-1,2-Dichloroethene
Trichloroethene
1,1-Dichloroethene

Future Potential Exposure

(General Kysor Plume):
Trans-1,2-Dichloroethene
Trichloroethylene

(Four Star Plume):
Tetrachloroethene
Trichloroethene

NORTHERNAIRE PLUME

Present Exposure

Hexavalent Chromium
Trichloroethene
1,1-Dichloroethene

Future Potential Exposure

Hexavalent Chromium

There are many uncertainties inherent in the risk assessment process which are noted in Section 7.5. However, this risk characterization presents an accurate representation of the present and future potential for risk to public health posed by contaminated groundwater in the Cadillac area, within these inherent limitations. High risks to potentially exposed human populations are posed by all plumes evaluated. In addition, as noted in Section 7.3, none of the plumes currently meet applicable health-based requirements.

A brief ecological risk characterization is also presented in Section 7.5. As noted in the exposure assessment, there are terrestrial organisms present at the site. However, there are no identifiable routes of exposure for these organisms to site contamination. Therefore, the risk of adverse effects to these organisms due to exposure to chemical contaminants from the site is minimal to non-existent.

GLOSSARY OF ACRONYMS

ACLs	Alternative Concentration Limits
ARARs	Applicable or Relevant and Appropriate Requirements
ASI	Aquatic Systems, Inc.
AWQC	Ambient Water Quality Criteria
BDAT	Best Demonstrated Available Technology
CERCLA	Comprehensive Environmental Response, Compensation, and Liability Act
CLP	Contract Laboratory Program
COP	Caucus on Organic Protocols
CRC	Cadillac Molded Rubber Company
CWA	Clean Water Act
DCA	dichloroethane
DCE	dichloroethene
DOT	Department of Transportation
EDI	EDI Engineering & Science
EP	Extraction Procedure
FS	Feasibility Study
GC	gas chromatograph
gpd	gallons per day
HASP	Health and Safety Plan
HSL	Hazardous Substance List
HSWA	Hazardous and Solid Waste Amendments
MCLs	Maximum Contaminant Levels
MCLGs	Maximum Contaminant Level Goals
MDNR	Michigan Department of Natural Resources
MSL	Mean Sea Level
NCP	National Contingency Plan
NPDES	National Pollution Discharge Elimination System
NPL	National Priorities List
OSHA	Occupational Safety and Health Administration
PCE	tetrachloroethene
PI	photoionization
POTW	Publicly Owned Treatment Works
ppb	parts per billion
ppm	parts per million
PRP	Potentially Responsible Party
QAPP	Quality Assurance Project Plan

RA	Risk Assessment
RAMP	Remedial Action Master Plan
RCRA	Resource Conservation and Recovery Act
RI	Remedial Investigation
SARA	Superfund Amendments and Reauthorization Act
SDWA	Safe Drinking Water Act
TBC	To Be Considered
TCA	trichloroethane
TCE	trichloroethene
TSP	trisodium phosphate
USCS	Unified Soils Classification System
USEPA	U.S. Environmental Protection Agency
USGS	U.S. Geological Survey
VOCs	Volatile Organic Compounds